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MOHSIN IBRA HOSSEIN
AIUB, DLAC NOTE'S.

FINAL
TERM:-

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SEQUENTIAL
CIRCUIT
DESIGN

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LATCHES

Sub: _____

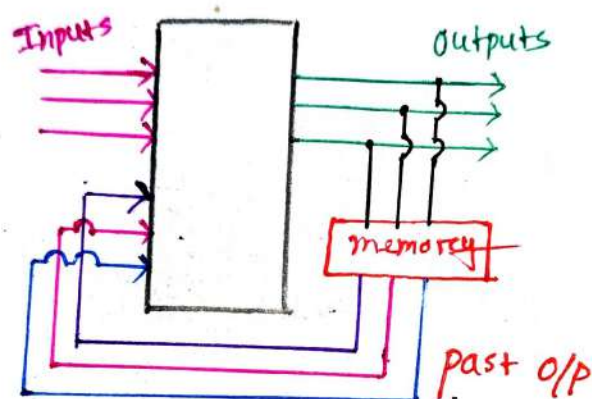
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Time: _____

Date: / /

Sequential Circuits:—

⇒ In Sequential Circuits, the present output depends on the present input as well as past output/outputs.



SR Latch:—

⇒ The basic storage element is called "LATCH".

There are two types:—

1. NOR SR Latch.

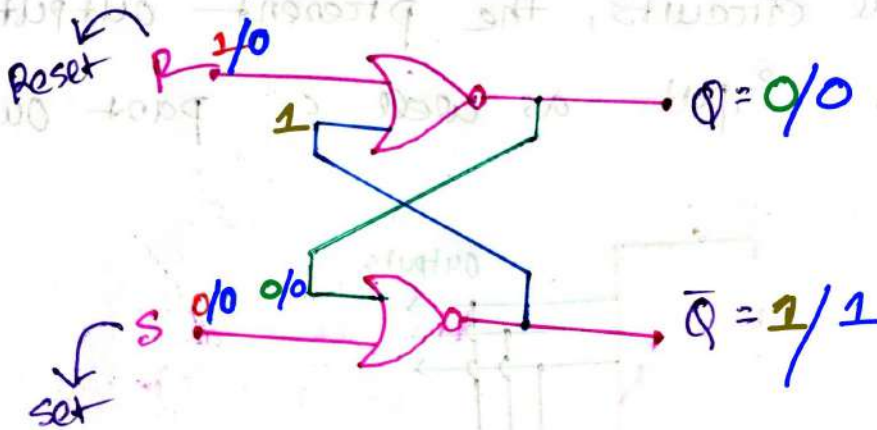
2. NAND SR Latch

Sub: _____

Day _____

Time: _____

Date: / /

SR NOR Latch:-Note:-Reset $Q = 0$ Set $Q = 1$ ☒ TRUTH TABLE: For NOR Gate:-

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

☒ Analyse the SR Latch:-☒ Case 1:-if $S = 0, R = 1$; then $Q = 0 \& \bar{Q} = 1$ Ex:- if $S = 0, R = 0$ then $Q = 0 \& \bar{Q} = 1$ $\rightarrow$ memory☒ Case: 2:-if $S = 1, R = 0$; then $Q = 1 \& \bar{Q} = 0$ $\therefore$ Ex:- if $S = 0, R = 0$; then $Q = 1 \& \bar{Q} = 0$ $\rightarrow$ memory

Sub: _____

Day

Time: _____

Date: / /

Case 3: —
common

if $S=1, R=1$ then:- $Q=0$ & $\bar{Q}=0$

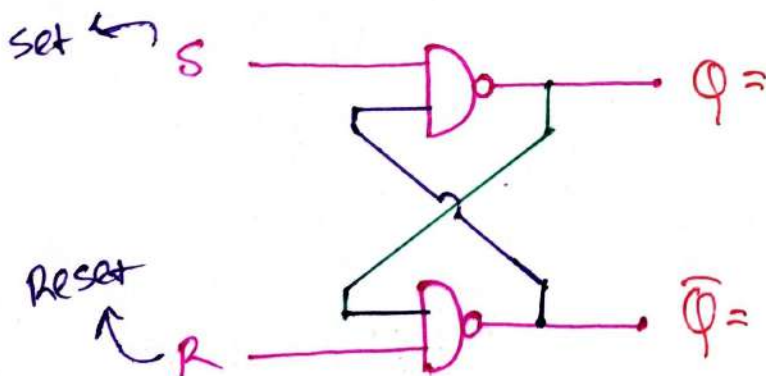
Ex if:- $S=0, R=0$; then $Q=0$ & $\bar{Q}=1$
 $Q=1$ & $\bar{Q}=0$

"Not used"
as it's break
the rules.

Truth Table for SR Latch: (NOR)

S	R	Q	$\bar{Q}$
0	0	Memory	Memory
0	1	0	1
1	0	1	0
1	1	Not used	Not used

SR NAND Latch: —



Truth table for NAND Gate: —

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

Sub: _____

Time: _____

Date: / /

Analyse the Latch:-

Case 1:-

if $S=0, R=1$ then, $Q=1$ & $\bar{Q}=0$

Ex:- if $S=1, R=1$ then, $Q=1$ & $\bar{Q}=0$ → memory

Case 2:-

if $S=1, R=0$ then:- $Q=0$ & $\bar{Q}=1$

Ex:- if $S=1, R=1$ then:- $Q=0$ & $\bar{Q}=1$ → memory

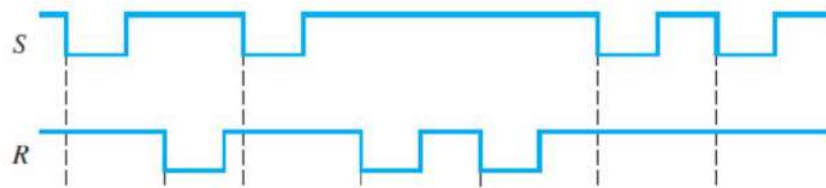
Case 3:-

if $S=0, R=0$, then:- $Q=1$ & $\bar{Q}=1$ → Not used

Truth table for SR latch (NAND):-

S	R	Q	$\bar{Q}$
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	memory	

Exercise 1:

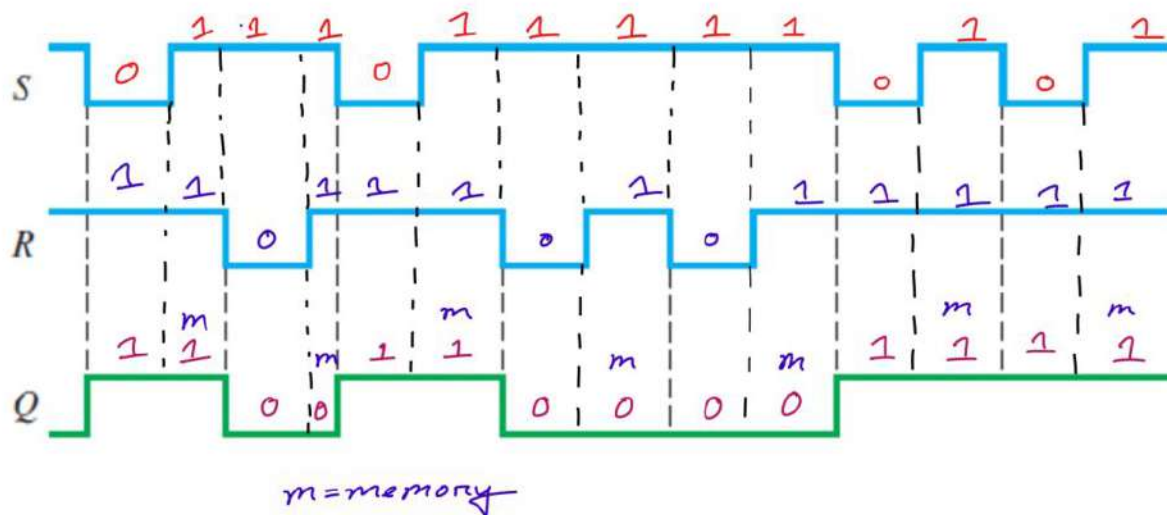


Determine the **Output(Q) Waveform** for **NAND SR Latches** that are initially **LOW**.

➔ We Know,

Truth table for SR latch (NAND):

S	R	Q	$\bar{Q}$
0	0	Not Used	
0	1	1	0
1	0	0	1
1	1	memory	



Sub: _____

Day

Time: _____

Date: / /

#Clock

Sub: _____

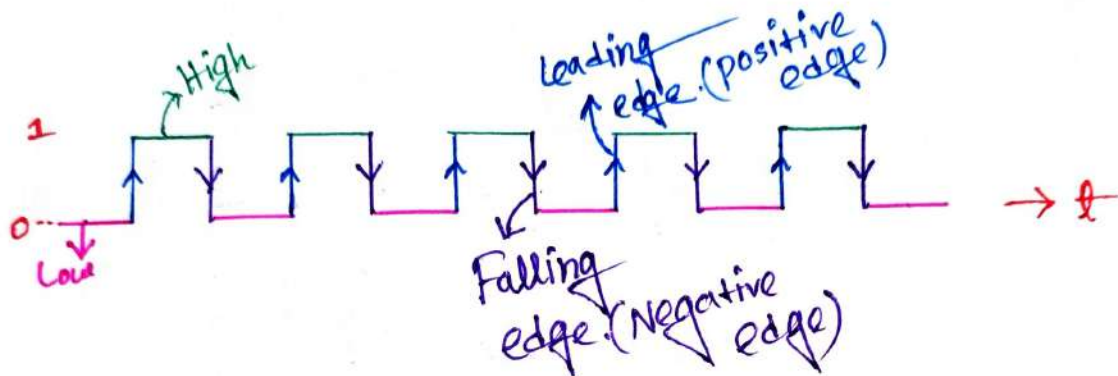
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Clock:-

⇒ It's a signal that goes from low to High then again low and repeats.



Triggering Methods:-

⇒ It's usually refer to how flip-flops and other Sequential elements change their states.

✱ There are two types:-

1. Level-triggered.

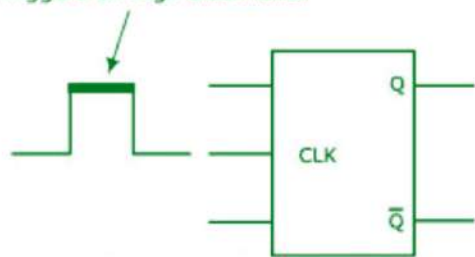
2. Edge-triggered.

Level-triggered:-

⇒ This flip-flops respond to the level of the clock signal, either high (1) or Low (0).

Block Diagram for high level:-

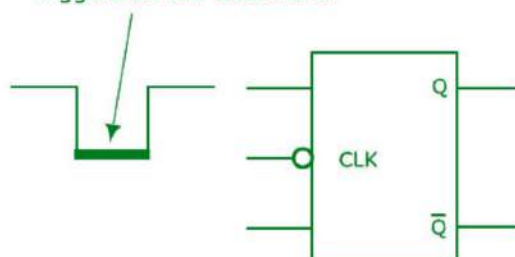
Triggers on high clock level



High Level Triggering

Block Diagram for Low level:-

Triggers on low clock level



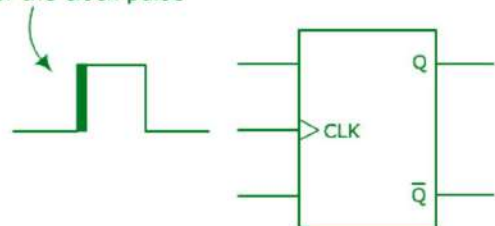
Low Level Triggering

Edge-triggered:-

⇒ These flip-flops change state on a specific edge of the clock signal, either the rising edge (positive edge) or the falling edge (Negative edge)

Block Diagram For (+ve edge)

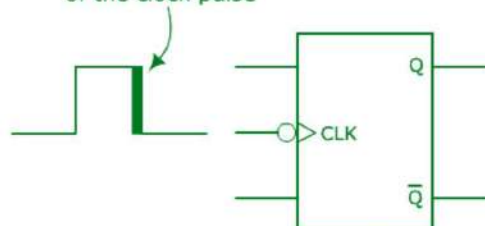
Triggers on this edge of the clock pulse



Positive Edge Triggering

Block Diagram for (-ve edge)

Triggers on this edge of the clock pulse



Negative Edge Triggering

Sub: _____

Time: _____

Date: / /

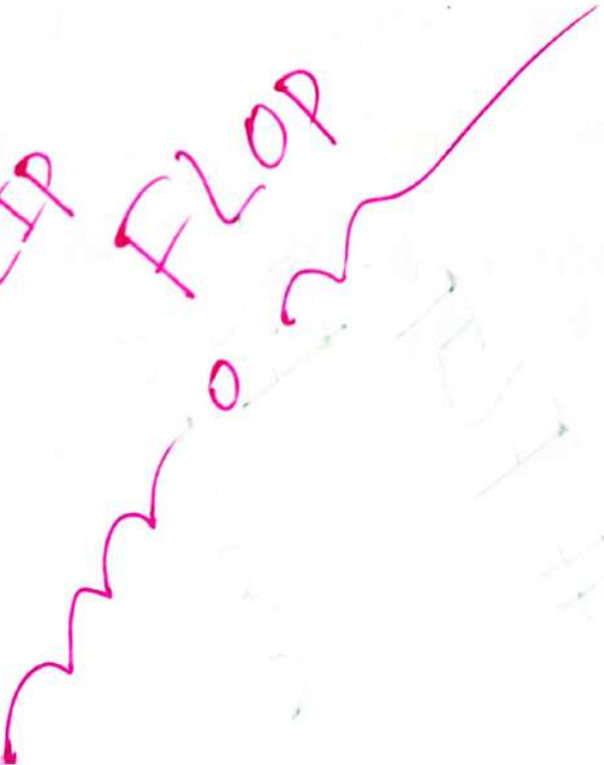
FLIP
Flop

Sub :

Time :

Date : / /

SR FLIP FLOP

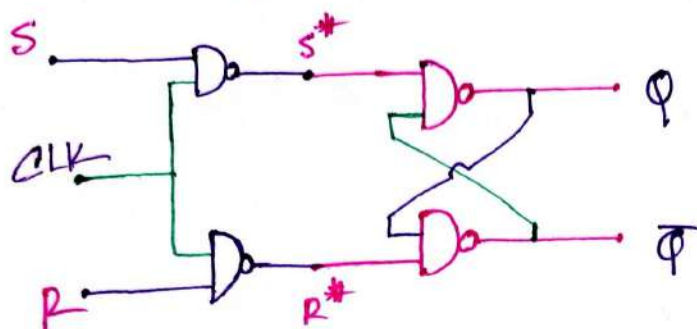


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Time: _____

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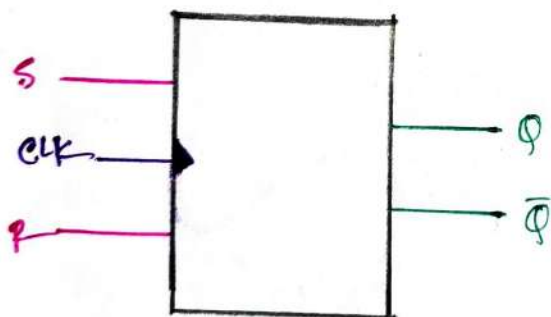
SR Flip Flop

Truth table of SR Latch (NAND)

S*	R*	Q	Q̄
0	0	Not used	
0	1	1	0
1	0	0	1
1	1	memory	

$$\therefore S^* = (\overline{S \cdot CLK}) = \overline{S} + \overline{CLK}$$

$$\therefore R^* = (\overline{R \cdot CLK}) = \overline{R} + \overline{CLK}$$

Block Diagram of SR Flip-Flop:-Truth table for SR Flip-Flop:-

CLK	S	R	Q	Q̄
0	X	X	memory	
1	0	0	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	Not used	

Analyse the truth table:-Case 1:-

$$CLK = 0;$$

$$S^* = \overline{S} + 1 = X + 1 = 1$$

$$R^* = \overline{R} + 1 = X + 1 = 1$$

$$\therefore Q \text{ \& } \overline{Q} = \text{memory}$$

Case 2:-

$$CLK = 1;$$

$$S^* = \overline{S}$$

$$R^* = \overline{R}$$

$$\text{Let, } S=0, R=0 \rightarrow$$

$$\therefore S^* = 1$$

$$R^* = 1$$

$$\therefore Q \text{ \& } \overline{Q} = \text{memory}$$

Case 3:-

from eq (1):-

$$\text{Let, } S=0, R=1$$

$$\therefore S^* = 1$$

$$R^* = 0$$

$$\therefore Q = 0 \text{ \& } \overline{Q} = 1$$

Case 4:-

from eq (1):-

$$\text{Let, } S=1, R=0$$

$$\therefore S^* = 0$$

$$R^* = 1$$

$$\therefore Q = 1 \text{ \& } \overline{Q} = 0$$

Characteristic Truth Table for SR Flip-Flop:-

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Invalid(x)
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Invalid(x)

Truth table for SR-Flip-flop

clk	S	R	Q_{n+1} ^{→ next state}
0	X	X	Q_n ^{→ p. state}
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

Excitation Table:-

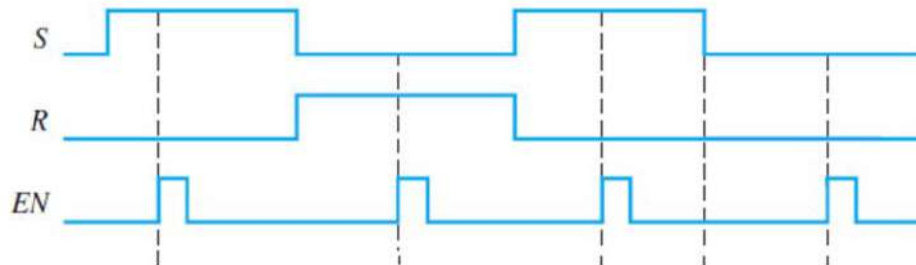
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

K-map for Q_{n+1} :-

		SR			
		00	01	11	10
Q_n	0	0	0	X	1
	1	1	0	X	1

$$\therefore Q_{n+1} = Q_n \bar{R} + S$$

Exercise 2:

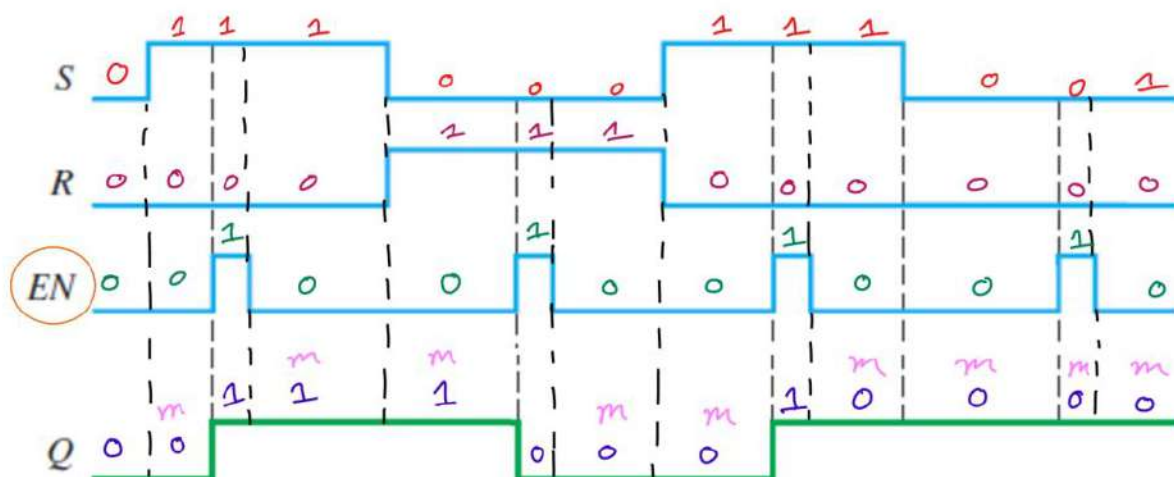


Determine the **Output(Q) Waveform** for **SR FLIP FLOPS** that are initially **Reset**

→ We Know,

Truth table for SR-flip-flop

clk	S	R	Q_{n+1} state
0	X	X	$Q_n \rightarrow$ P. state
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid



m = memory

D-Flip Flop

Sub: _____

Day

Time: _____

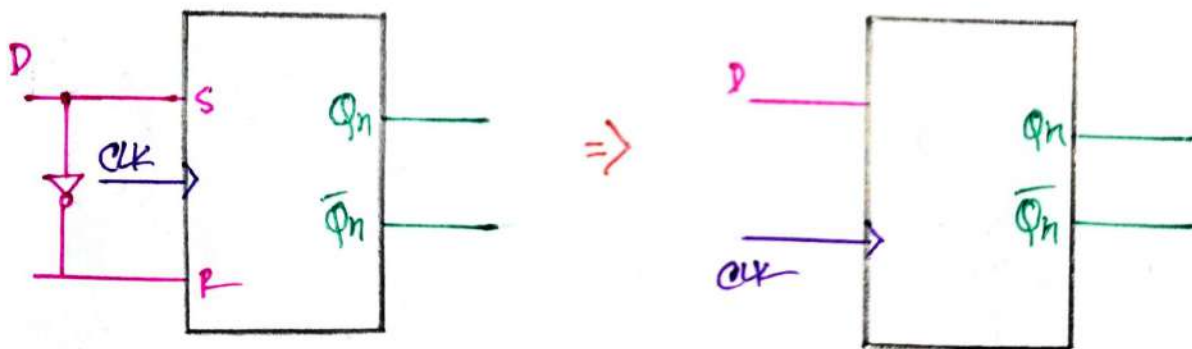
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D Flip Flop:-

⇒ In "NAND" SR Latch the invalid input condition of $S=0, R=0$. It is the drawback of the SR Latch.

To avoid the drawback we need an inverter to connect ~~the~~ between the set and Reset inputs. It ensure that at the same time S and R never equal to 1.

Logic Circuit with Block Diagram



T.T for D.F.F:-

clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Truth Table for SR Flip-Flop:-

clk	S	R	Q	$\bar{Q}$
0	X	X	memory	
1	0	0	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	Not Used	

Sub: _____

Time: _____

Date: / /

Characteristic Table for D-flip-flop:-

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

$$\therefore Q_{n+1} = D$$

we know:-

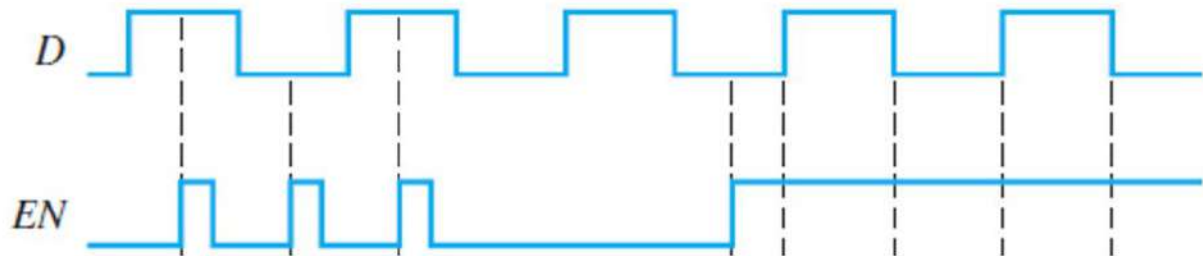
T.T for D.F.F

clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Excitation Table for D-flip-flop:-

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Exercise 3:

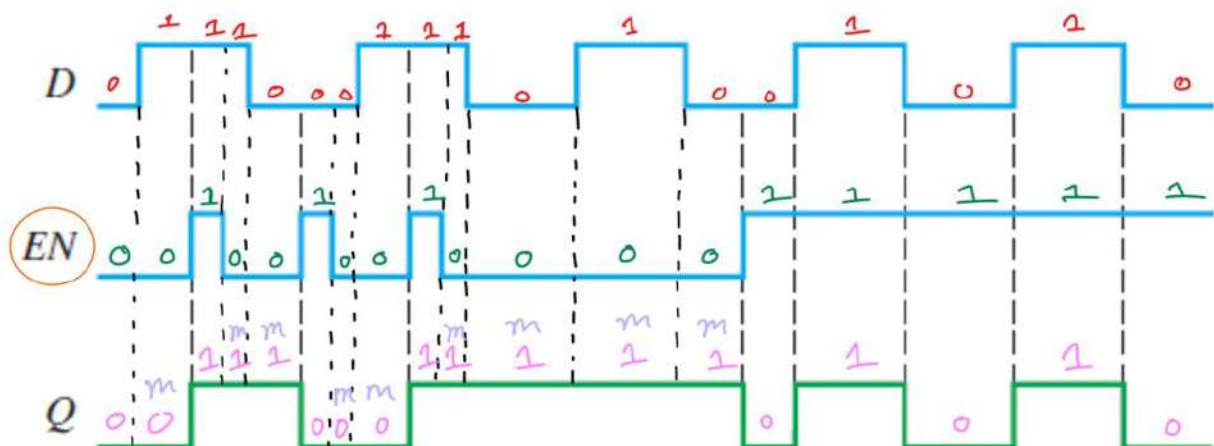


Determine the **Output(Q) Waveform** for **D FLIP FLOPS** that are initially **Reset**

→ We Know,

T.T for D.F.F.

clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1



m = memory

Exercise 4:

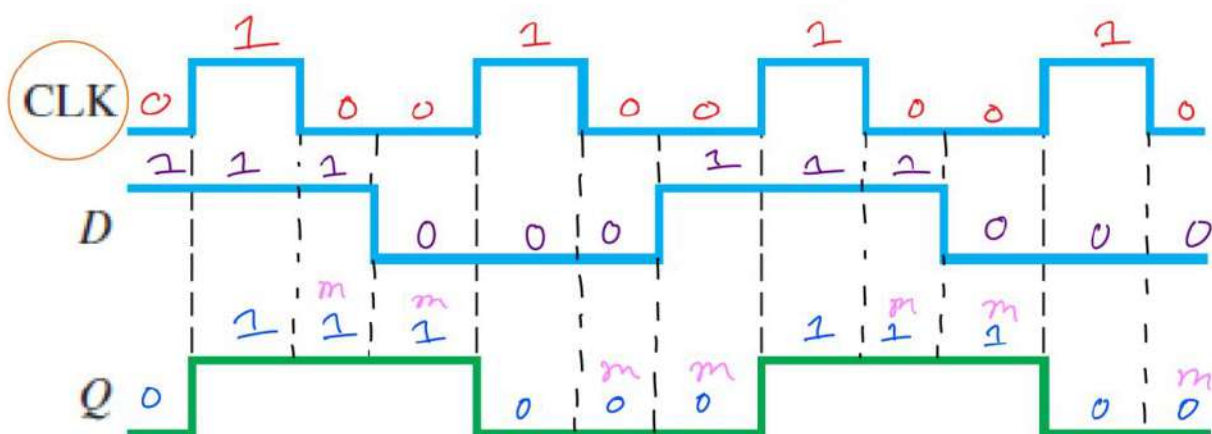


Determine the **Output(Q) Waveform** for **D FLIP FLOPS** that are initially **Reset**

→ We Know,

T.T for D.F.F.

CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1



Answer = m

Sub: _____

Time: _____

Date: / /

JK FLIP FLOP

Sub: _____

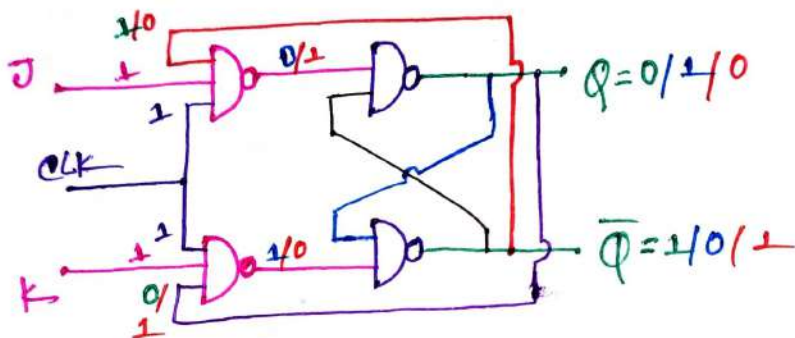
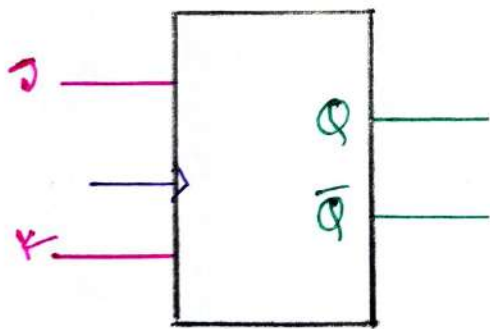
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Time: _____

Date: / /

JK Flip-Flop:-

⇒ The JK flip flop is similar to the SR flip flop but there is no change in state when $J=0$ & $K=0$.

Logic CircuitBlock Diagram:-T.T. for JK f.f

Similar to SR f.f

clk	J	K	Q_{n+1}
0	x	x	Q_n (memory)
1	0	0	Q_n (memory)
1	0	1	0
1	1	0	1
1	1	1	$\overline{Q_n}$ (toggle)

Analyse the T.T.Case 1:- $clk = 1$ Let $Q=0, \overline{Q}=1$ Now $J=1, K=1$ $\therefore Q = 0, 1, 0, 1, \dots$ $\overline{Q} = 1, 0, 1, 0, \dots$ $\therefore Q_{n+1} =$

1
0
1
⋮
 $\overline{Q_n}$

Sub :

Time :

Date : / /

Characteristic Table:-

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

We know,
T.T. for JK ff

clk	J	K	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	$\overline{Q_n}$

☑ K-map for Q_{n+1} :-

$Q_n \backslash JK$	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$$\therefore Q_{n+1} = Q_n \overline{K} + \overline{Q_n} J$$

☑ Excitation Table:-

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

☑ K-map for J

$Q_n \backslash Q_{n+1}$	0	1
0	0	1
1	X	X

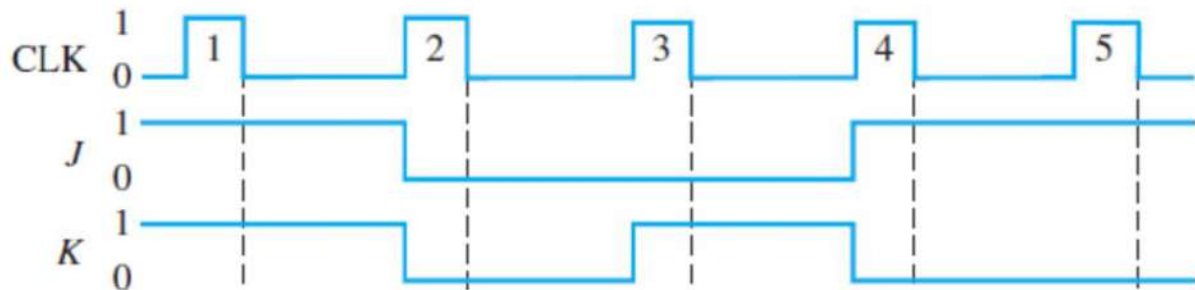
$$J = Q_{n+1}$$

☑ K-map for K

$Q_n \backslash Q_{n+1}$	0	1
0	X	X
1	1	0

$$K = \overline{Q_{n+1}}$$

Exercise 5:

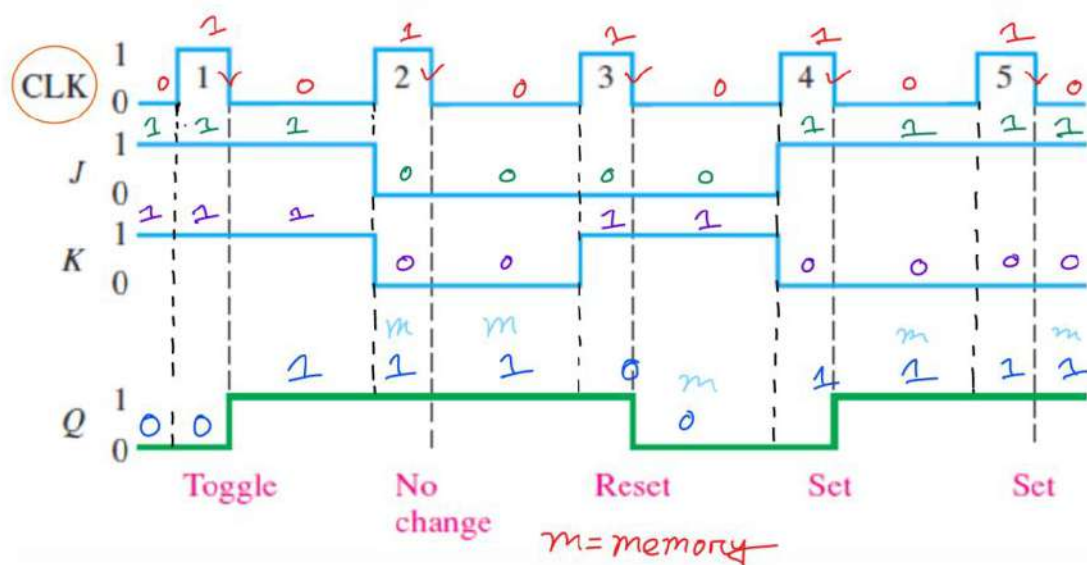


Determine the **Output(Q) Waveform** for **JK FLIP FLOPS** that are initially **Reset** and the flip flop is **negative edge triggered**.

→ We Know,

T.T for J.K f.f

clk	J	K	Q _{n+1}
0	x	x	Q _n (memory)
1	0	0	Q _n (memory)
1	0	1	0
1	1	0	1
1	1	1	Q _n (toggle)



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Time: _____

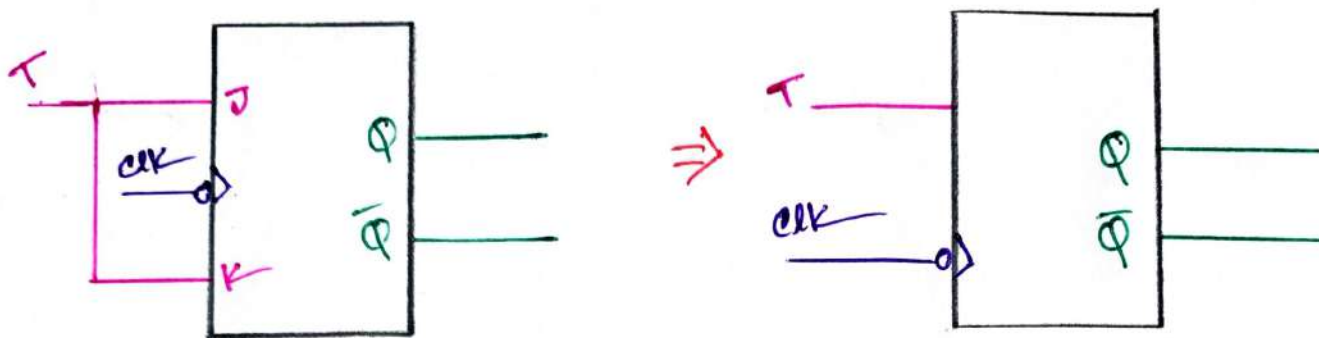
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T- Flip Flop
min

Sub: _____

Time: _____

Date: / /

T-Flip-Flop:-Logic Circuit & Block Diagram:-T.T for T-BB:-

clk	T	Q_{n+1}
0	X	Q_n (memory)
1	0	Q_n (memory)
1	1	$\bar{Q}_n$ (Toggle)

Characteristic Table:-

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$$\therefore Q_{n+1} = Q_n \oplus T$$

Excitation Table:-

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

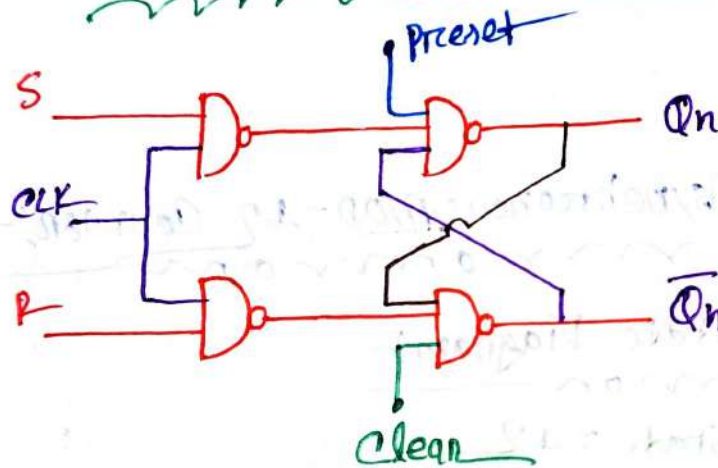
$$T = Q_n \oplus Q_{n+1}$$

Preset & Clear Inputs

≡ They are the direct inputs or overriding inputs on asynchronous inputs.

≡ The Synchronous inputs are S, R, D, K, D, & T

Logic Circuit:-



if

$\text{preset} = 0$, Then $Q_n = 1$ & $\bar{Q}_n = 0$

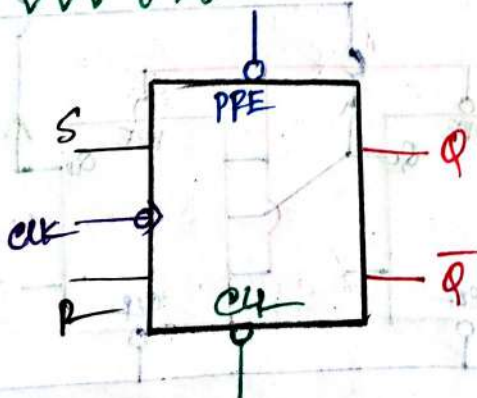
if

$\text{clear} = 0$, Then $Q_n = 0$ & $\bar{Q}_n = 1$

What ever be the value of clock & synchronous inputs.

Block Diagram:-

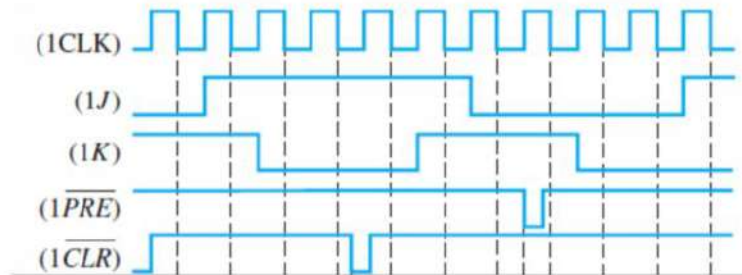
for Active Low:-



Truth Table

PR	CLK	Qn
0	0	Not used
0	1	1
1	0	0
1	1	F.F will Perform Normally

Exercise 6:



Determine the **Output(Q) Waveform** for **JK FLIP FLOPS** that are initially **Reset** and the flip flop is **negative edge triggered**.

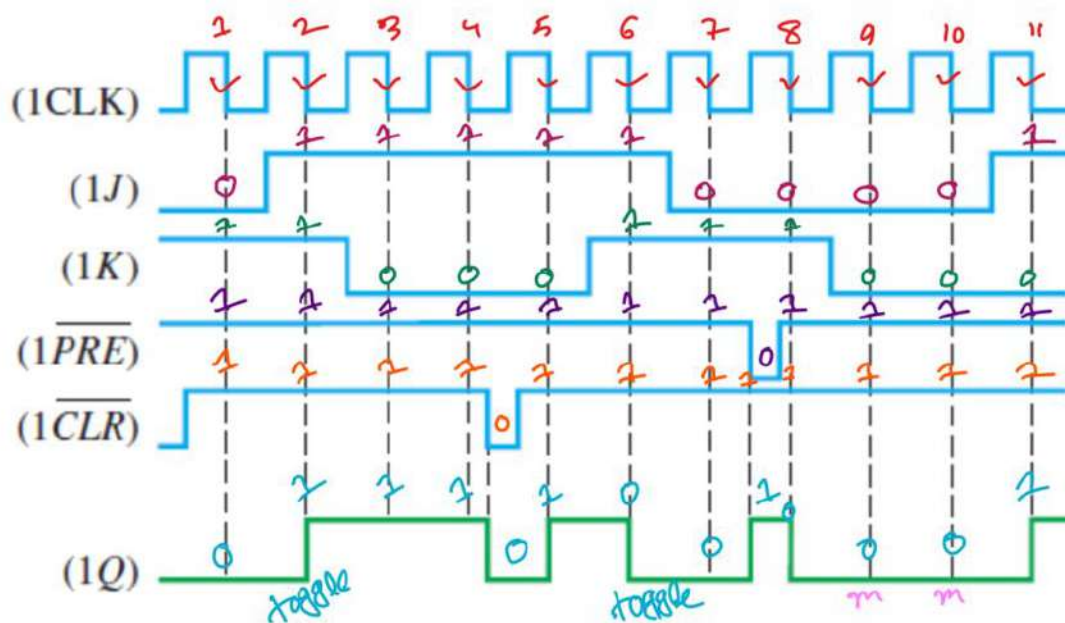
→ We Know,

T.T for J.K.F.F

clk	J	K	Q _{n+1}
0	x	x	Q _n (memory)
1	0	0	Q _n (memory)
1	0	1	0
1	1	0	1
1	1	1	Q _n (toggle)

Truth Table

PR	clk	Q _n
0	0	Not used
0	1	1
1	0	0
1	1	F.F will perform normally



Sub: _____

Time: _____

Date: / /

Counters
mon

Sub: _____

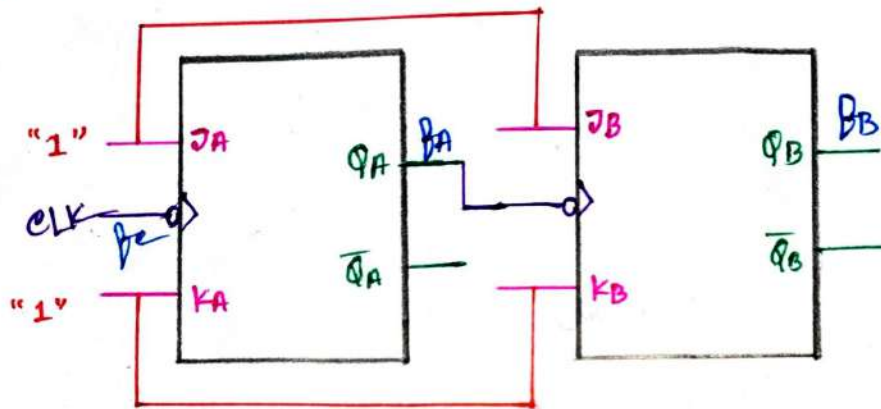
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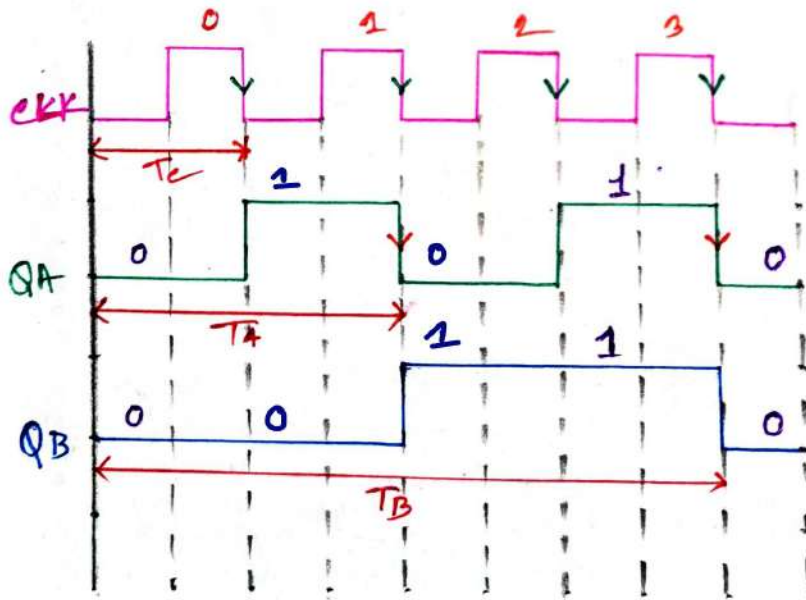
Date: / /

Counter! —

⇒ It's a Sequential circuit and it just simply counts.

Example! —

Analyse the Input & output clock frequency

## Truth Table! —

clk	Q _B	Q _A
0	0	0
1	0	1
2	1	0
3	1	1

Types of Counters:-

⇒ There are two types:-

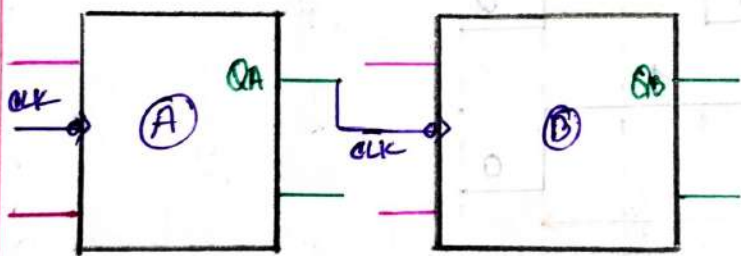
1. Asynchronous Counters (Ripple Counters)

2. Synchronous Counters.

Asynchronous Vs Synchronous:-

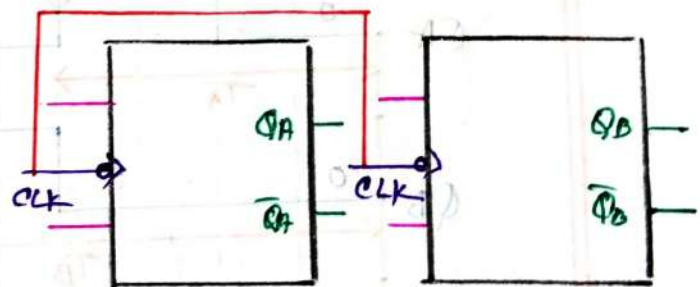
Asynchronous

⇒ Flip Flop are Connected in such a way that the O/P of 1st flip flop drives the clock of next flip flop



Synchronous

⇒ There is no connection between o/p of 1st flip flop and clock of next flip flop. F.F are clocked simultaneously

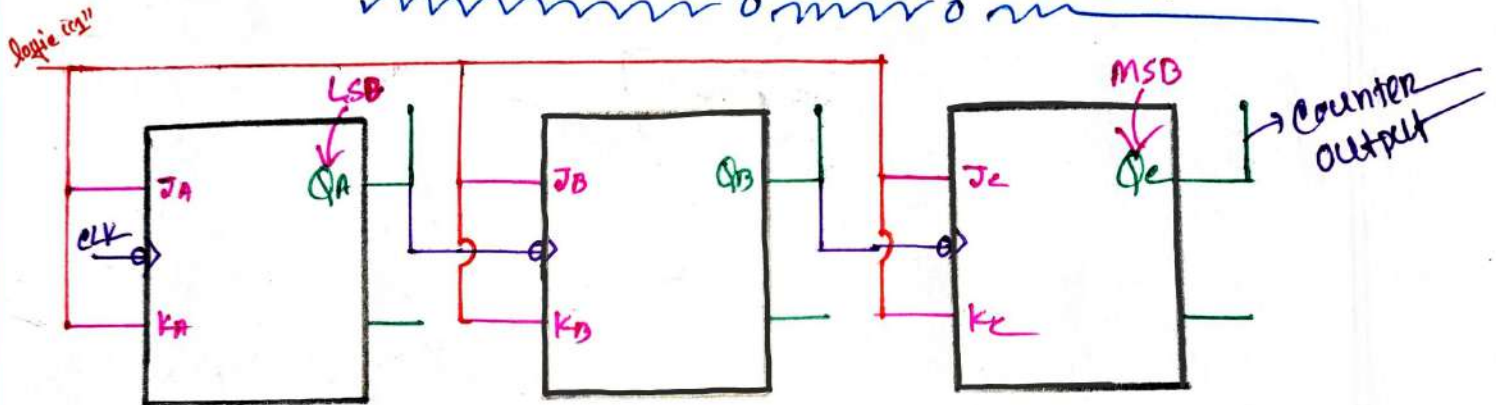


A	B	C
0	0	0
1	0	1
0	1	1
1	1	0

Classification of Counters:-

1. Up Counter
2. Down Counter
3. Up/Down Counter.

3 Bit Asynchronous Up Counter:-



☐ Analyse the Clock Diagram:-

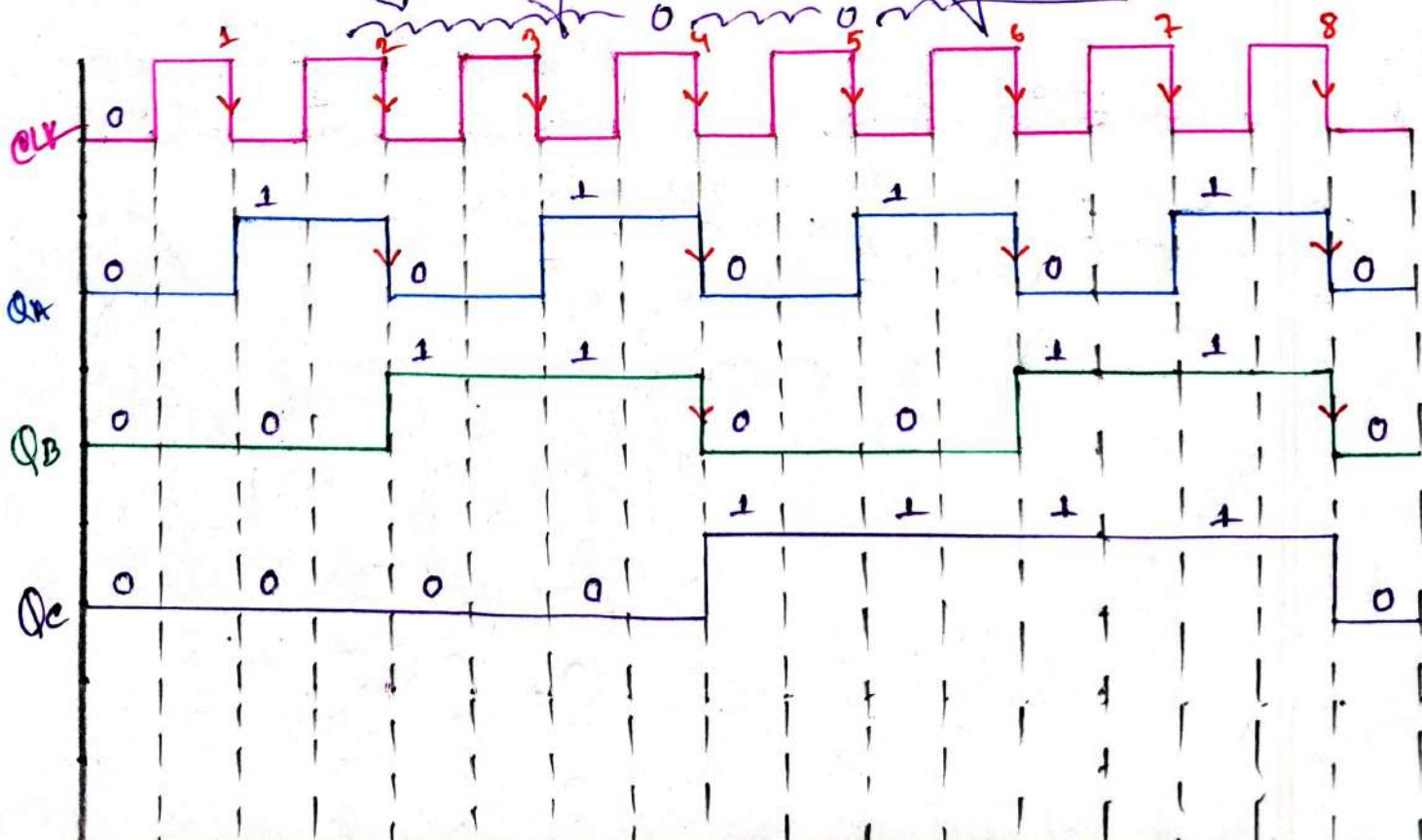
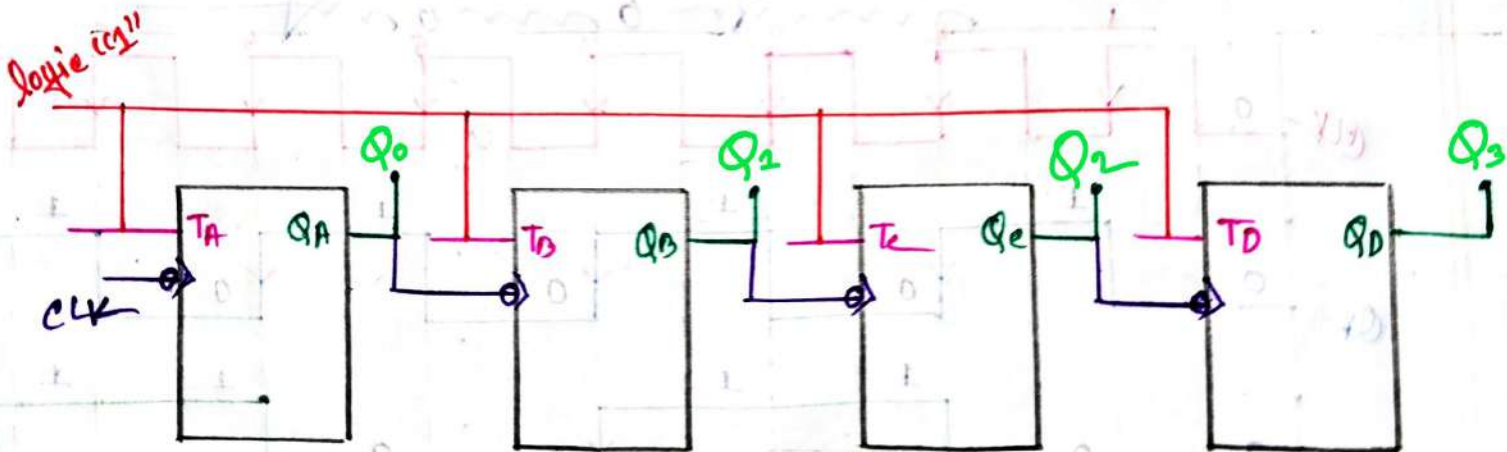


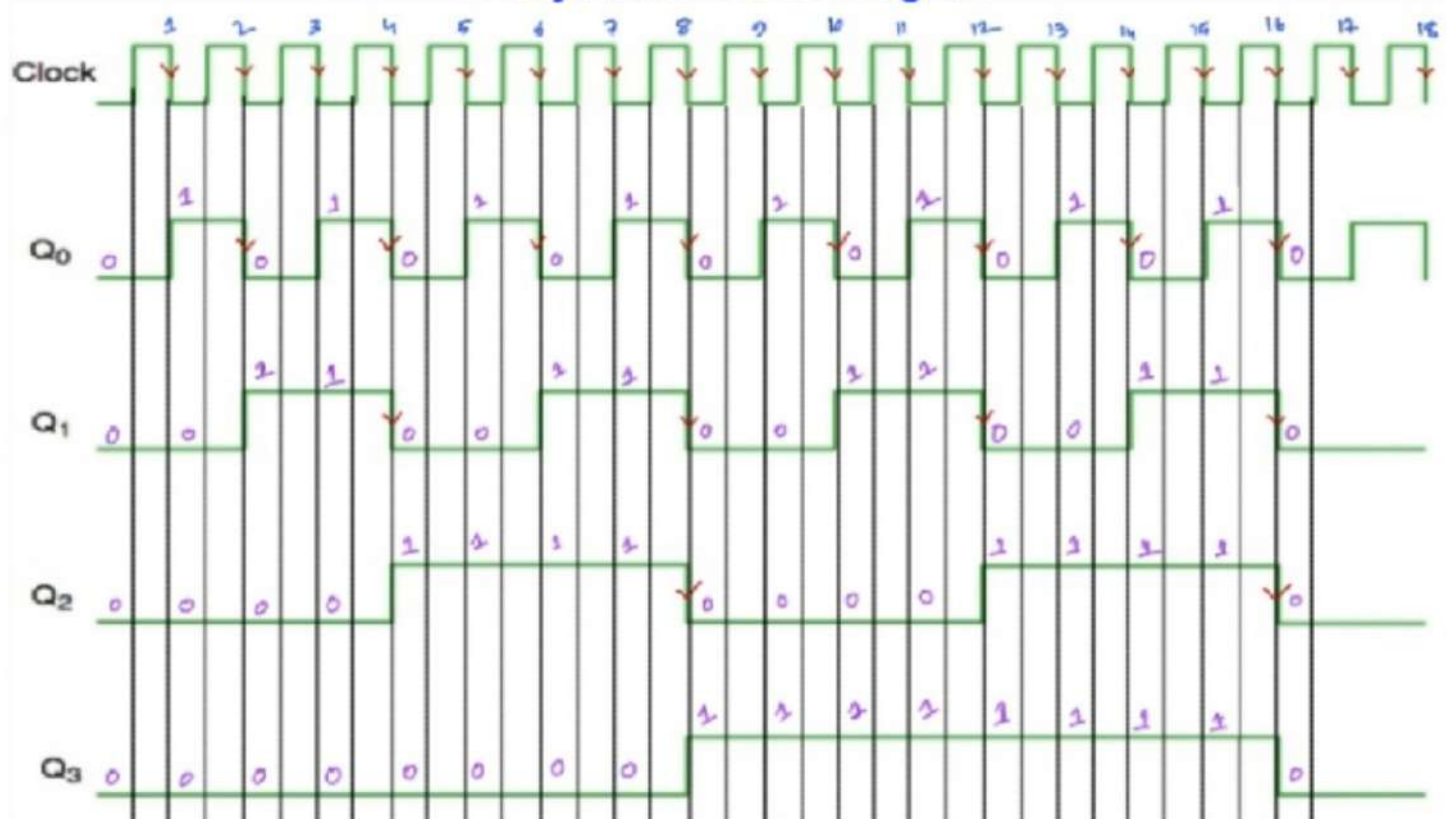
Table for 3 bit Async. Up Counter:-

Clock	Q_C	Q_B	Q_A	Decimal eq
Initially	0	0	0	0
1 st ($\downarrow$)	0	0	1	1
2 nd ($\downarrow$)	0	1	0	2
3 rd ($\downarrow$)	0	1	1	3
4 th ($\downarrow$)	1	0	0	4
5 th ($\downarrow$)	1	0	1	5
6 th ($\downarrow$)	1	1	0	6
7 th ($\downarrow$)	1	1	1	7

4-Bit Asynchronous Up Counter:-



Analyse the Clock Diagram



Truth table for 4 bit asynchronous Up Counter

Clock Cycles	Output Bits				Decimal Value
	QA	QB	QC	QD	
1	0	0	0	0	0
2	0	0	0	1	1
3	0	0	1	0	2
4	0	0	1	1	3
5	0	1	0	0	4
6	0	1	0	1	5
7	0	1	1	0	6
8	0	1	1	1	7
9	1	0	0	0	8
10	1	0	0	1	9
11	1	0	1	0	10
12	1	0	1	1	11
13	1	1	0	0	12
14	1	1	0	1	13
15	1	1	1	0	14
16	1	1	1	1	15

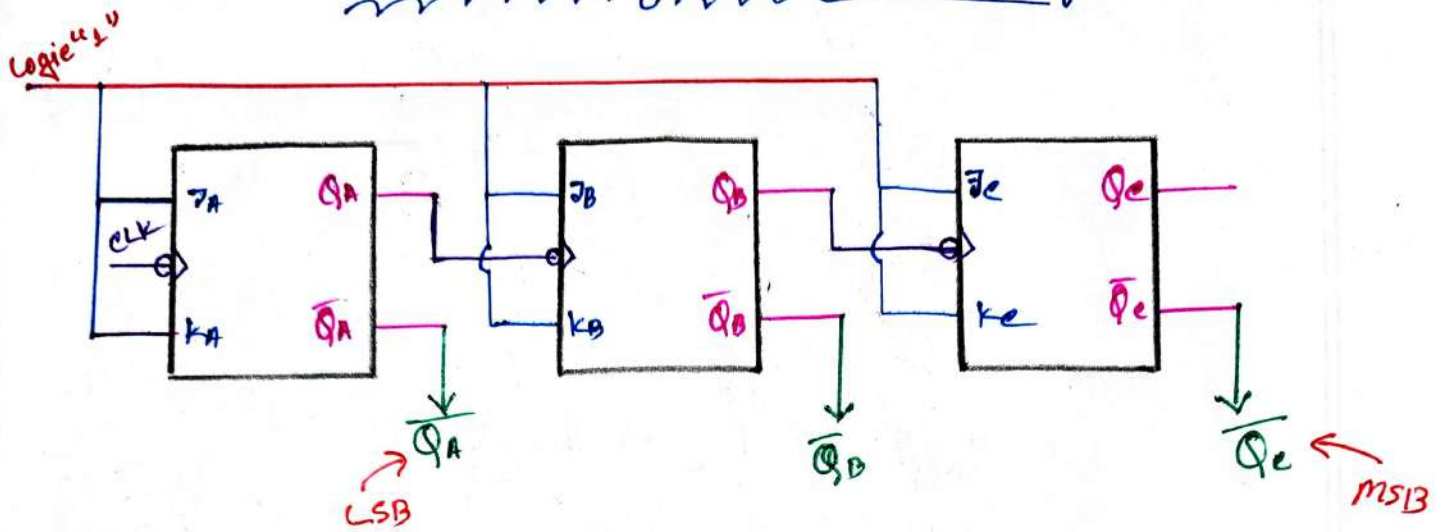
Sub: _____

Day _____

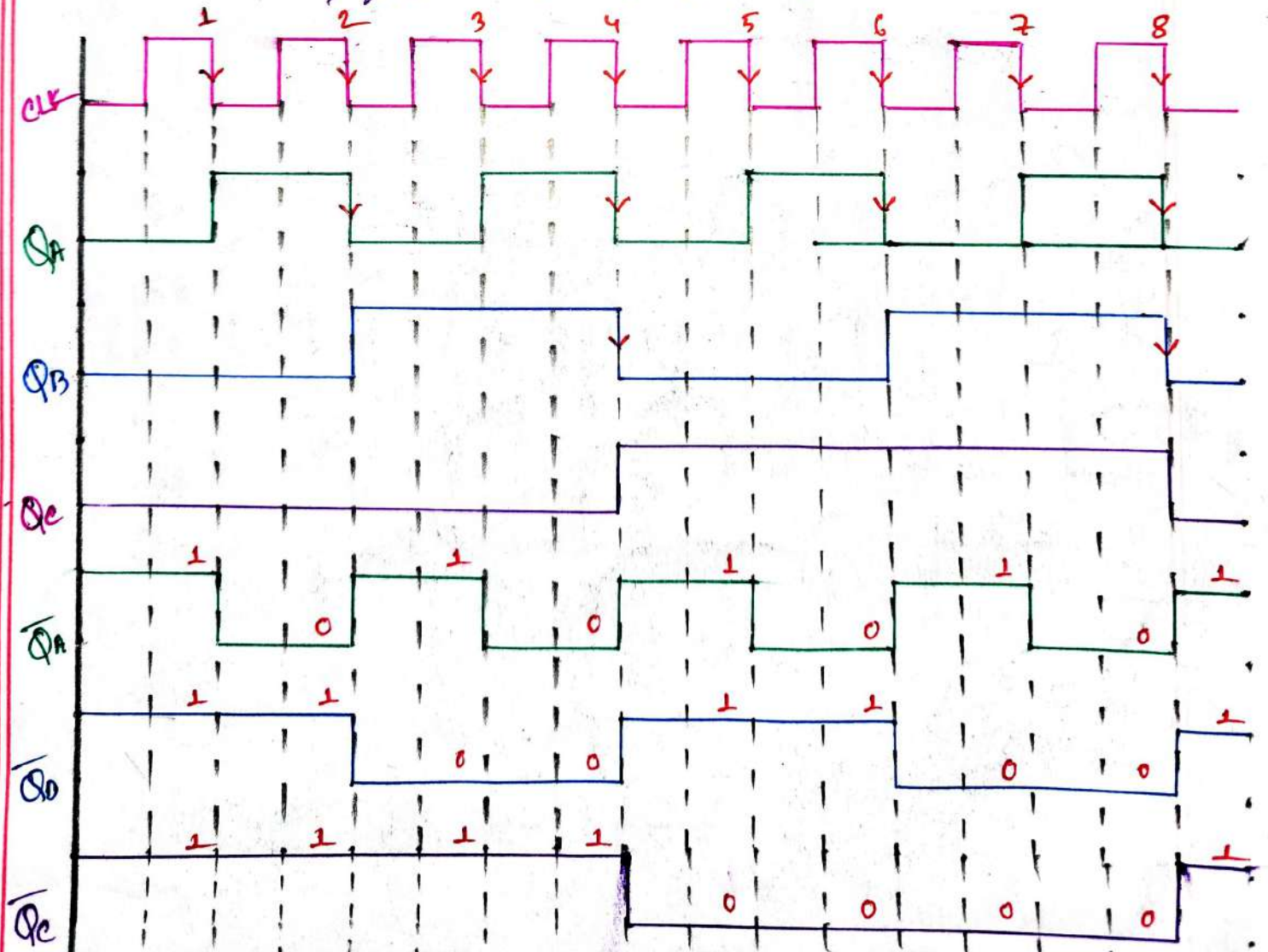
Time: _____

Date: / /

#3 Bit Down Counter: (Asynchronous)



Analyse the clock Diagram:-



Sub: _____

Day _____

Time: _____

Date: / /

State Diagram of a Counter:-

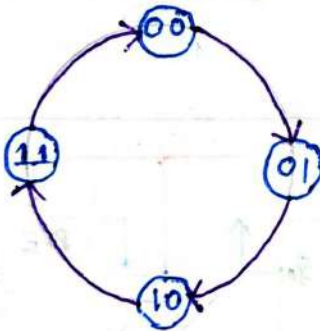
⇒ 2-Bit Up Counter:-

- Total state for 2 bits = 4
- MAX Count = $4 - 1 = 3$

The combinations are:-

Q _B	Q _A	Count
0	0	0
0	1	1
1	0	2
1	1	3

State Diagram:-



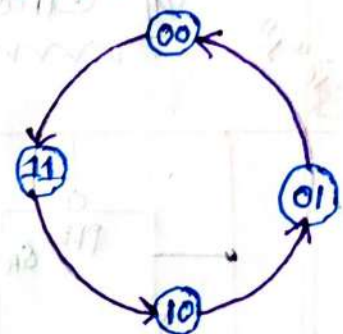
⇒ 2-Bit Down Counter:-

- Total state for 2 bit = 4
- max Count = $4 - 1 = 3$

The combinations are:-

Q _B	Q _A	Count
1	1	3
1	0	2
0	1	1
0	0	0

State Diagram:-



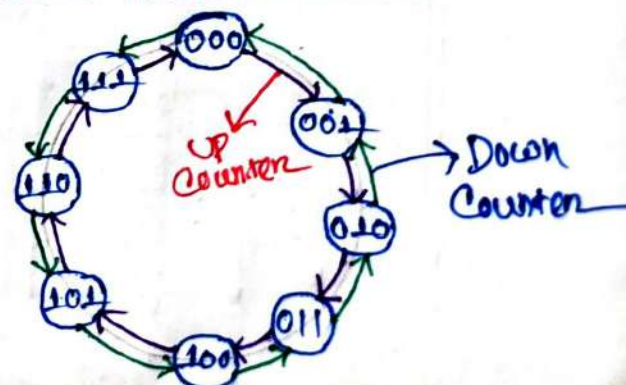
3-bit Up-down Counter:-

- ⇒ Total state for 3 bits = 8
- ⇒ MAX Count = $8 - 1 = 7$

The combinations are:-

Q _C	Q _B	Q _A	Count
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

State Diagram:-



MOD Counters:-

⇒ MOD Counters are cascaded counter circuits which count to a set modulus value before resetting.

Some of them are:-

1. Asynchronous Counter for 2 bit (known as MOD-4)
2. Asynchronous Counter for 3 bit (known as MOD-8)
3. Asynchronous Counter for Decade (known as MOD-10)
4. MOD-12 ripple counter
5. MOD-6 ripple counter

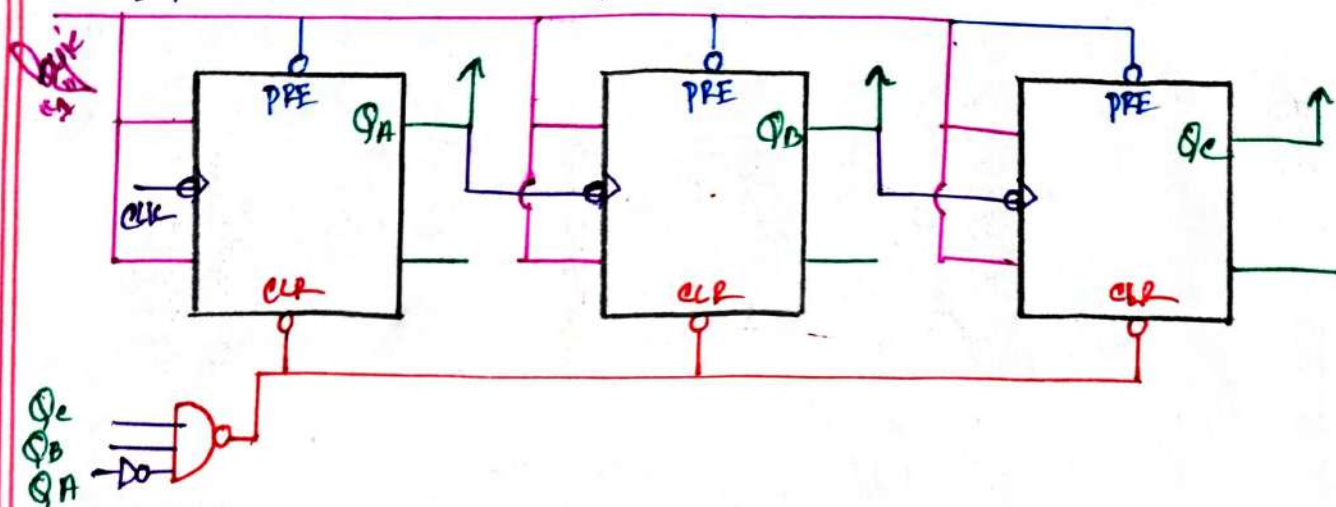
MOD-6 Ripple Counter using MOD-8 Counter:-

T.T for MOD-8

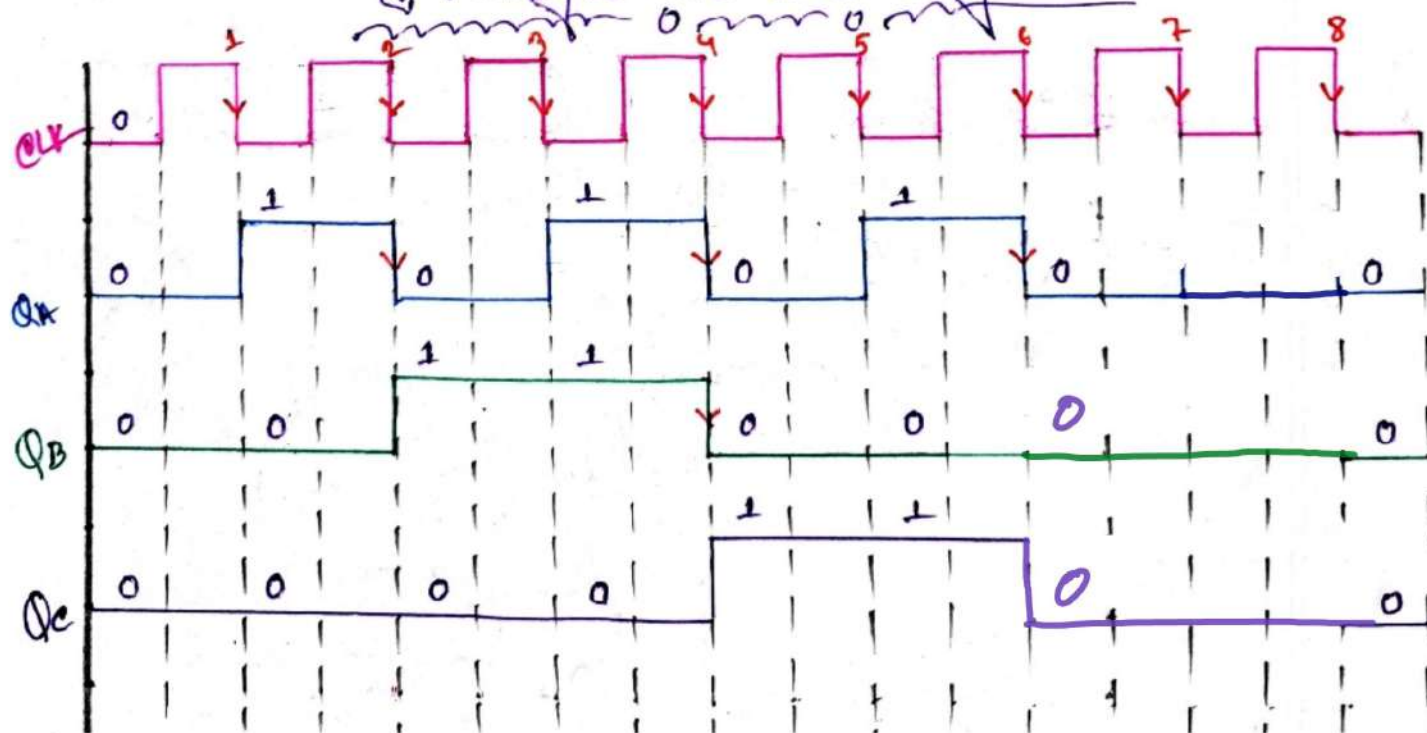
C	B	A	Decimal Value
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

$$\text{Max Count} = 8 - 1 = 7$$

Logic circuit



Analyse the Clock Diagram:-



Important points:-

1. Negative edge triggered $\rightarrow Q$ is clock $\rightarrow$ Up Counter
2. Positive edge triggered $\rightarrow \bar{Q}$ is clock $\rightarrow$ Up Counter
3. Negative edge triggered $\rightarrow \bar{Q}$ is clock $\rightarrow$ Down Counter
4. Positive edge triggered $\rightarrow Q$ is clock $\rightarrow$ Down Counter

Day

Time:

Date: / /

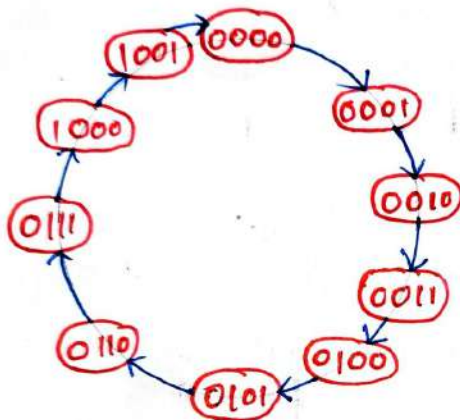
Sub:

Decode (DED) Ripple Counter -

(MOD 10)

⇒ Draw the State Diagram:-

here, No. of State = 10

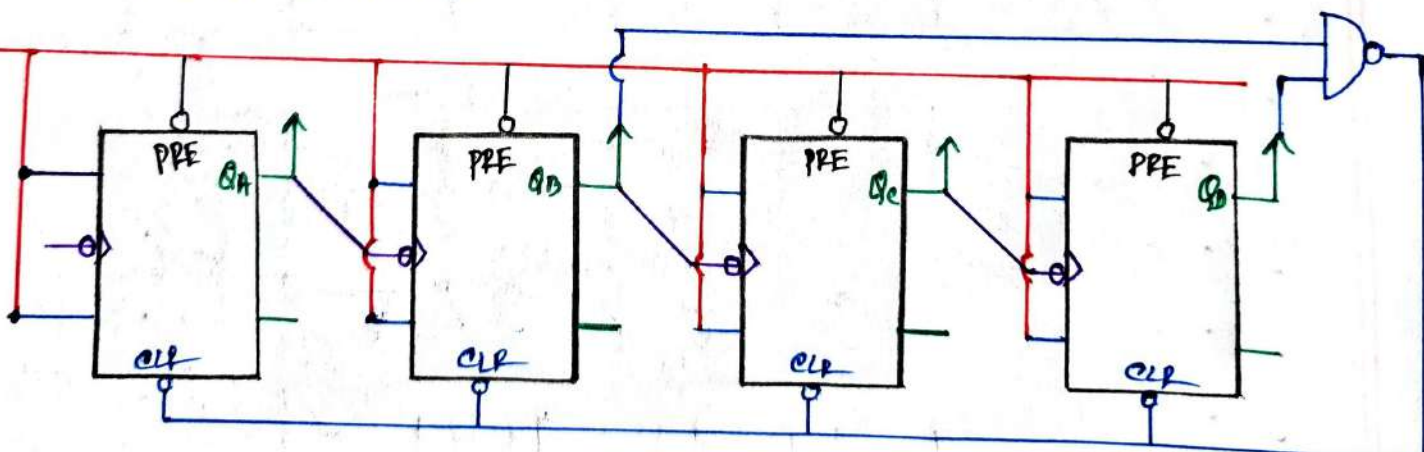


$$\text{Max Count} = 10 - 2 = 8$$

Truth Table for MOD 16

clk	Q _D	Q _C	Q _B	Q _A	BCD
1	0	0	0	0	0
2	0	0	0	1	1
3	0	0	1	0	2
4	0	0	1	1	3
5	0	1	0	0	4
6	0	1	0	1	5
7	0	1	1	0	6
8	0	1	1	1	7
9	1	0	0	0	8
10	1	0	0	1	9
11	1	0	1	0	10
12	1	0	1	1	
13	1	1	0	0	
14	1	1	0	1	
15	1	1	1	0	

Circuit Design:-

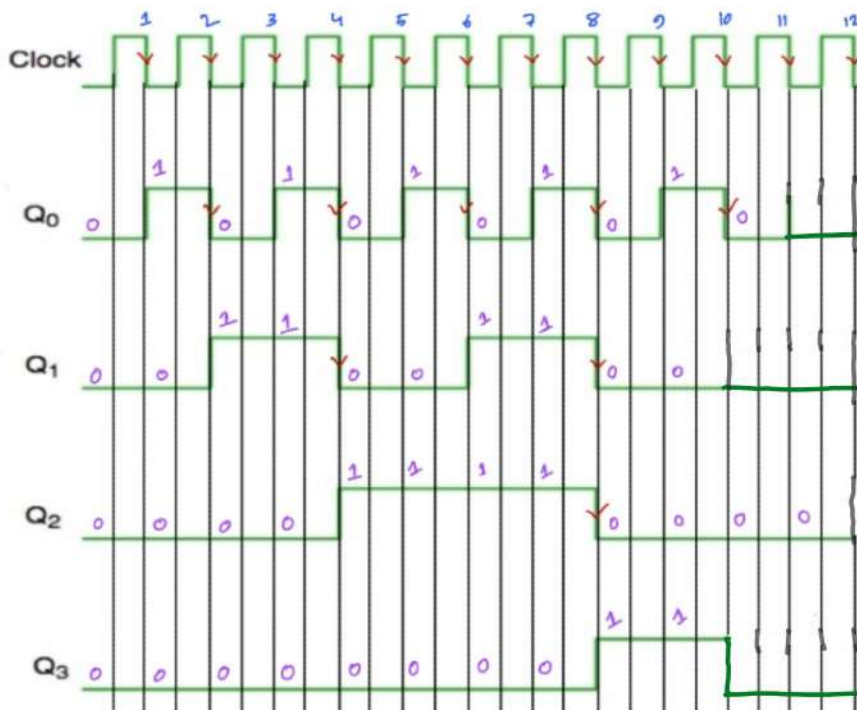


Day

Time:

Date: / /

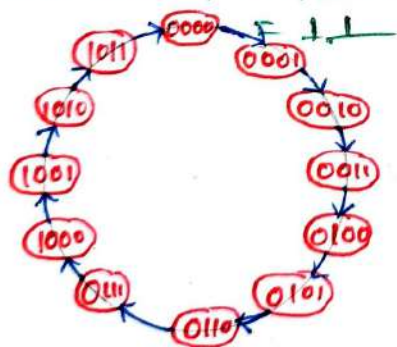
Sub:



Asynchronous MOD-12 Counter:-

⇒ Draw the state Diagram:-

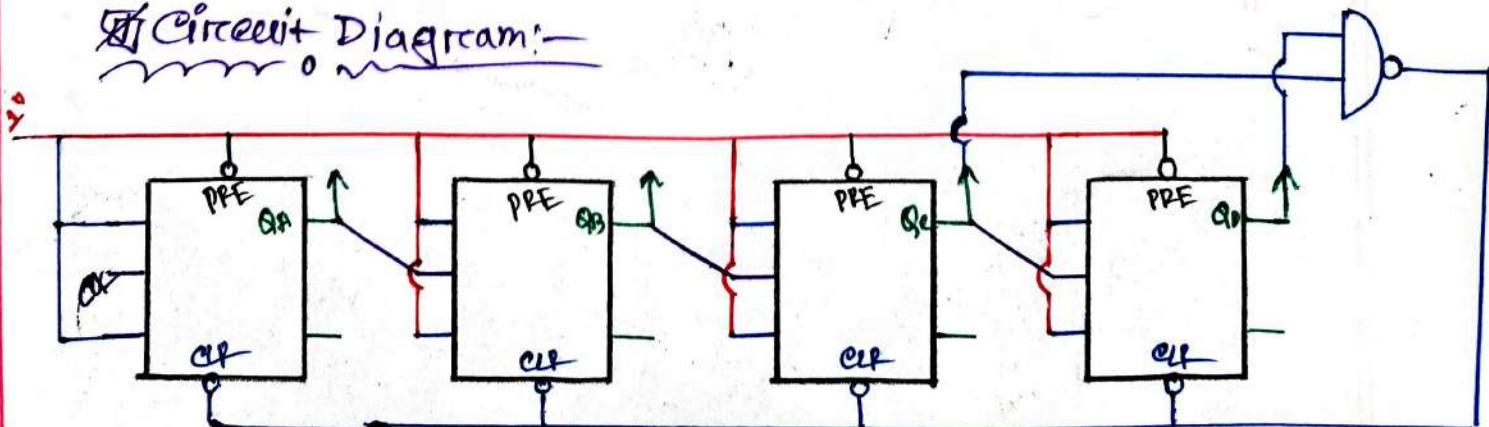
here, No. of state = 12
Max count = 12 - 1



Truth Table for MOD16

clk	Q _D	Q _C	Q _B	Q _A	Dec
1	0	0	0	0	0
2	0	0	0	1	1
3	0	0	1	0	2
4	0	0	1	1	3
5	0	1	0	0	4
6	0	1	0	1	5
7	0	1	1	0	6
8	0	1	1	1	7
9	1	0	0	0	8
10	1	0	0	1	9
11	1	0	1	0	10
12	1	0	1	1	11
13	1	1	0	0	12
14	1	1	0	1	
15	1	1	1	0	

Circuit Diagram:-



Synchronous Counter!

Sub: _____

Day _____

Time: _____

Date: / /

How to Design Synchronous Counter:-

- ⇒ Step 1:- Decide the number of flip-flops & which one?
- ⇒ Step 2:- Excitation table for the flip-flops.
- ⇒ Step 3:- Draw state diagram & circuit excitation Table
- ⇒ Step 4:- Obtain simplified equations using "K" map.
- ⇒ Step 5:- Draw the Logic diagram.

2-bit synchronous UP Counter:-

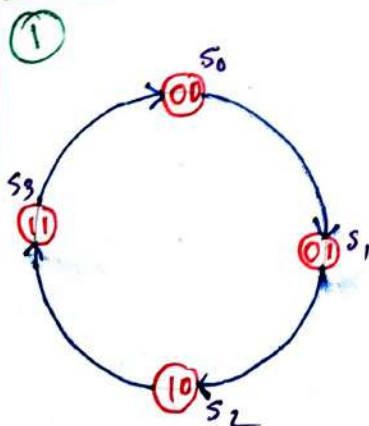
Step: 1:-

⇒ For 2bit we need two Flip Flop (JK FF)

Step: 2:-

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step: 3:-



②

ckt excitation Table:-

Q_1	Q_2	Q_1^*	Q_2^*	J_1	K_1	J_2	K_2
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

Step: 4:-For J_1 :-
map

$Q_1 \backslash Q_2$	0	1
0	0	1
1	X	X

$$J_1 = Q_2$$

for K_1 :-
map

$Q_1 \backslash Q_2$	0	1
0	X	X
1	0	1

$$K_1 = Q_2$$

for J_2 :-
map

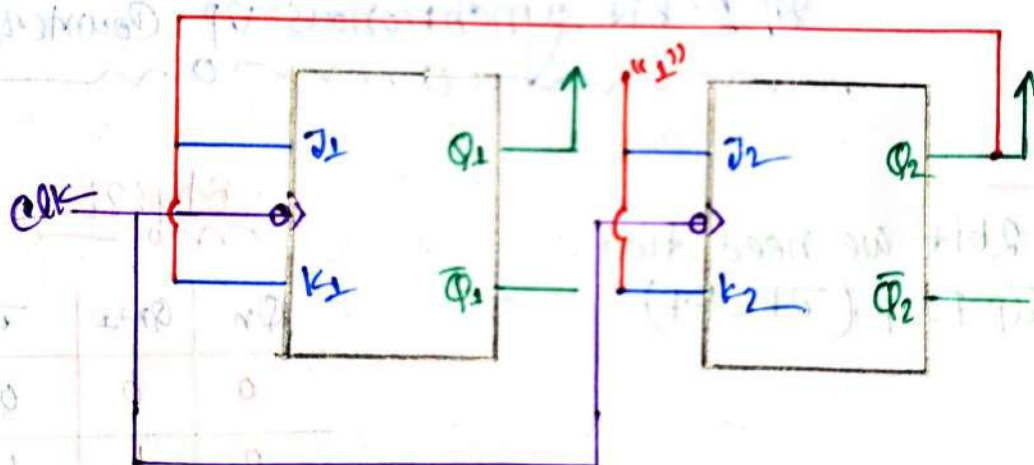
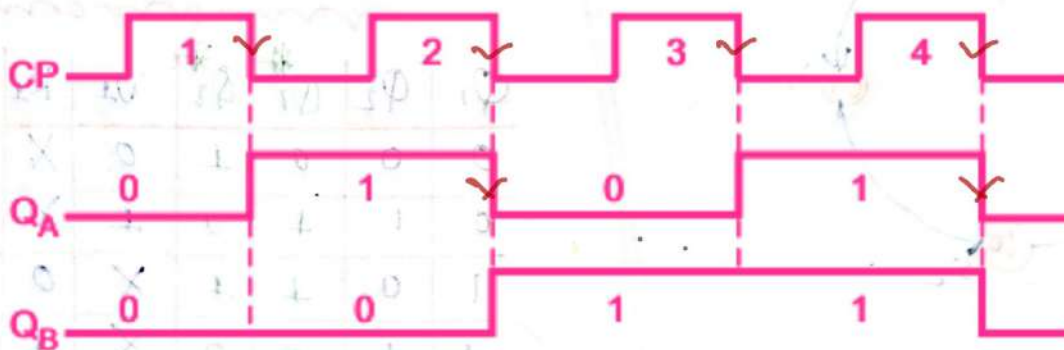
$Q_1 \backslash Q_2$	0	1
0	1	X
1	1	X

$$J_2 = 1$$

for K_2 :-
map

$Q_1 \backslash Q_2$	0	1
0	X	1
1	X	1

$$K_2 = 1$$

Step: 5:-Analyse the output waveforms

Sub: _____

Day _____

Time: _____

Date: / /

3-Bit Synchronous Up Counter Using JK flip flop:-

⇒ Step 1:-

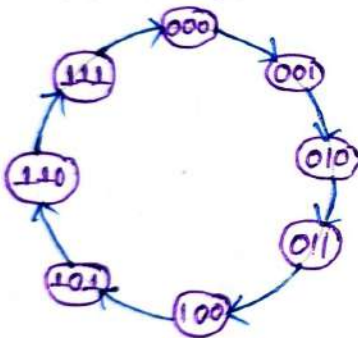
For 3 bit we need 3
JK flip flop

⇒ Step 2:-

Q_n	Q_{n-1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

⇒ Step 3:-

① State diagram:-



②

Excitation table:-

Q_c	Q_B	Q_A	Q_c^*	Q_B^*	Q_A^*	J_c	K_c	J_B	K_B	J_A	K_A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

⇒ Step 4:-

K-map for J_c

Q_c	$Q_B Q_A$	00	01	11	10
0		0	0	1	0
1		X	X	X	X

$$\therefore J_c = Q_B Q_A$$

K-map for K_c

Q_c	$Q_B Q_A$	00	01	11	10
0		X	X	X	X
1		0	0	1	0

$$\therefore K_c = Q_B Q_A$$

K-map for J_B

Q_c	$Q_B Q_A$	00	01	11	10
0		0	1	X	X
1		0	1	X	X

$$\therefore J_B = Q_A$$

K-map for K_B

Q_c	$Q_B Q_A$	00	01	11	10
0		X	X	1	0
1		X	X	1	0

$$\therefore K_B = Q_A$$

Sub: _____

Day _____

Time: _____

Date: / /

$\therefore$ K-map for J_A

$Q_B Q_A$	00	01	11	10
0	1	X	X	1
1	1	X	X	1

$\therefore J_A = 1$

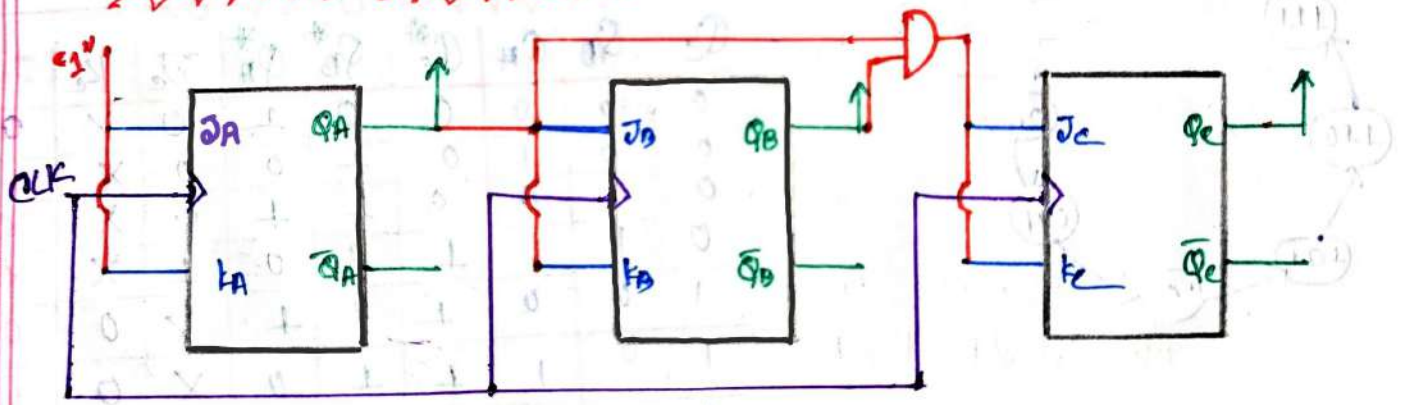
$\therefore$ K-map for K_A

$Q_B Q_A$	00	01	11	10
0	X	1	1	X
1	X	1	1	X

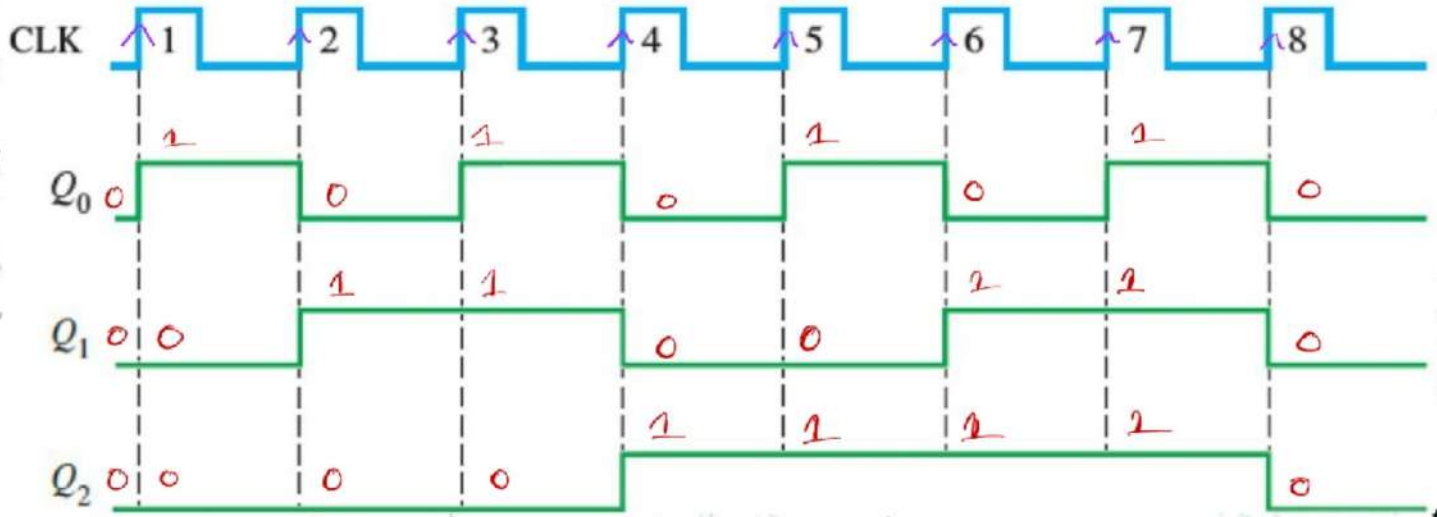
$\therefore K_A = 1$

$\therefore$ Step 5:-

$\Rightarrow$ Draw the Logic circuit:-



Clock Diagram:-



4-Bit Synchronous Up Counter Using JK flip flop

Step 1:

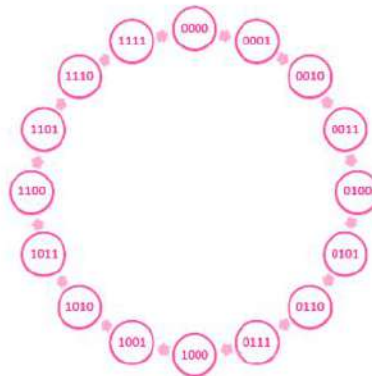
→ For 4 bit we need 4 JK flip flop

Step 2:

→ Draw the JK Excitation Table:

Qn	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:



Step 3.2:

CKT Excitation Table

Q _D	Q _C	Q _B	Q _A	Q _D [*]	Q _C [*]	Q _B [*]	Q _A [*]	J _D	K _D	J _C	K _C	J _B	K _B	J _A	K _A
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X
1	0	0	1	1	0	1	0	X	0	0	X	1	X	X	1
1	0	1	0	1	0	1	1	X	0	0	X	X	0	1	X
1	0	1	1	1	1	0	0	X	0	1	X	X	1	X	1
1	1	0	0	1	1	0	1	X	0	X	0	0	X	1	X
1	1	0	1	1	1	1	0	X	0	X	0	1	X	X	1
1	1	1	0	1	1	1	1	X	0	X	0	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	1	X	1	X	1

Step 4:

→ K-Map for J_D :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

$$\therefore J_D = Q_C Q_B Q_A$$

→ K-Map for J_C :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	0	0	1	0
01	X	X	X	X
11	X	X	X	X
10	0	0	1	0

$$\therefore J_C = Q_B Q_A$$

→ K-Map for J_B :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	0	1	X	X
10	0	1	X	X

$$\therefore J_B = Q_A$$

→ K-Map for K_D :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	0	0	1	0
10	0	0	0	0

$$\therefore K_D = Q_C Q_B Q_A$$

→ K-Map for K_C :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	X	X	X	X
01	0	0	1	0
11	0	0	1	0
10	X	X	X	X

$$\therefore K_C = Q_B Q_A$$

→ K-Map for K_B :

$Q_D Q_C / Q_B Q_A$	00	01	11	10
00	X	X	1	0
01	X	X	1	0
11	X	X	1	0
10	X	X	1	0

$$\therefore K_B = Q_A$$

→ K-Map for J_A :

00 01 11 10

00	1	x	x	1
01	1	x	x	1
11	1	x	x	1
10	1	x	x	1

$$\therefore \overline{v}_A = 1$$

→ K-Map for K_A :

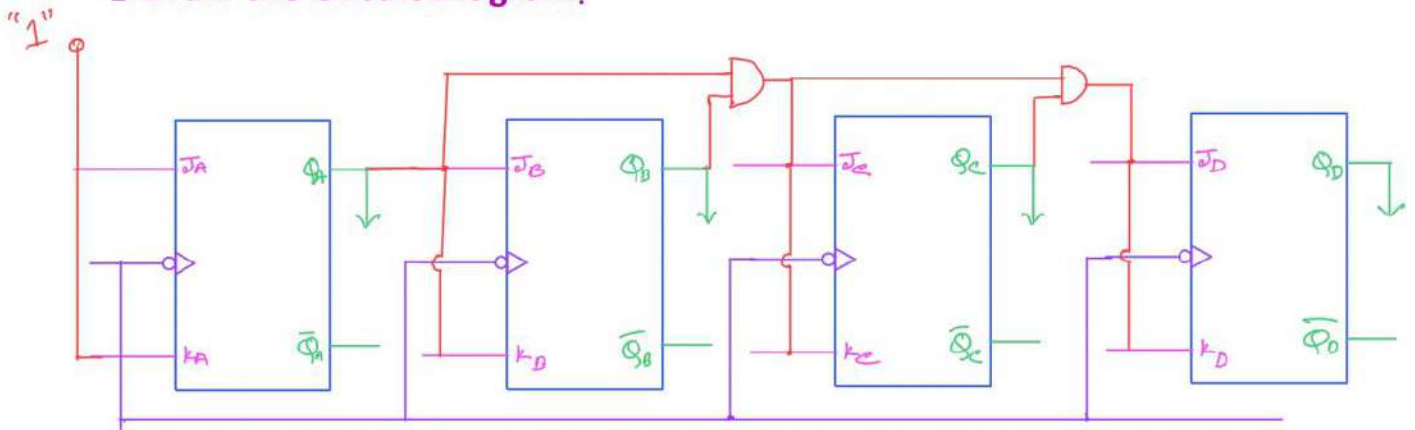
Q₁Q₂Q₃Q₄

	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

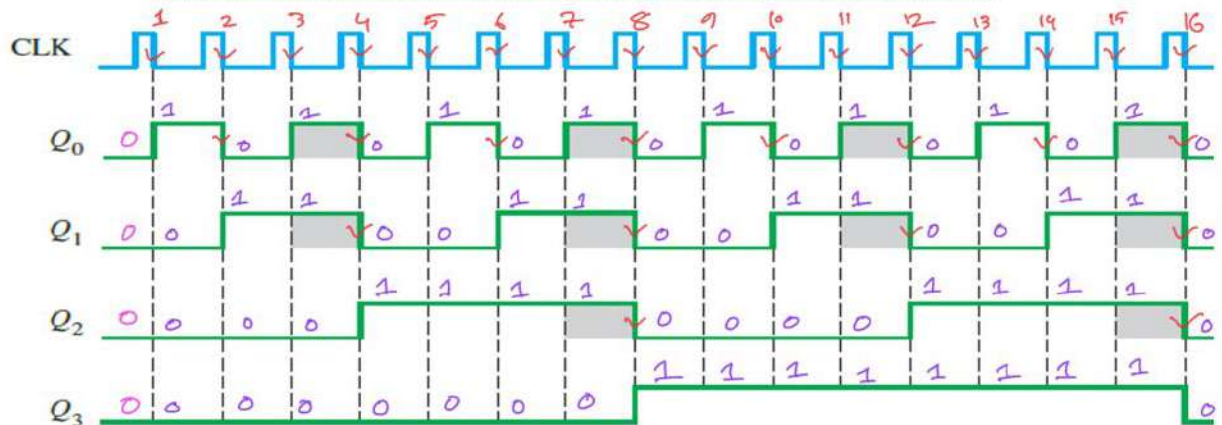
$$K_A = 1$$

Step 5:

→ Draw the Circuit Diagram.



Timing diagram of 4-Bit Synchronous Up Counter.



4-Bit Synchronous Decade BCD Counter Using JK flip flop

Step 1:

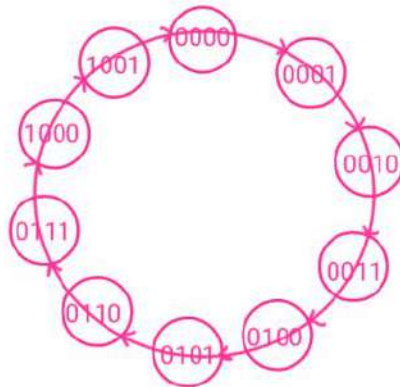
→ For 4 bit we need 4 JK flip flop

Step 2:

→ [Draw the JK Excitation Table:](#)

Qn	Qn+1	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:



Step 3.2:

CKT Excitation Table

Q _D	Q _C	Q _B	Q _A	Q _D [*]	Q _C [*]	Q _B [*]	Q _A [*]	J _D	K _D	J _C	K _C	J _B	K _B	J _A	K _A
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X
1	0	0	1	0	0	0	0	X	1	0	X	0	X	X	1

Step 4:

→ K-Map for J_D :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	0	0	0	0	0
01	0	0	1	0	0
11	X	X	X	X	X
10	X	X	X	X	X

$$\therefore J_D = \overline{Q_C} Q_B Q_A$$

→ K-Map for K_D :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	X	X	X	X	X
01	X	X	X	X	X
11	X	X	X	X	X
10	0	1	X	X	X

$$\therefore K_D = Q_A$$

→ K-Map for J_C :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	0	0	1	0	0
01	X	X	X	X	X
11	X	X	X	X	X
10	0	0	X	X	X

$$\therefore J_C = Q_B Q_A$$

→ K-Map for K_C :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	X	X	X	X	X
01	0	0	1	0	0
11	X	X	X	X	X
10	X	X	X	X	X

$$\therefore K_C = Q_B Q_A$$

→ K-Map for J_B :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	0	1	X	X	X
01	0	1	X	X	X
11	X	X	X	X	X
10	0	0	X	X	X

$$\therefore J_B = \overline{Q_D} Q_A$$

→ K-Map for K_B :

$Q_D Q_C$	$Q_B Q_A$	00	01	11	10
00	X	X	1	0	0
01	X	X	1	0	0
11	X	X	X	X	X
10	X	X	X	X	X

$$\therefore K_B = Q_A$$

→ K-Map for J_A :

Page 10/11

	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	X	X	X	X
10	1	X	X	X

$$\sigma_A = 1$$

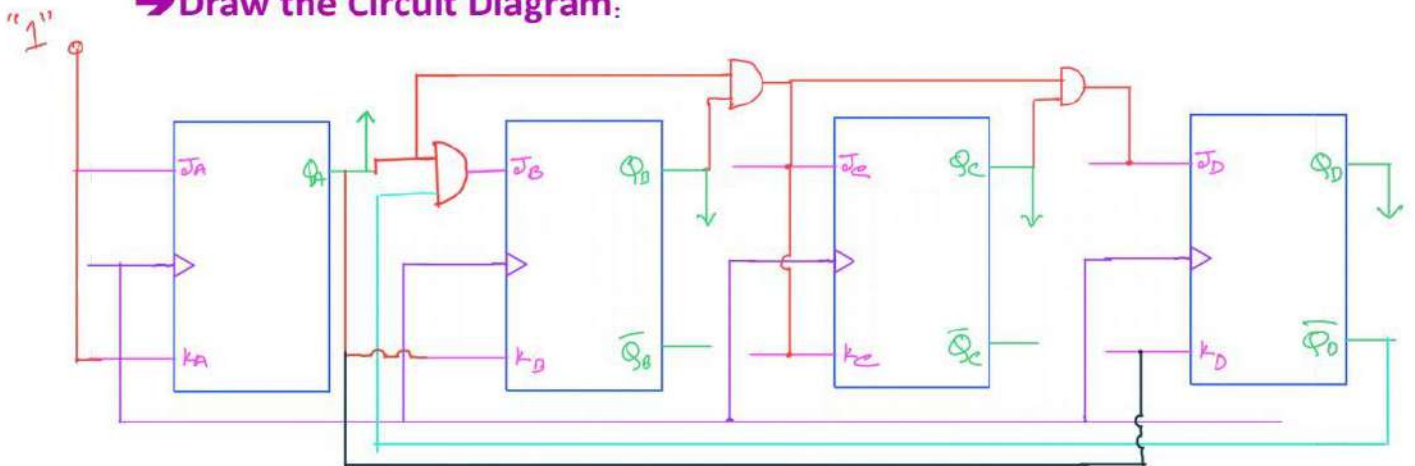
→ K-Map for K_A :

00 01 11 10
 00 X 1 1 X
 01 X 1 1 X
 11 X X X X
 10 X 1 X X

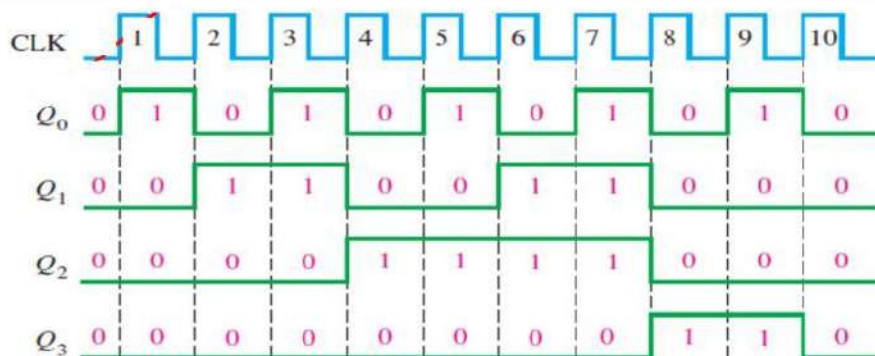
$$K_A = 1$$

Step 5:

➔ Draw the Circuit Diagram:



Timing diagram of 4-Bit Synchronous Up Counter



Sub: _____

Time: _____

Date: / /

Up/Down
Counters!

3-Bit Asynchronous UP/DOWN Counter Using JK flip flop

1. We need control input **M**
2. Let, if $M=0 \rightarrow$ Up Counting $\rightarrow$ Q connected to the Clock
if $M=1 \rightarrow$ Down Counting $\rightarrow$ Q' connected to the Clock

$\rightarrow$ Truth Table:

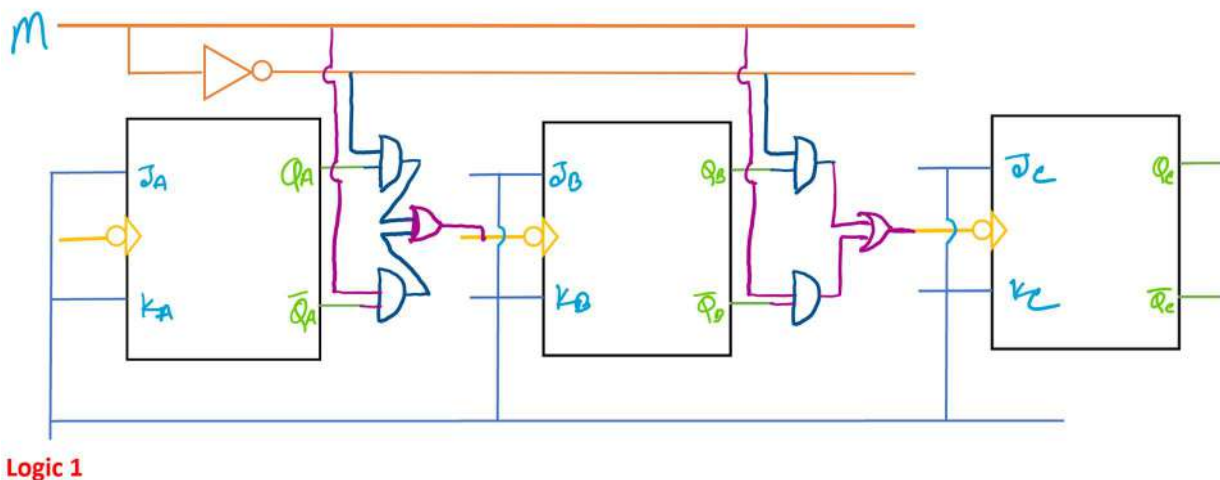
M	Q	Q'	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$\rightarrow$ K-Map for Y:

		00	01	11	10
0	0	0	0	1	1
1	0	0	1	1	0

$$Y = MQ' + \bar{M}Q$$

$\rightarrow$ Draw the logic circuit:



3-Bit Synchronous UP/DOWN Counter Using JK flip flop

1. We need control input **M**
2. Let, if $M=0 \rightarrow$ Up Counting
if $M=1 \rightarrow$ Down Counting

Step 1:

$\rightarrow$ For 3 bit we need 3 JK flip flop

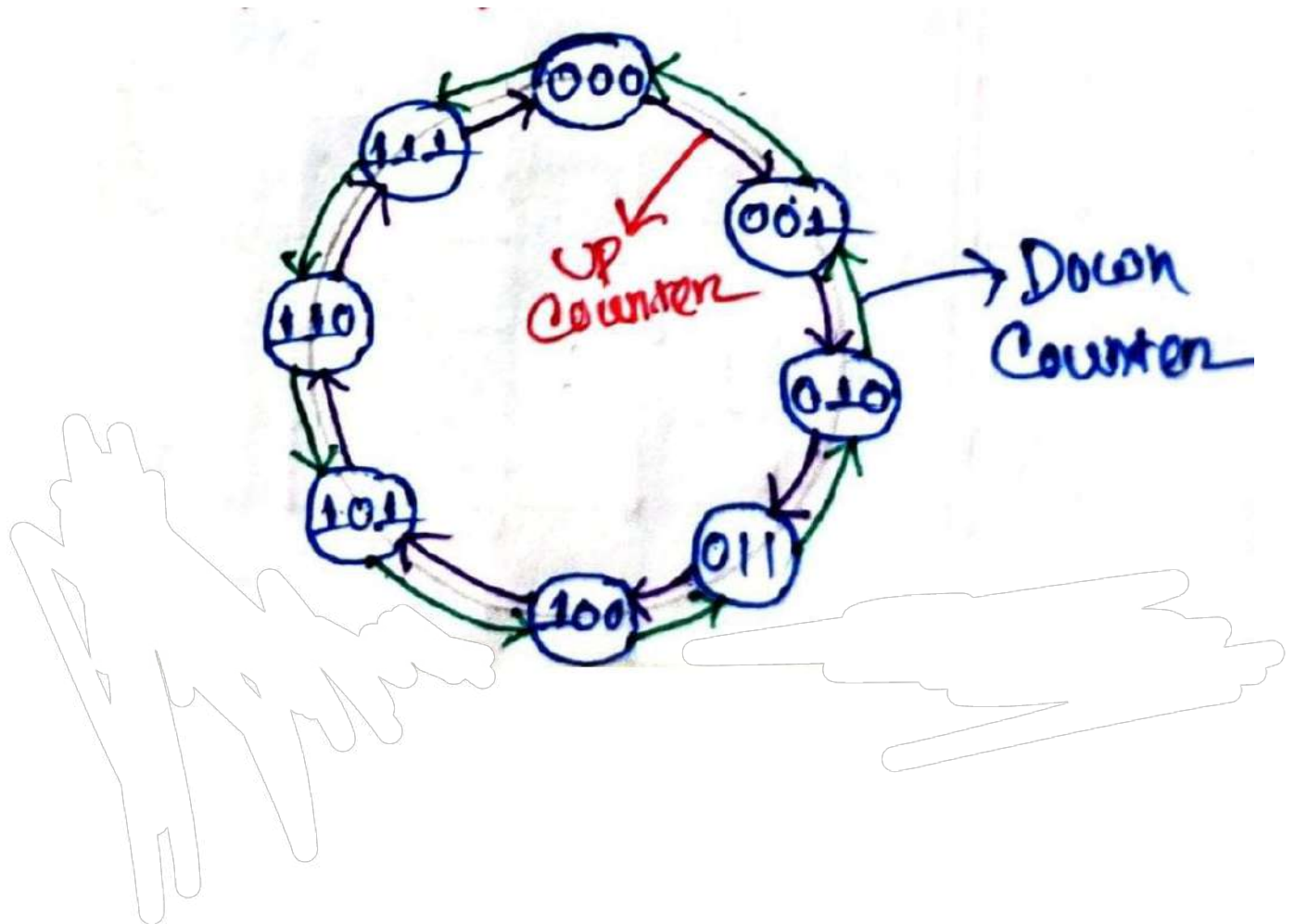
Step 2:

$\rightarrow$ [Draw the JK Excitation Table:](#)

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:

$\rightarrow$ **Draw State Diagram:**



Step 3.2:

CKT Excitation Table

M	Q_C	Q_B	Q_A	Q_C^*	Q_B^*	Q_A^*	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	0	0	0	X	1	X	1	X	1
1	0	0	0	1	1	1	1	X	1	X	1	X
1	0	0	1	0	0	0	0	X	0	X	X	1
1	0	1	0	0	0	1	0	X	X	1	1	X
1	0	1	1	0	1	0	0	X	X	0	X	1
1	1	0	0	0	1	1	X	1	1	X	1	X
1	1	0	1	1	0	0	X	0	0	X	X	1
1	1	1	0	1	0	1	X	0	X	1	1	X
1	1	1	1	1	1	0	X	0	X	0	X	1

Step 4:

→ K-Map for J_C :

		$Q_B Q_A$			
		00	01	11	10
$M Q_C$	00	0	0	1	0
	01	X	X	X	X
	11	X	X	X	X
	10	1	0	0	0

$$J_C = M Q'_B Q'_A + M' Q_B Q_A$$

→ K-Map for K_C :

		$Q_B Q_A$			
		00	01	11	10
$M Q_C$	00	X	X	X	X
	01	0	0	1	0
	11	1	0	0	0
	10	X	X	X	X

$$K_C = M Q'_B Q'_A + M' Q_B Q_A$$

→ K-Map for J_B :

$Q_B Q_A$	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	1	0	X	X
10	1	0	X	X

$$J_B = MQ'_A + M'Q_A$$

→ K-Map for K_B :

$Q_B Q_A$	00	01	11	10
00	X	X	1	0
01	X	X	1	0
11	X	X	0	1
10	X	X	0	1

$$K_B = MQ'_A + M'Q_A$$

→ K-Map for J_A :

$Q_B Q_A$	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$$J_A = 1$$

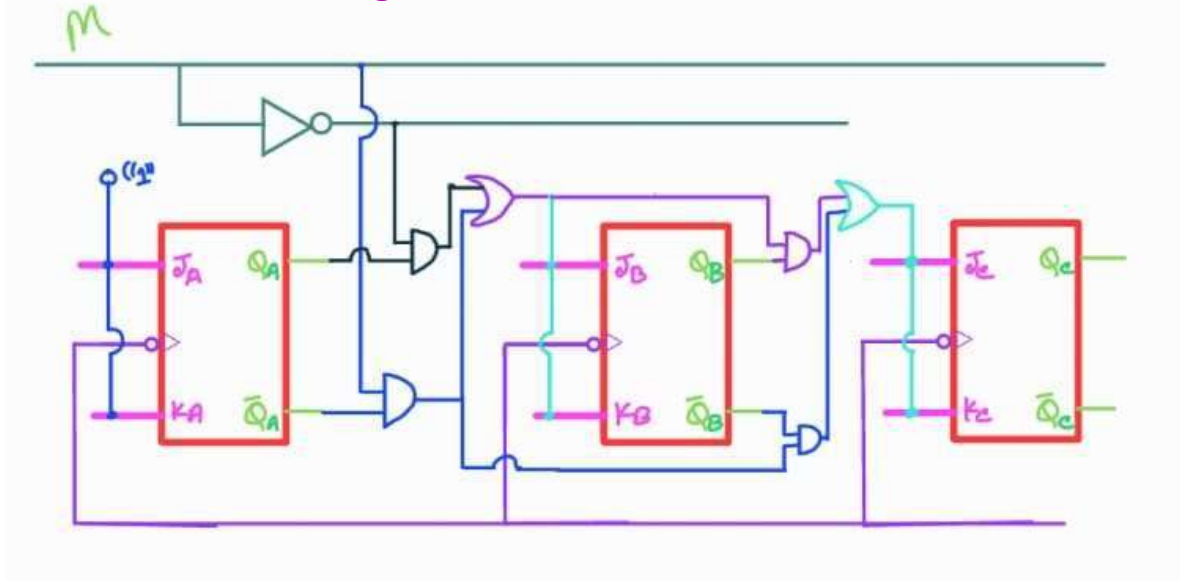
→ K-Map for K_A :

$Q_B Q_A$	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$$K_A = 1$$

Step 5:

→ Draw the Circuit Diagram:



Irregular Sequence Counter Design

Design a counter with the sequence 0,1,3,2,6,7,5 and 4 using JK flip flop

Step 1:

→ For 3 bit we need 3 JK flip flop

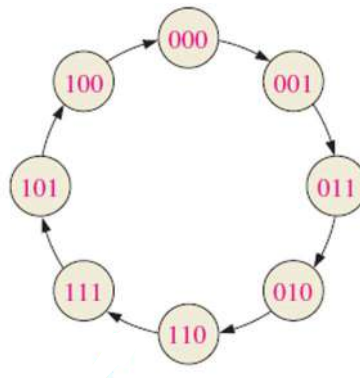
Step 2:

→ Draw the JK Excitation Table:

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:

→ **Draw State Diagram:**



Step 3.2:

CKT Excitation Table

Q _C	Q _B	Q _A	Q _C [*]	Q _B [*]	Q _A [*]	J _C	K _C	J _B	K _B	J _A	K _A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	1	1	0	1	X	X	0	0	X
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	1	0	1	X	0	X	1	X	0
1	0	1	1	0	0	X	0	0	X	X	1
1	0	0	0	0	0	X	1	0	X	0	X

Step 4:

→ K-Map for J_C :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	0	0	0	1
1	X	X	X	X

$$J_C = Q_B Q'_A$$

→ K-Map for K_C :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	X	X	X	X
1	1	0	0	0

$$K_C = Q'_B Q'_A$$

→ K-Map for J_B :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	0	1	X	X
1	0	0	X	X

$$J_B = Q'_C Q_A$$

→ K-Map for K_B :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	X	X	0	0
1	X	X	1	0

$$K_B = Q_C Q_A$$

→ K-Map for J_A :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	1	X	X	0
1	0	X	X	1

$$J_A = Q'_C Q'_B + Q_C Q_B$$

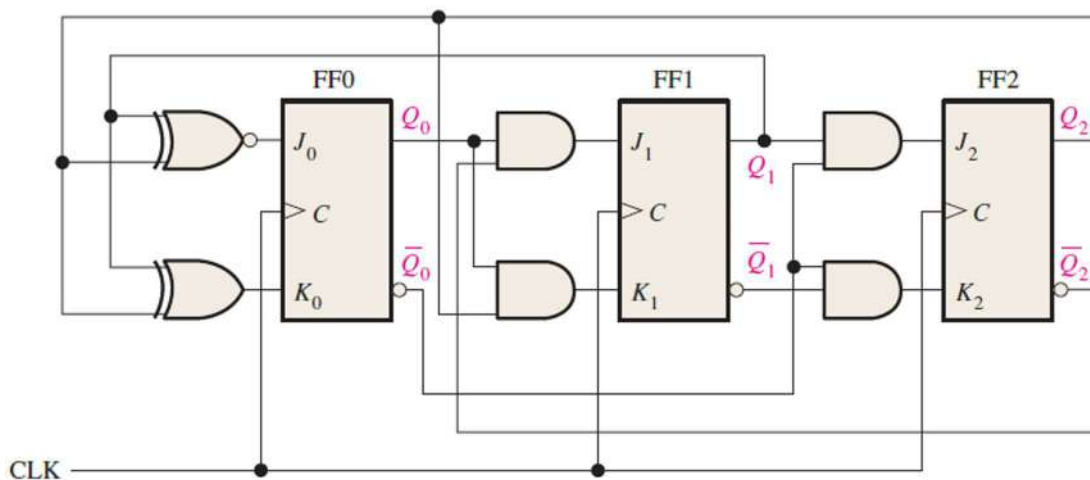
→ K-Map for K_A :

$Q_C \backslash Q_B Q_A$	00	01	11	10
0	X	0	1	X
1	X	1	0	X

$$K_A = Q_C Q'_B + Q'_C Q_B$$

Step 5:

→ Draw the Circuit Diagram:



Design a counter with the sequence 1,2,5 and 7 using JK flip flop

Step 1:

→ For 3 bit we need 3 JK flip flop

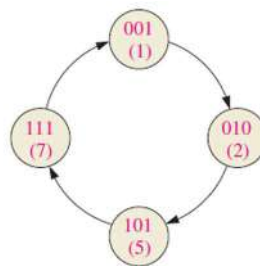
Step 2:

→ [Draw the JK Excitation Table:](#)

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:

→ **Draw State Diagram:**



Step 3.2:

CKT Excitation Table

Q _C	Q _B	Q _A	Q _C [*]	Q _B [*]	Q _A [*]	J _C	K _C	J _B	K _B	J _A	K _A
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	1	0	1	1	X	X	1	1	X
1	0	1	1	1	1	X	0	1	X	X	0
1	1	1	0	0	1	X	1	X	1	X	0

Step 4:

→ **K-Map for J_C:**

Q _C \ Q _B Q _A	00	01	11	10
0	X	0	X	1
1	X	X	X	X

$$J_C = Q_B$$

→ **K-Map for K_C:**

Q _C \ Q _B Q _A	00	01	11	10
0	X	X	X	X
1	X	0	1	X

$$K_C = Q_B$$

→ K-Map for J_B :

$Q_C \backslash Q_A$	00	01	11	10
0	X	1	X	X
1	X	1	X	X

$$J_B = 1$$

→ K-Map for K_B :

$Q_C \backslash Q_A$	00	01	11	10
0	X	X	X	1
1	X	X	1	X

$$K_B = 1$$

→ K-Map for J_A :

$Q_C \backslash Q_A$	00	01	11	10
0	X	X	X	1
1	X	X	X	X

$$J_A = 1$$

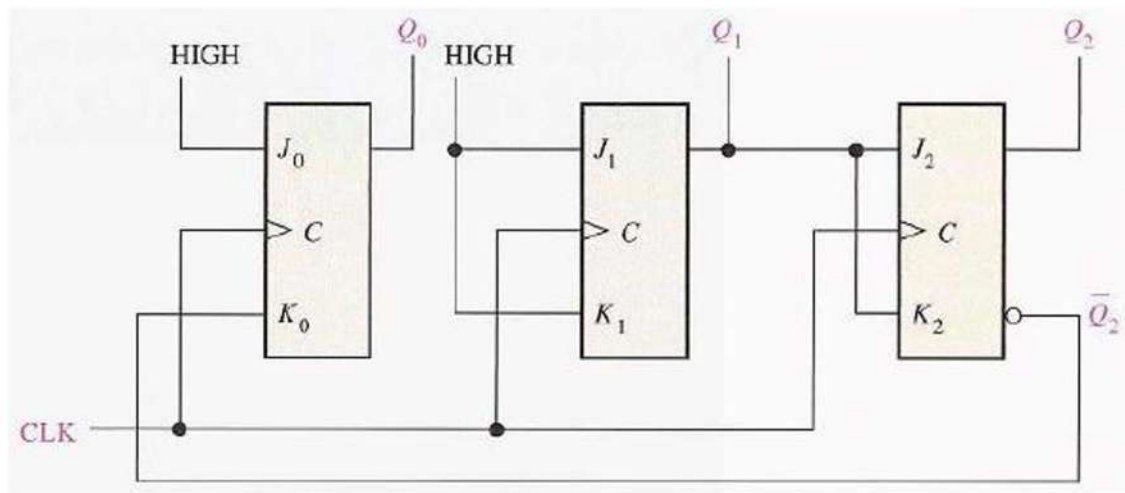
→ K-Map for K_A :

$Q_C \backslash Q_A$	00	01	11	10
0	X	1	X	X
1	X	0	0	X

$$K_A = Q'_C$$

Step 5:

→ Draw the Circuit Diagram:



Develop a synchronous 3-bit up/down counter with a Gray code sequence using J-K flip-flops. The counter should count up when an UP/DOWN control input is 1 and count down when the control input is 0. (0,1,3,2,6,7,5,4)

1. We need control input **Y**
2. Is given, if $Y=1 \rightarrow$ Up Counting
if $Y=0 \rightarrow$ Down Counting

Step 1:

$\rightarrow$ For 3 bit we need 3 JK flip flop

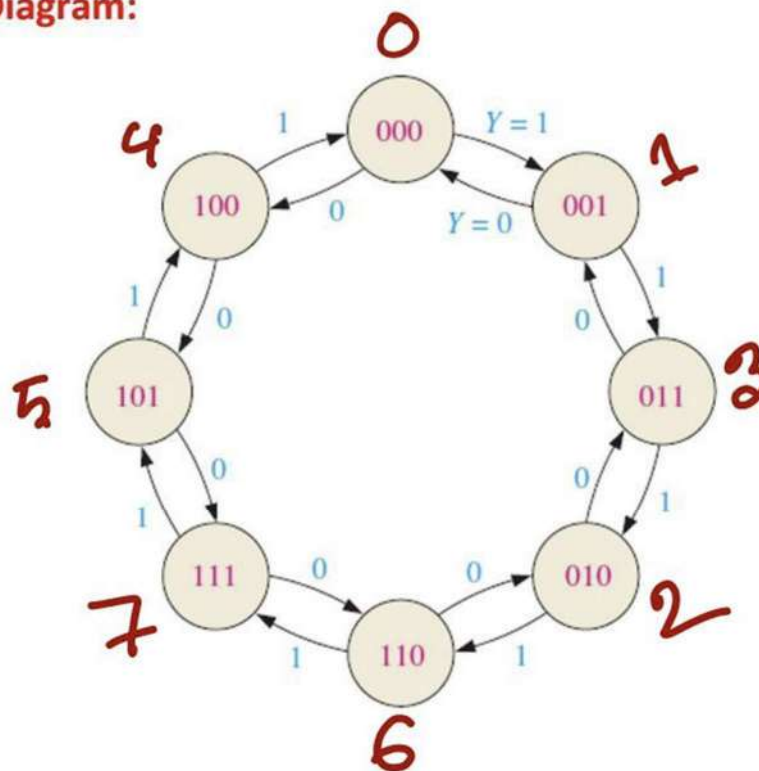
Step 2:

$\rightarrow$ Draw the JK Excitation Table:

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Step 3.1:

$\rightarrow$ Draw State Diagram:



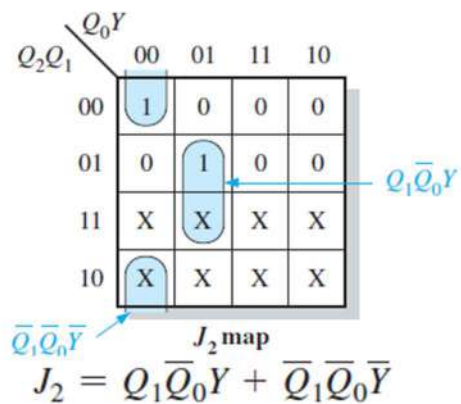
Step 3.2:

CKT Excitation Table

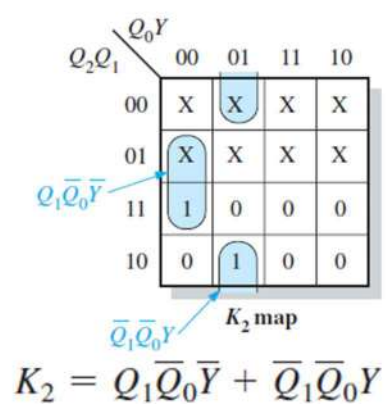
Q_2	Q_1	Q_0	Y	Q_2^*	Q_1^*	Q_0^*	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	1	0	0	1	X	0	X	0	X
0	0	1	0	0	0	0	0	X	0	X	X	1
0	1	1	0	0	0	1	0	X	X	1	X	0
0	1	0	0	0	1	1	0	X	X	0	1	X
1	1	0	0	0	1	0	X	1	X	0	0	X
1	1	1	0	1	1	0	X	0	X	0	X	1
1	0	1	0	1	1	1	X	0	1	X	X	0
1	0	0	0	1	0	1	X	0	0	X	1	X
0	0	0	1	0	0	1	0	X	0	X	1	X
0	0	1	1	0	1	1	0	X	1	X	X	0
0	1	1	1	0	1	0	0	X	X	0	X	1
0	1	0	1	1	1	0	1	X	X	0	0	X
1	1	0	1	1	1	1	X	0	X	0	1	X
1	1	1	1	1	0	1	X	0	X	0	X	0
1	0	1	1	1	0	0	X	0	0	X	X	1
1	0	0	1	0	0	0	X	1	0	X	X	0

Step 4:

→ K-Map for J_C/J_2 :



→ K-Map for K_C/K_2 :



→ K-Map for J_B/J_1 :

Q_2Q_1		Q_0Y			
		00	01	11	10
00	0	0	1	0	
01	X	X	X	X	
11	X	X	X	X	
10	0	0	0	1	

J_1 map

Annotations: $\bar{Q}_2Q_0Y$ (pointing to cell 011), $Q_2Q_0\bar{Y}$ (pointing to cell 101)

$$J_1 = \bar{Q}_2Q_0Y + Q_2Q_0\bar{Y}$$

→ K-Map for K_B/K_1 :

Q_2Q_1		Q_0Y			
		00	01	11	10
00	X	X	X	X	
01	0	0	0	1	
11	0	0	1	0	
10	X	X	X	X	

K_1 map

Annotations: $\bar{Q}_2Q_0Y$ (pointing to cell 011), Q_2Q_0Y (pointing to cell 101)

$$K_1 = \bar{Q}_2Q_0\bar{Y} + Q_2Q_0Y$$

→ K-Map for J_A/J_0 :

Q_2Q_1		Q_0Y			
		00	01	11	10
00	0	1	X	X	
01	1	0	X	X	
11	0	1	X	X	
10	1	0	X	X	

J_0 map

Annotations: $\bar{Q}_2\bar{Q}_1Y$ (pointing to cell 001), $\bar{Q}_2Q_1\bar{Y}$ (pointing to cell 010), Q_2Q_1Y (pointing to cell 111), $Q_2\bar{Q}_1\bar{Y}$ (pointing to cell 100)

$$J_0 = Q_2Q_1Y + Q_2\bar{Q}_1\bar{Y} + \bar{Q}_2\bar{Q}_1Y + \bar{Q}_2Q_1\bar{Y}$$

→ K-Map for K_A/K_0 :

Q_2Q_1		Q_0Y			
		00	01	11	10
00	X	X	0	1	
01	X	X	1	0	
11	X	X	0	1	
10	X	X	1	0	

K_0 map

Annotations: $\bar{Q}_2\bar{Q}_1\bar{Y}$ (pointing to cell 000), $\bar{Q}_2Q_1Y$ (pointing to cell 011), $Q_2Q_1\bar{Y}$ (pointing to cell 110), $Q_2\bar{Q}_1Y$ (pointing to cell 101)

$$K_0 = \bar{Q}_2\bar{Q}_1\bar{Y} + \bar{Q}_2Q_1Y + Q_2\bar{Q}_1Y + Q_2Q_1\bar{Y}$$

SHIFT REGISTERS

Shift Registers

⇒ It is a Digital circuit with Two basic Function.

1. Data Storage &
2. Data Movement

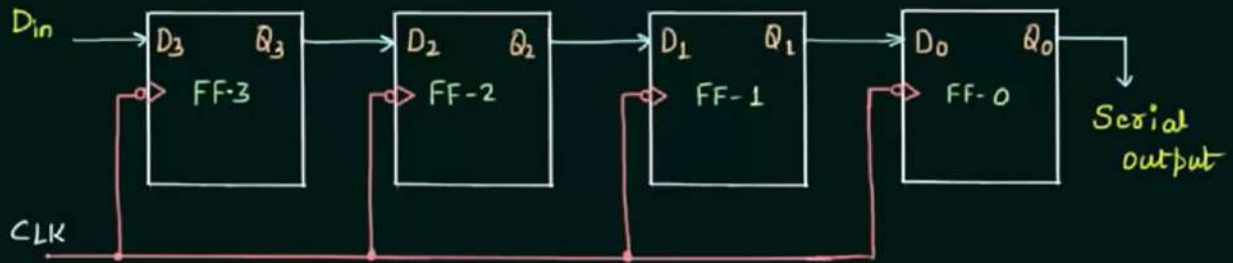
Note: Each flop-flop Stores 1 bit, & the number of flip-flops determine the registers Storage capacity.

Types of Shift Registers

1. Serial In Serial Out (SISO)
2. Serial In Parallel Out (SIPO)
3. Parallel In Serial Out (PISO)
4. Parallel In Parallel Out (PIPO)

Serial In Serial Out (SISO Mode)

⇒ We want to Store “1111” in this Shift Register.



Step 1: Identify the **LSB** and **MSB**.

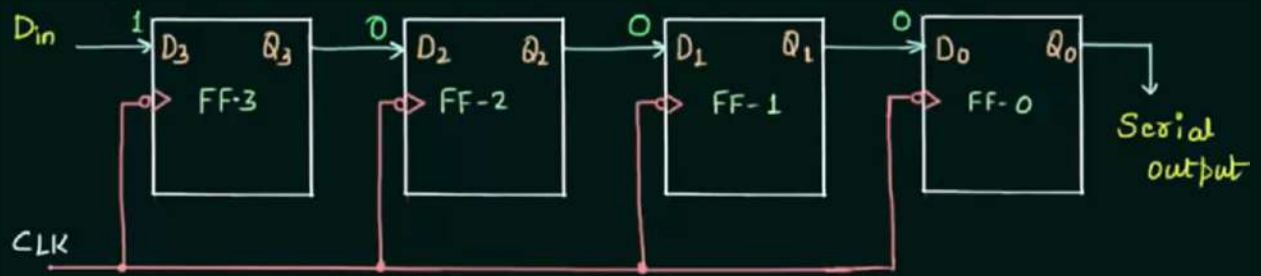
Step 2: Draw the D. FF TT:



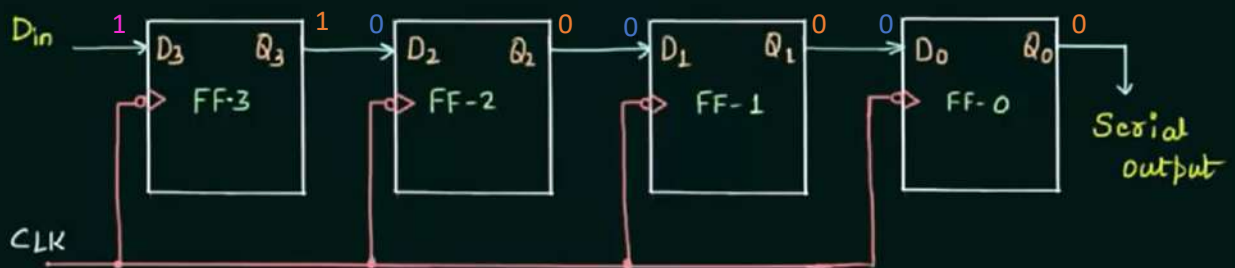
CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Step 3: Initially

CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓				
↓				
↓				
↓				

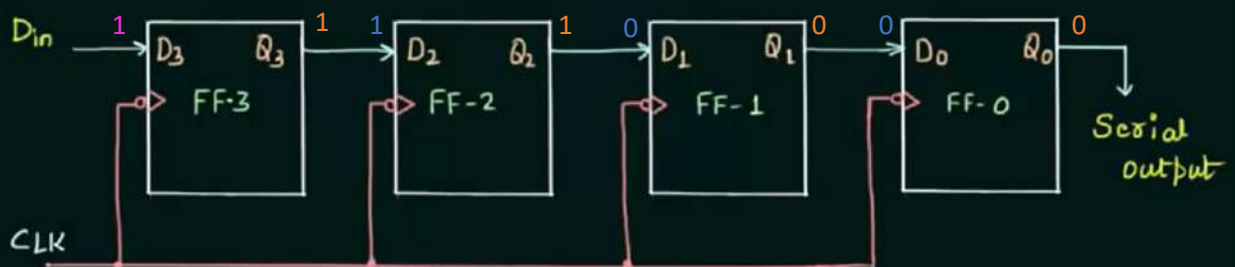


Step 4: for 1st falling edge:



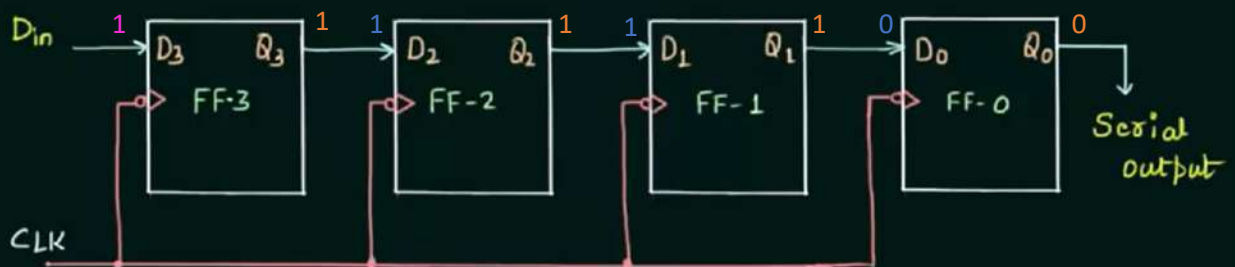
CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓	1	0	0	0
↓				
↓				
↓				

Step 5: for 2nd falling edge:



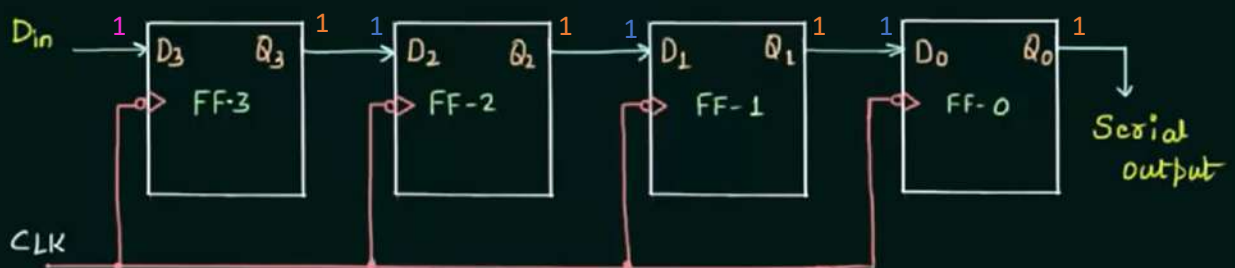
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓				
↓				

Step 5: for 3rd falling edge:



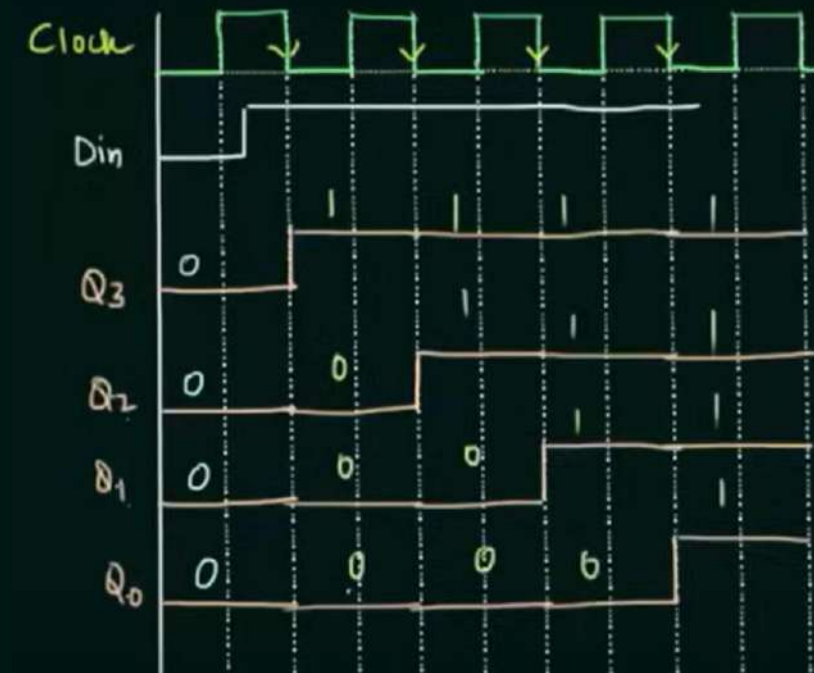
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓				

Step 6: for 4th falling edge:



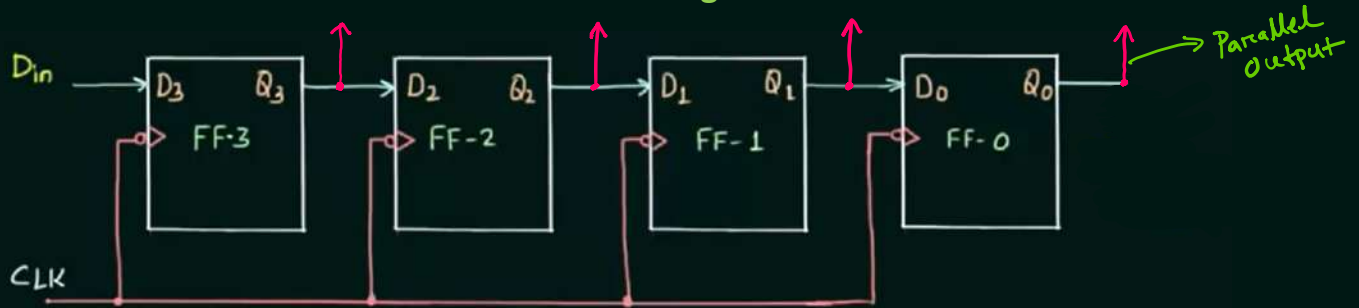
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓	1	1	1	1

Analyses the clock Diagram



Serial In Parallel Out (SIPO Mode)

⇒ We want to Store "1111" in this Shift Register.



Step 1: Identify the **LSB** and **MSB**.

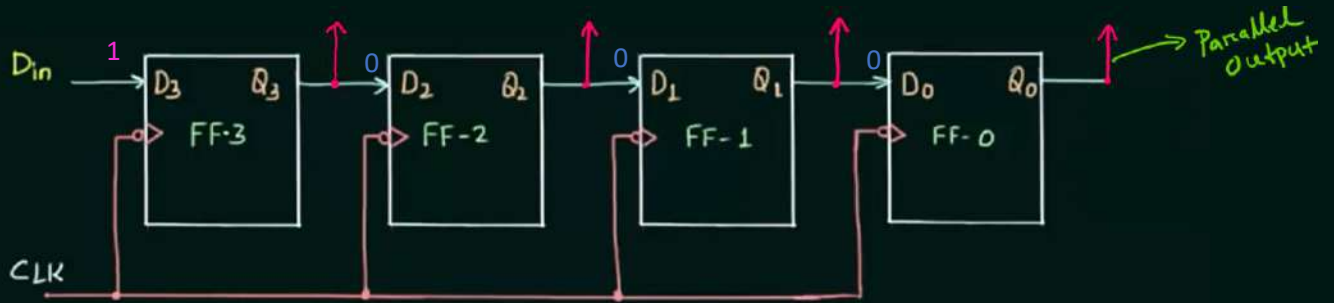
Step 2: Draw the D. FF TT:



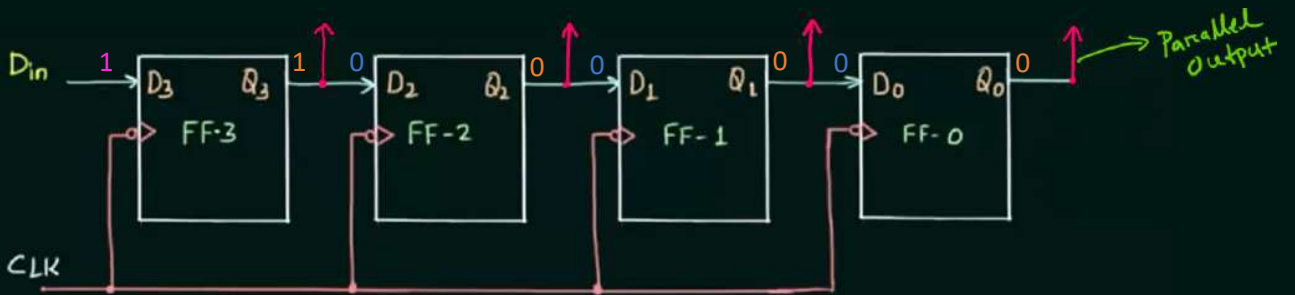
CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

Step 3: Initially

CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓				
↓				
↓				
↓				

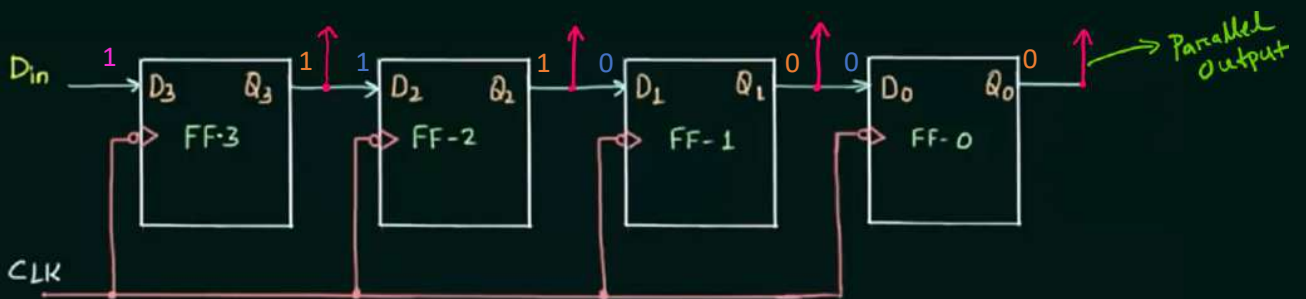


Step 4: for 1st falling edge:



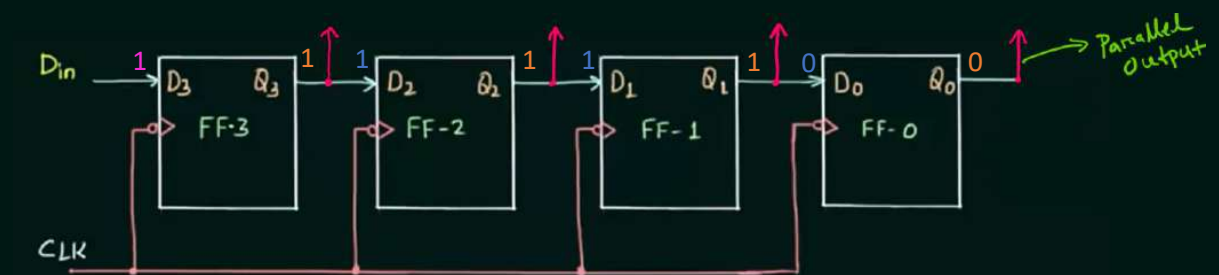
CLK	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
↓	1	0	0	0
↓				
↓				
↓				

Step 5: for 2nd falling edge:



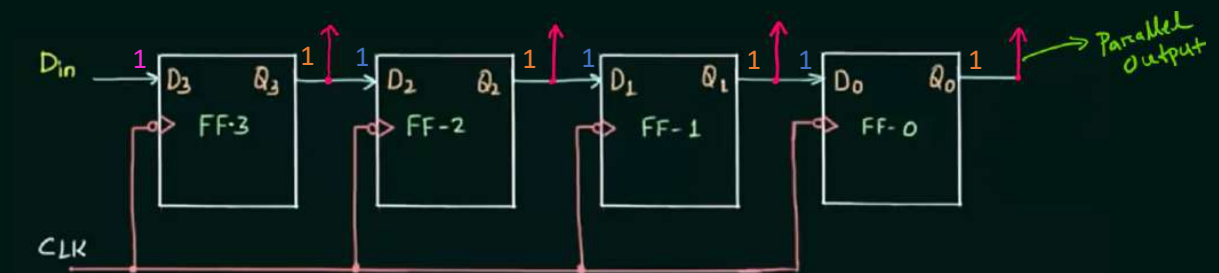
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓				
↓				

Step 5: for 3rd falling edge:



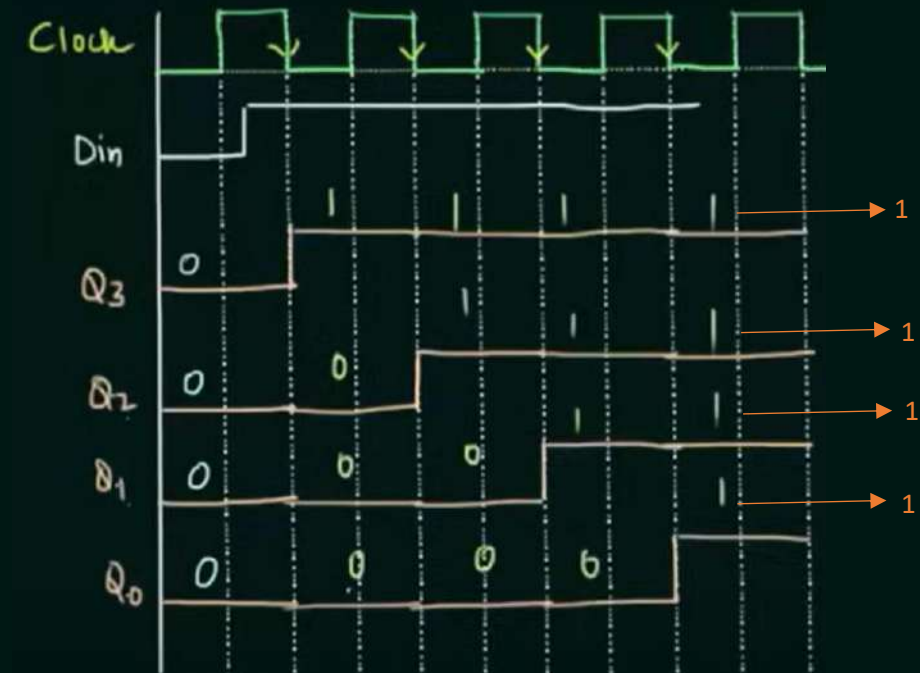
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓				

Step 6: for 4th falling edge:



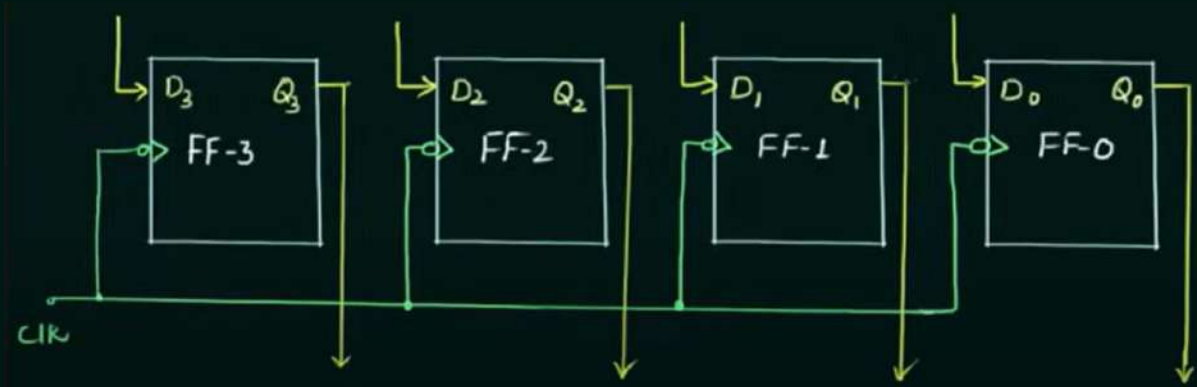
CLK	Q ₃	Q ₂	Q ₁	Q ₀
Initially	0	0	0	0
↓	1	0	0	0
↓	1	1	0	0
↓	1	1	1	0
↓	1	1	1	1

Analyses the clock Diagram



Parallel In Parallel Out (PIPO Mode)

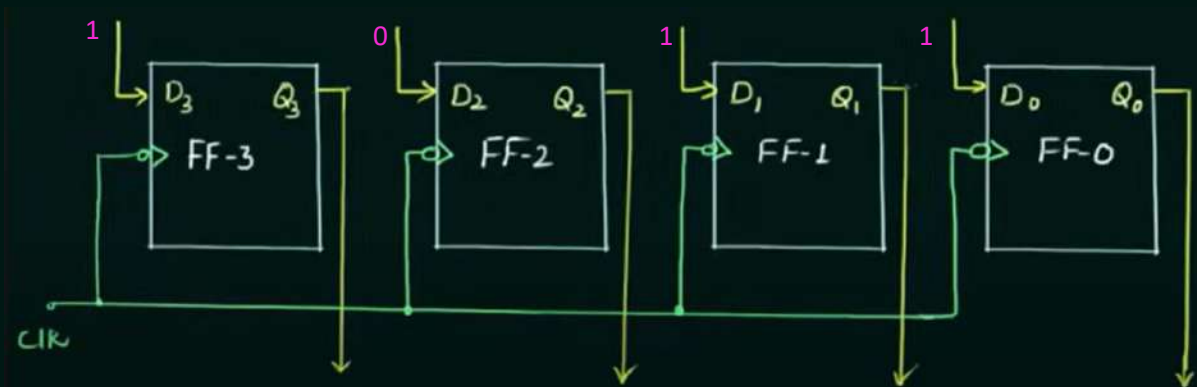
⇒ **Example 1:** We want to Store “1011” in this Shift Register.



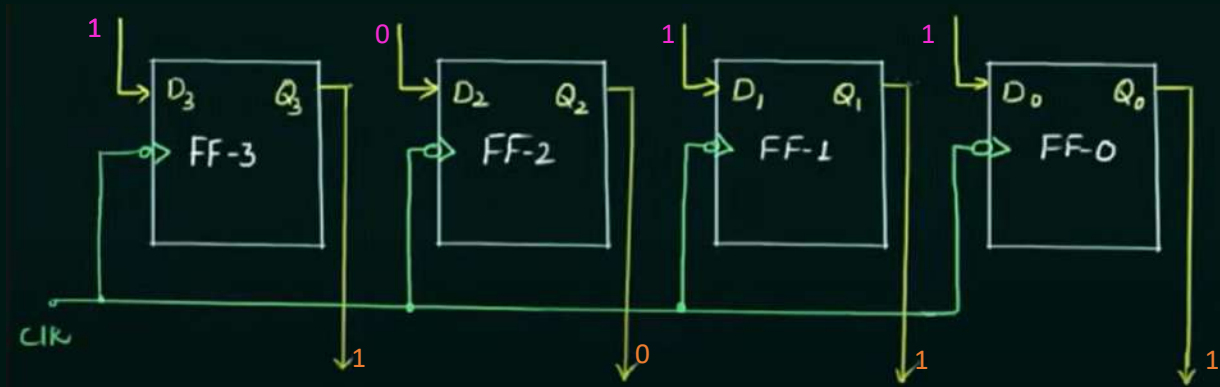
Step 1: Draw the D. FF TT:

CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

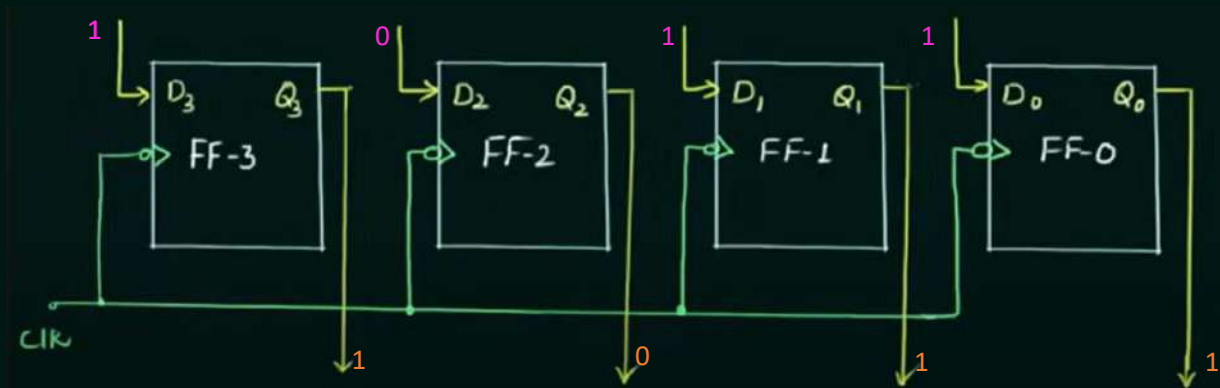
Step 2: Insert the Parallel Inputs:



Step 3: When CLK is “logic 1”

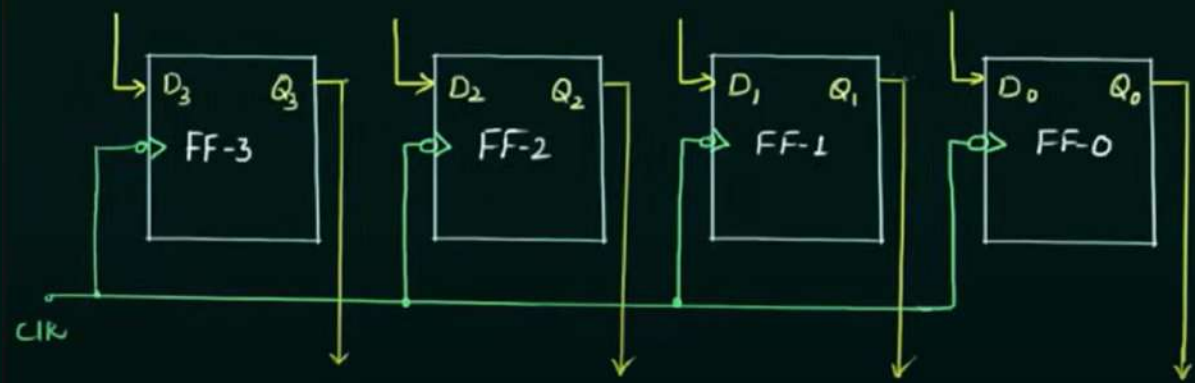


Step 4: When CLK is “logic 0”



⇒ Its Hold the Same value since if CLK = 0 then input = Don't care Output is Memory.

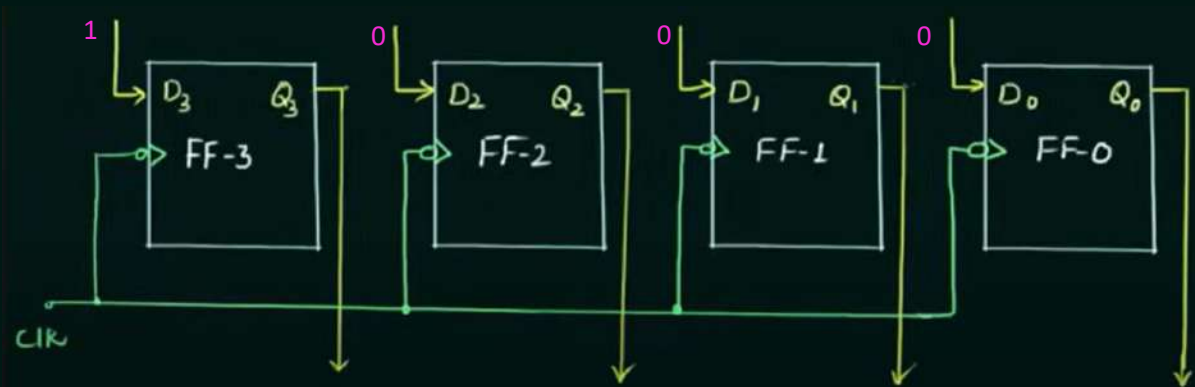
⇒ **Example 2:** We want to Store “1000” in this Shift Register.



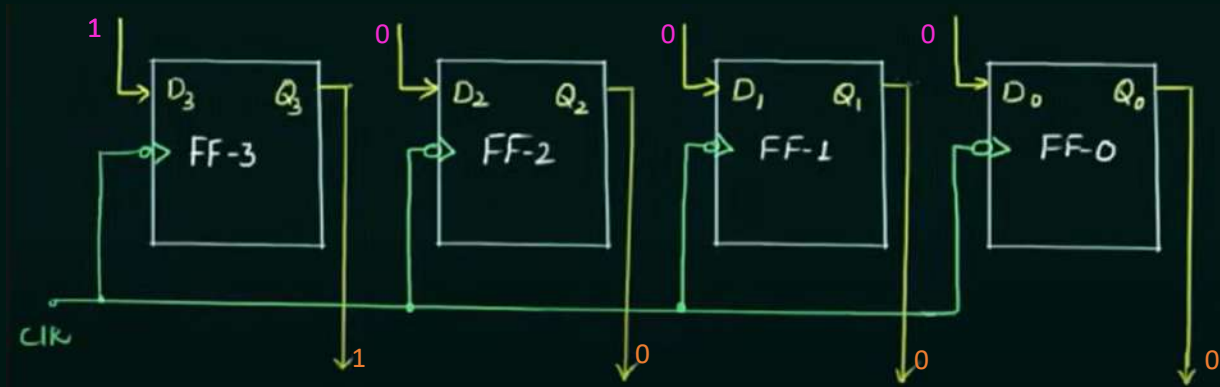
Step 1: Draw the D. FF TT:

CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

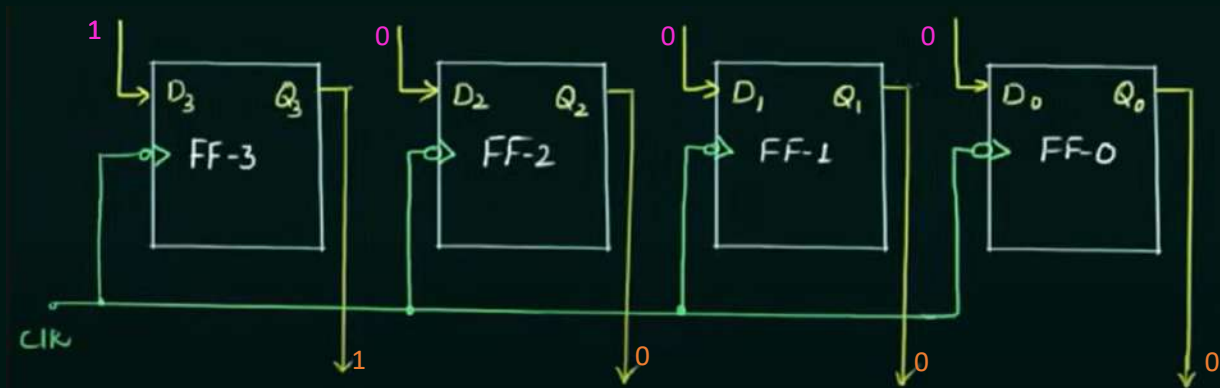
Step 2: Insert the Parallel Inputs:



Step 3: When CLK is “logic 1”



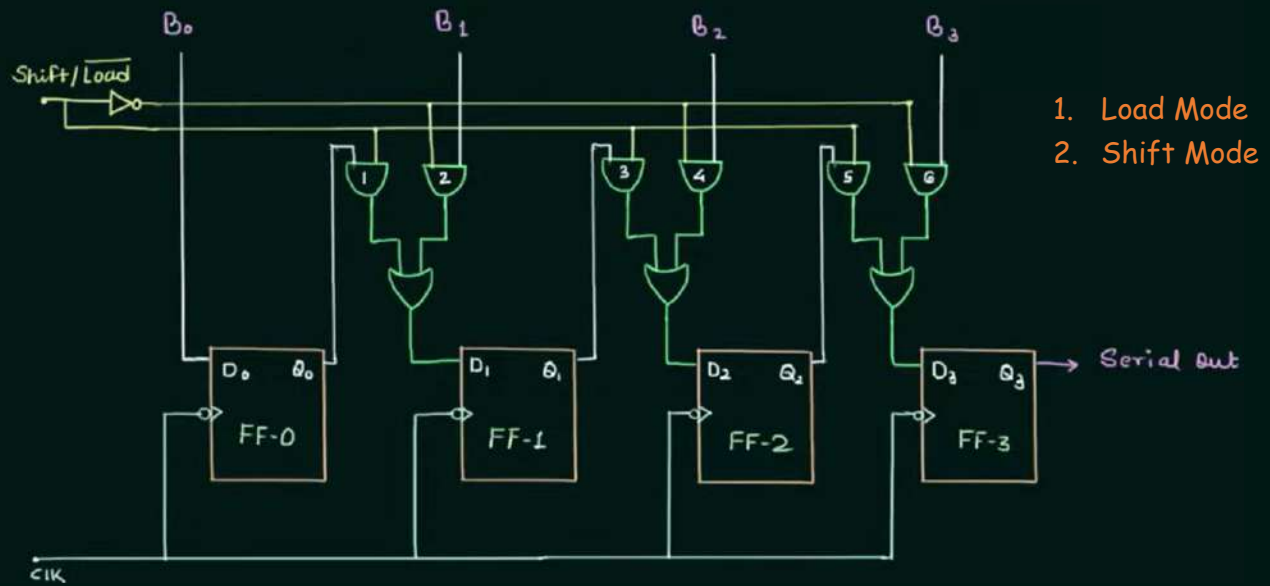
Step 4: When CLK is “logic 0”



⇒ Its Hold the Same value since if CLK = 0 then input = Don't care Output is Memory.

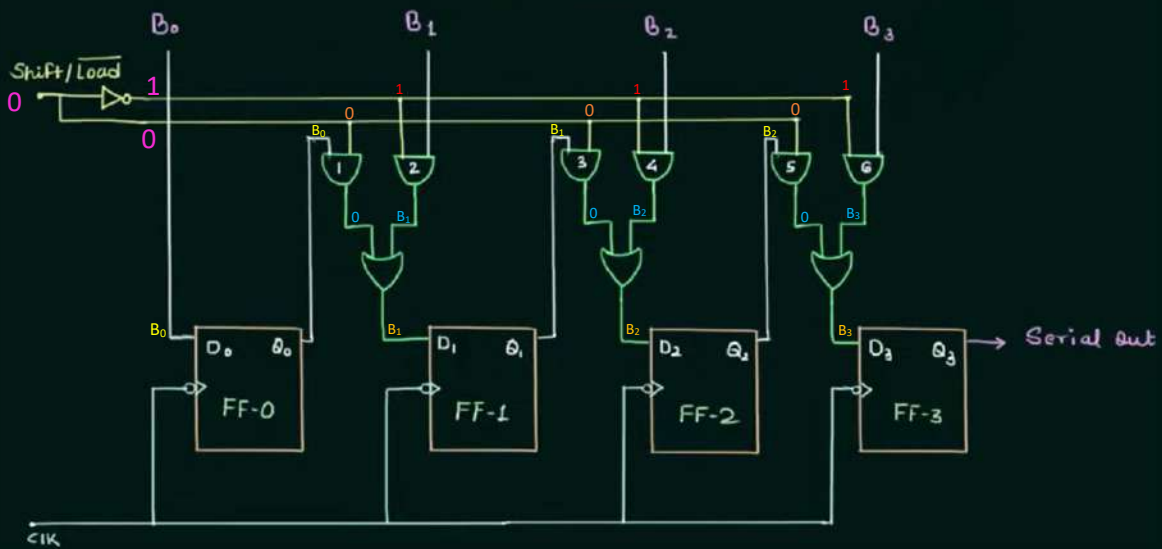
Parallel In Serial Out (PISO Mode)

⇒ The circuit Diagram is:



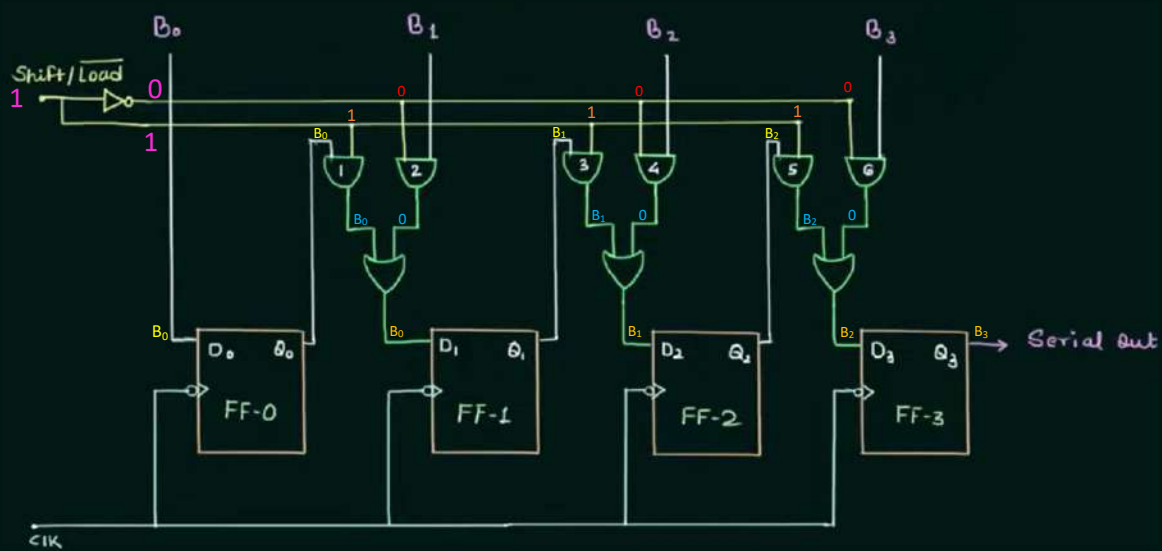
1. How is Load Mode Achieved?

⇒ For that $\text{Shift}/\overline{\text{Load}} = 0$



2. How is Shift Mode Achieved?

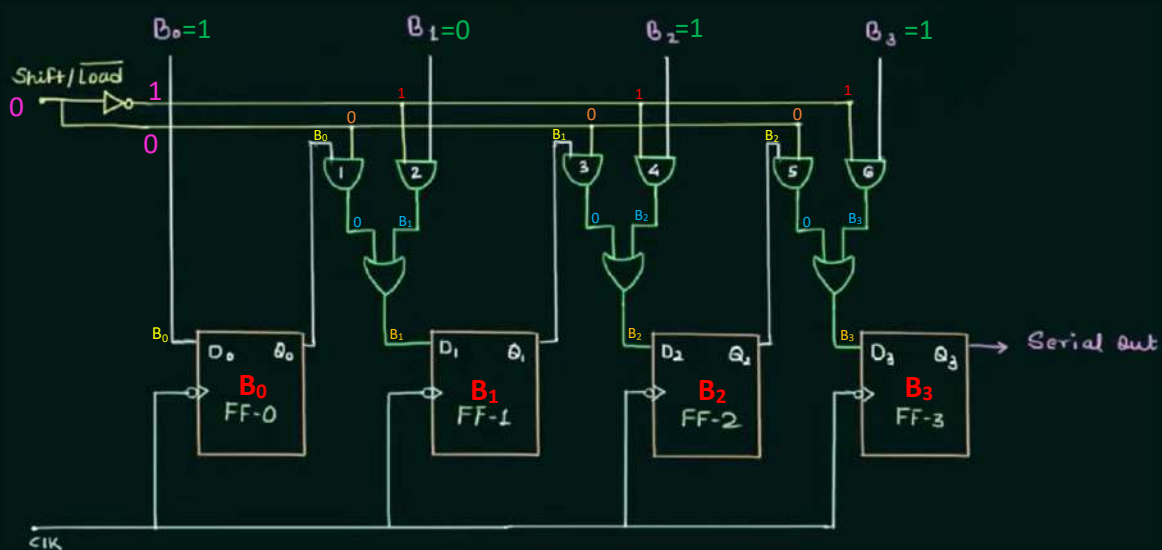
⇒ For that Shift/Load' is = 1



⇒ We want to Store "1011" in this Shift Register.

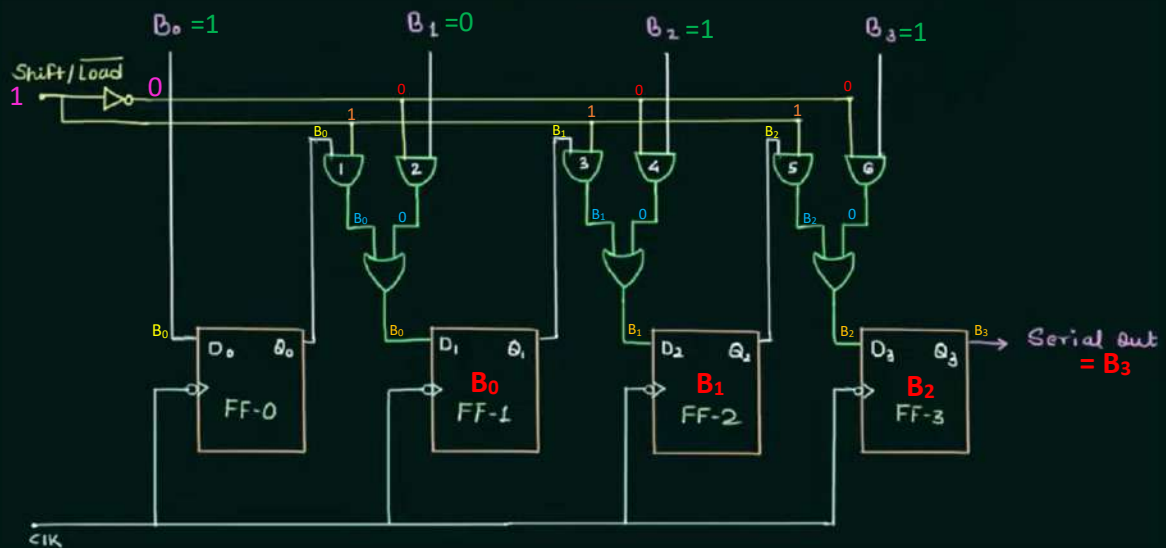
Step 1: 1st Achieved Load Mode?

⇒ For that Shift/Load' is = 0



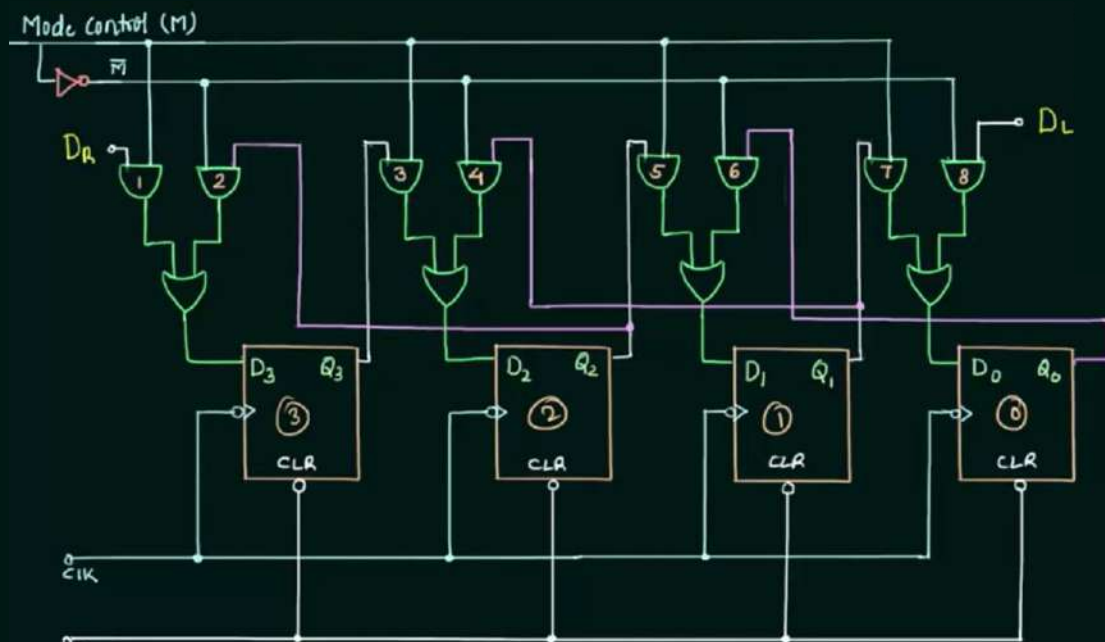
⇒ **Step 2: 2nd Achieved Shift Mode?**

⇒ **For that Shift/Load' is = 1**

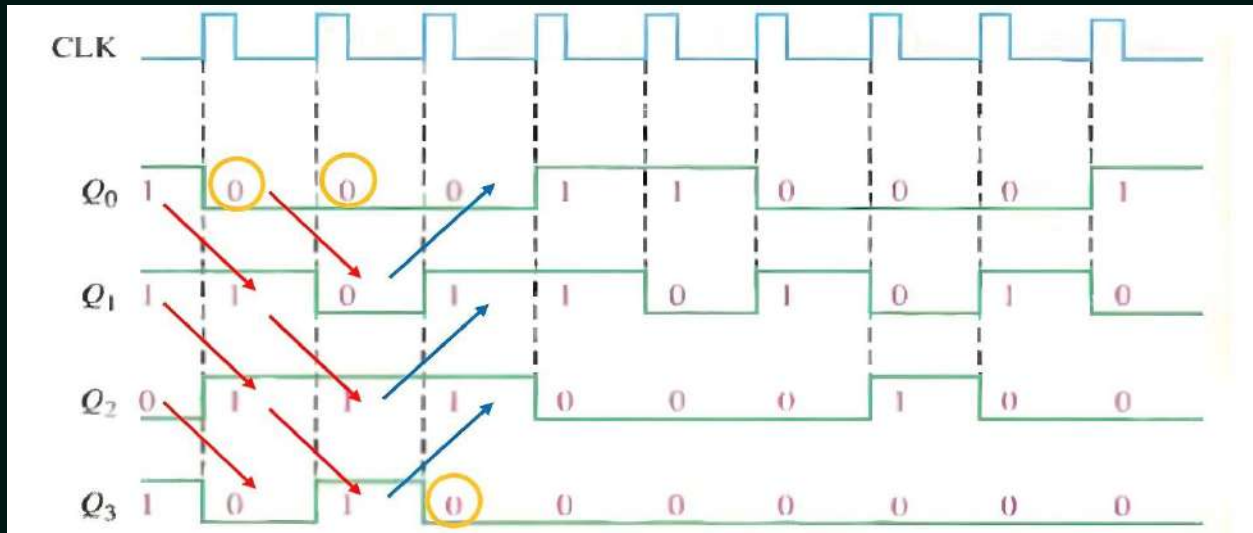


Bidirectional Shift Register

⇒ **The circuit Diagram is:**



Example 1: Determine the state of the shift register after each clock pulse for the given RIGHT/LEFT control input waveform. Assume $Q_0=1$, $Q_1=1$, $Q_2=0$, and $Q_3=1$ and the serial data input line is LOW.



Shift Register Counters

⇒ A shift register counter is a shift register with its serial output connected to its serial input, creating a specific sequence of states.

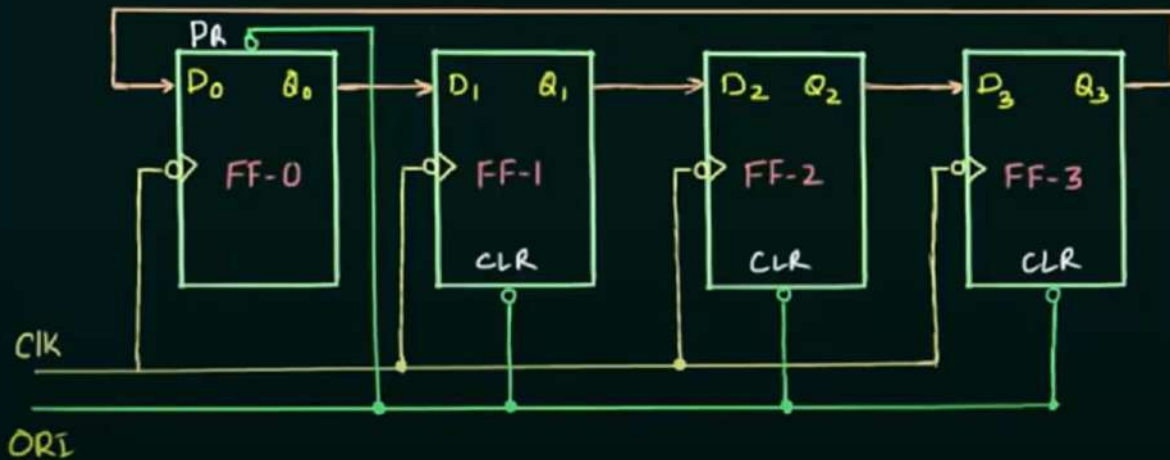
There are 2 types:

1. Ring Counter
2. Johnson Counters

RING COUNTER

Ring Counter

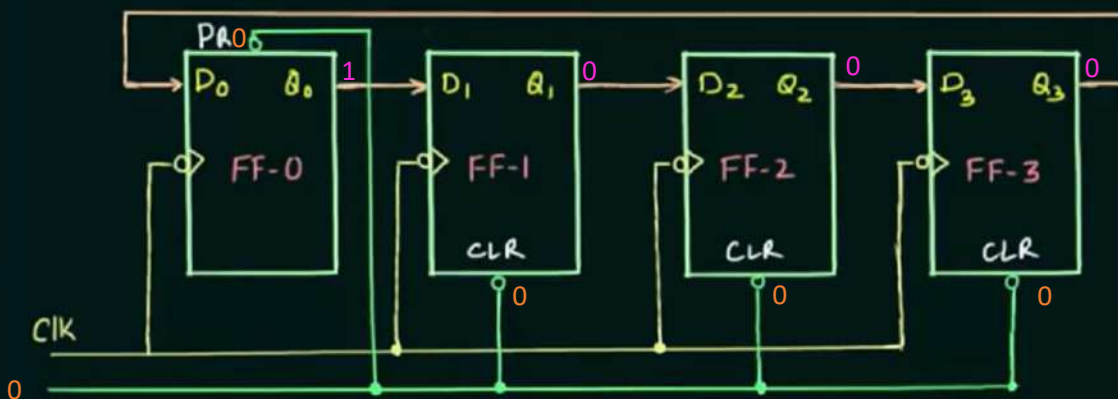
- >> Ring counter is a typical application of shift register
- >> The only change is the output of last ff is connected to the input of first ff.



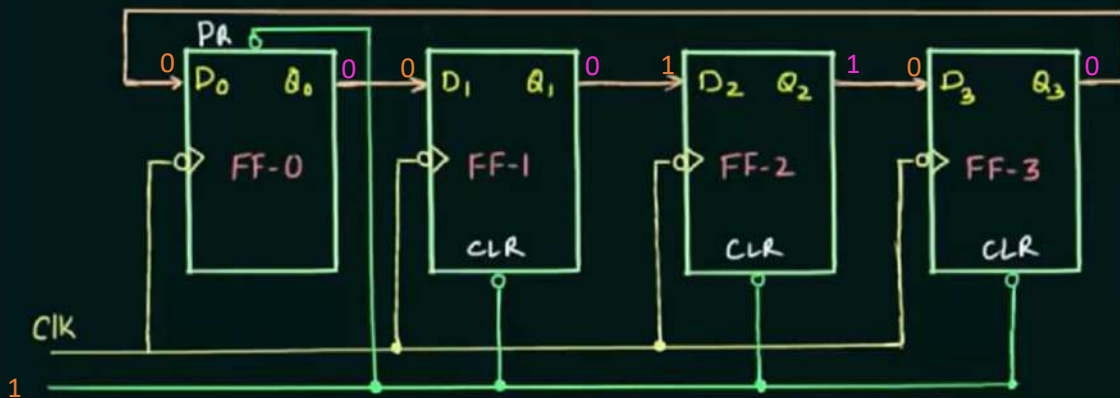
Note: NO of State = No. of flip-flops Used in the circuit.

4-bit Ring counter

Step 1: Assume that Initially ORI=0

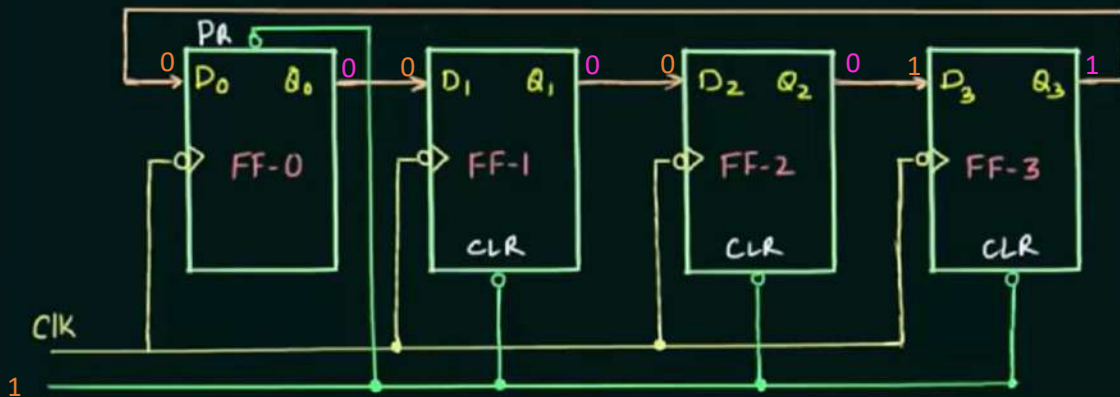


Step 2: Now for 2nd falling edge:



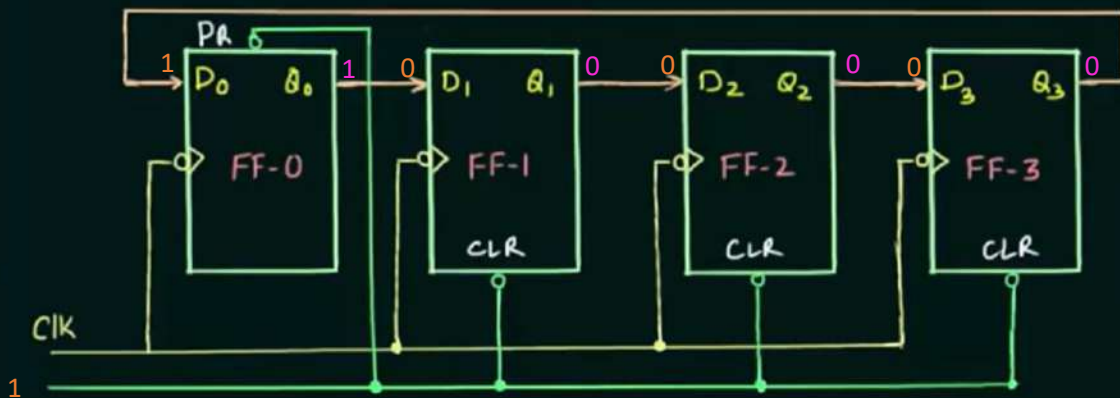
ORI	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓				
1	↓				

Step 3: Now for 3rd falling edge:



ORI	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓				

Step 4: Now for 4th falling edge:



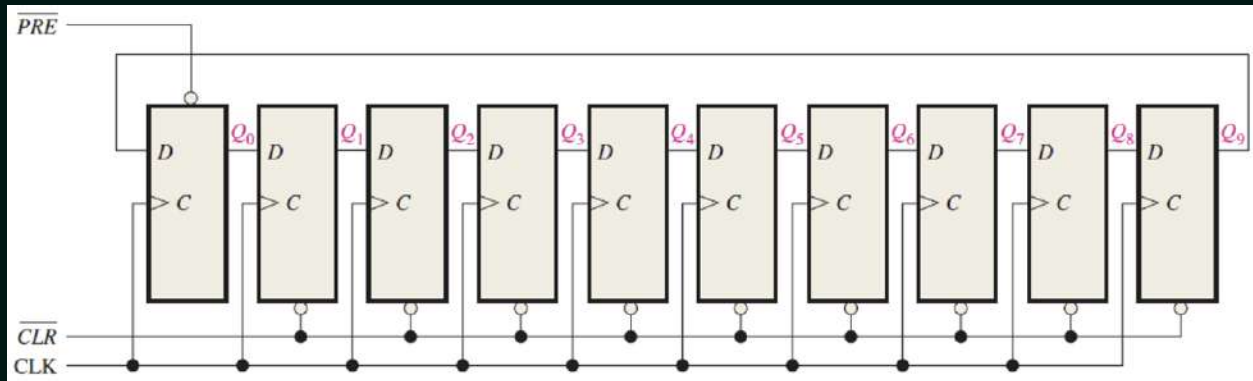
ORI	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

Analyses the clock Diagram



10-bit ring counter

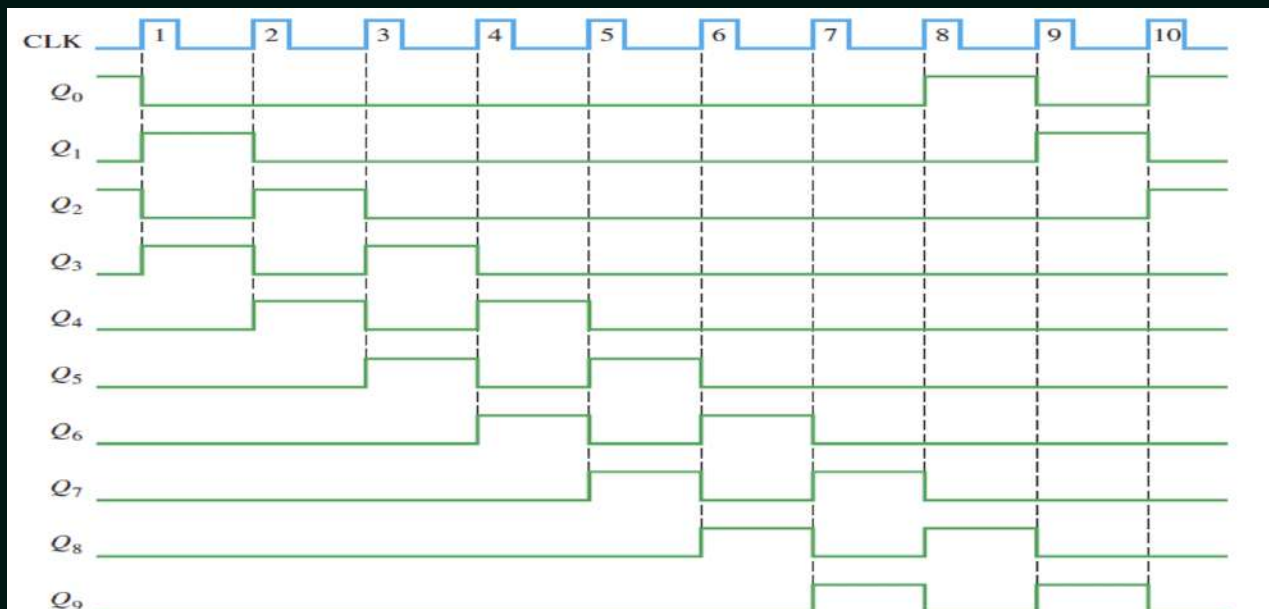
⇒ The circuit Diagram is:



⇒ The 10-bit Ring Counter Sequence is:

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9
0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	0	0	0
4	0	0	0	0	1	0	0	0	0	0
5	0	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	0	1	0	0	0
7	0	0	0	0	0	0	0	1	0	0
8	0	0	0	0	0	0	0	0	1	0
9	0	0	0	0	0	0	0	0	0	1

Analyses the clock Diagram



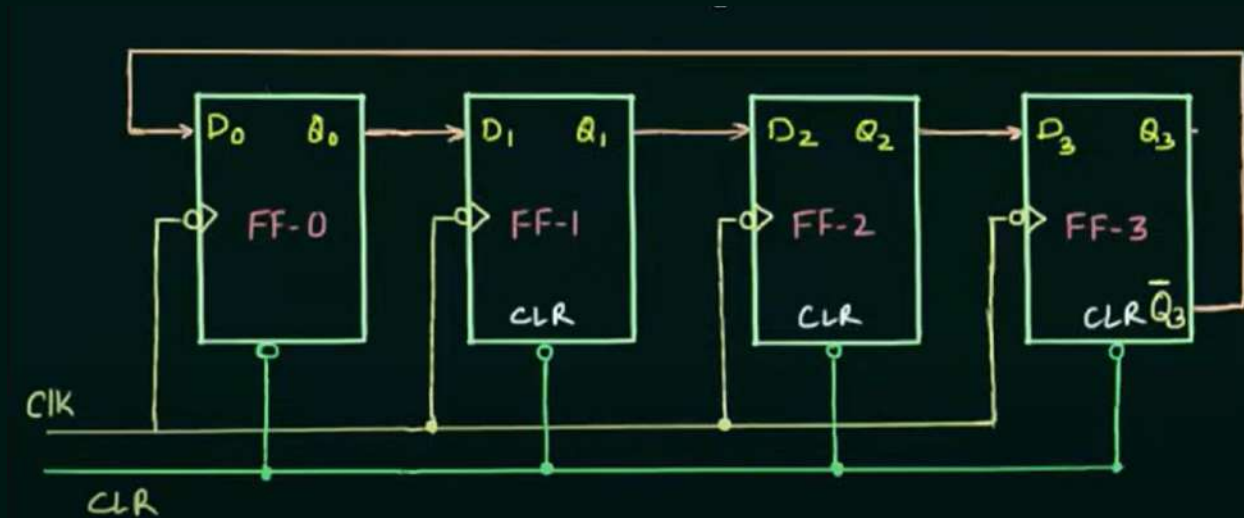
JOHNSON
COUNTER

Johnson Counter

>> Its also known as Twisted/Switch tail Ring Counter.

>> In a Johnson counter, the complement of the last flip-flop's output connects to the first flip-flop's input, generating a unique sequence. A Johnson counter has $2N$ states for N flip-flops (e.g., 4-bit = 8 states, 5-bit = 10 states).

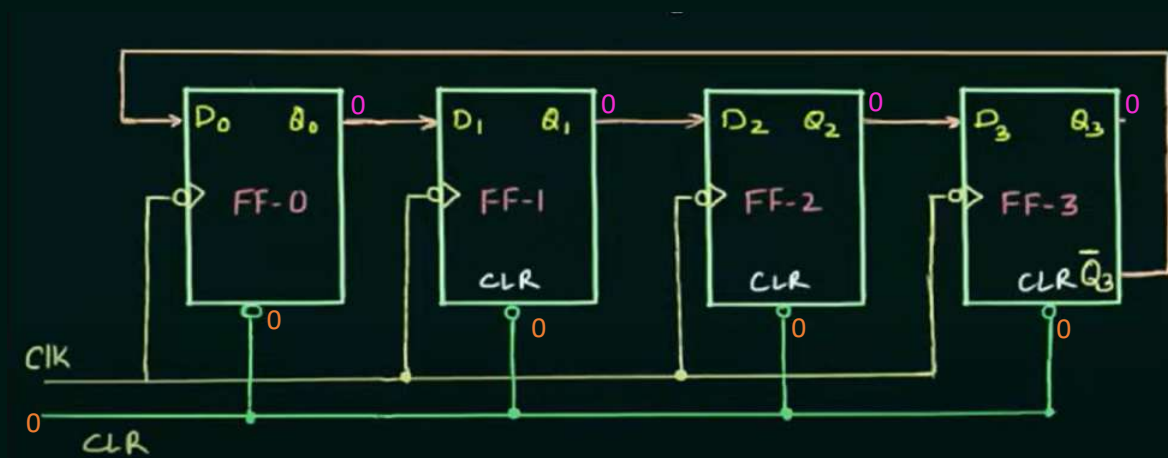
⇒ The circuit Diagram is:



Note: NO of State = $2 * \text{No. of flip-flops Used in the circuit.}$

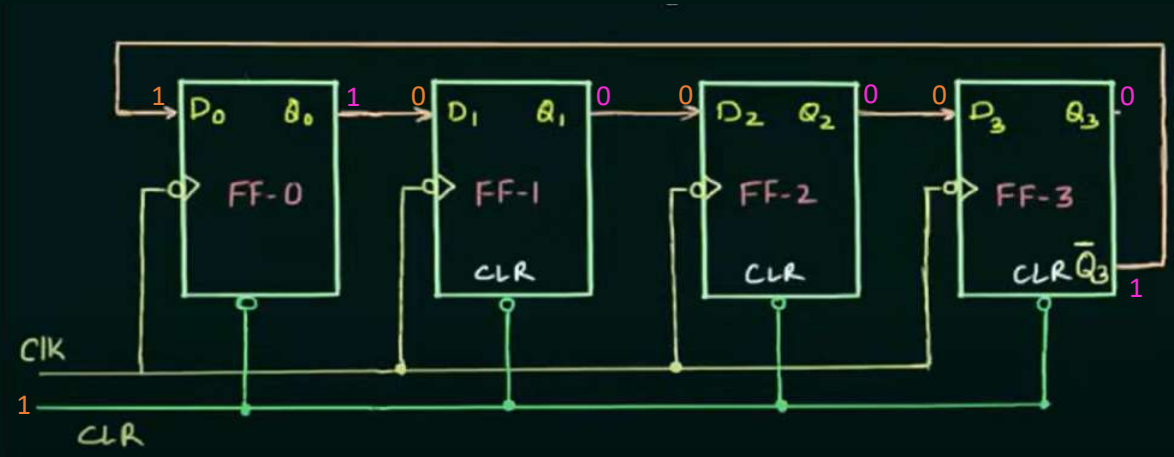
4-bit Johnson counter

Step 1: Assume that Initially CLR=0



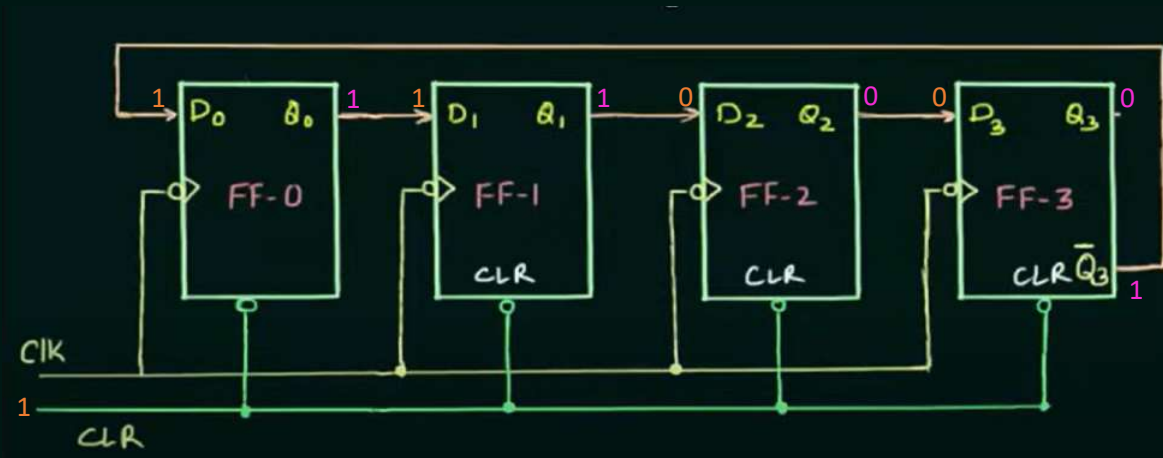
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓				
1	↓				
1	↓				
1	↓				

Step 1: Now for 1st falling edge & rest of the time that CLR=1:



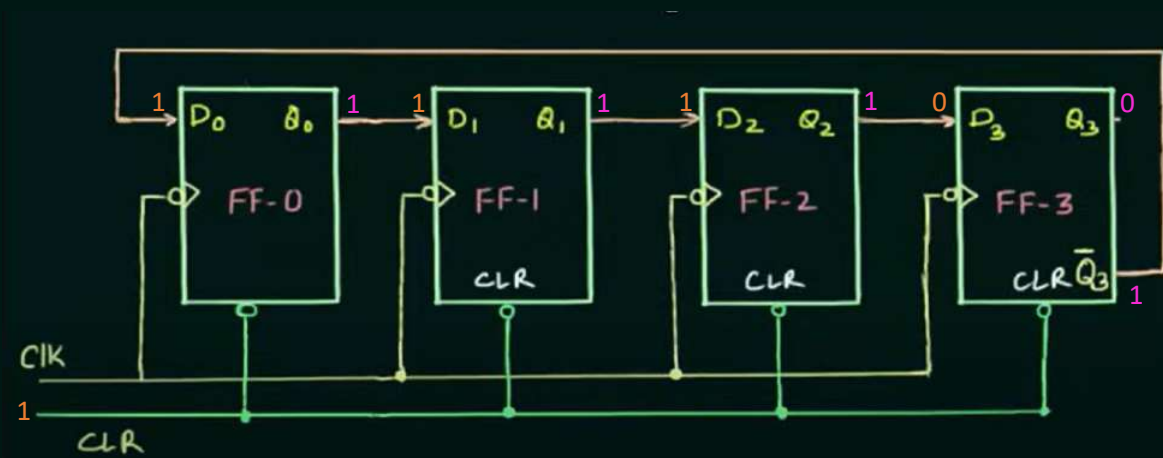
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓				
1	↓				
1	↓				

Step 2: Now for 2nd falling edge:



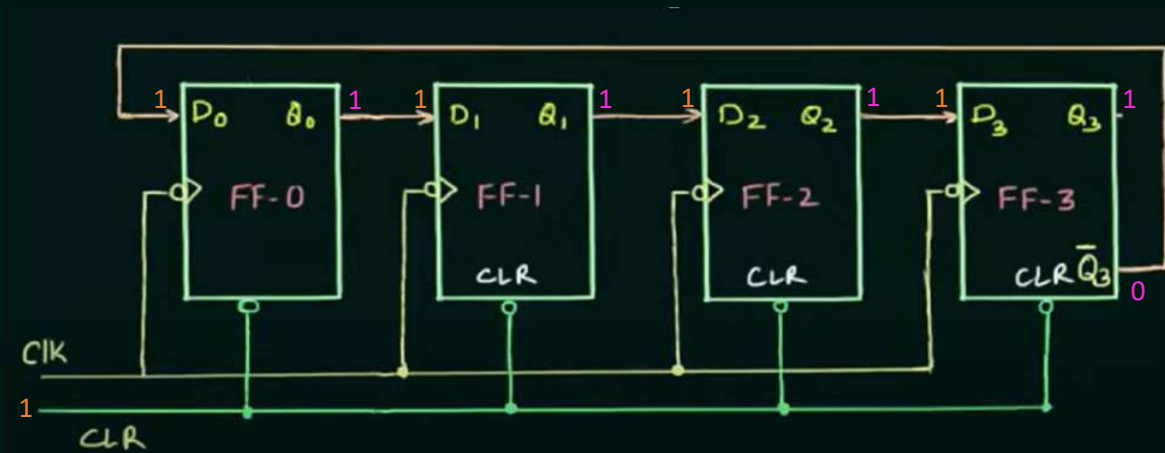
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓				
1	↓				

Step 3: Now for 3rd falling edge:



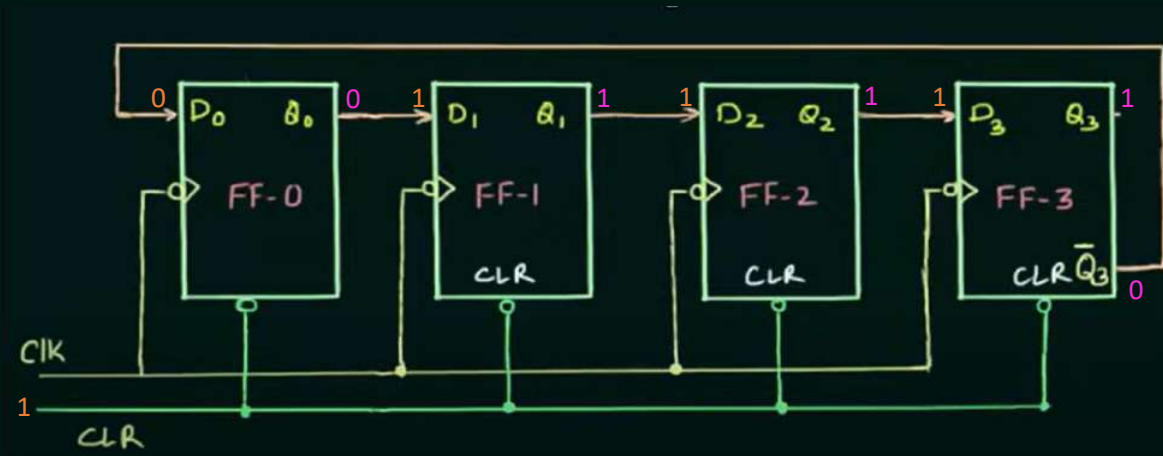
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓				

Step 4: Now for 4th falling edge:



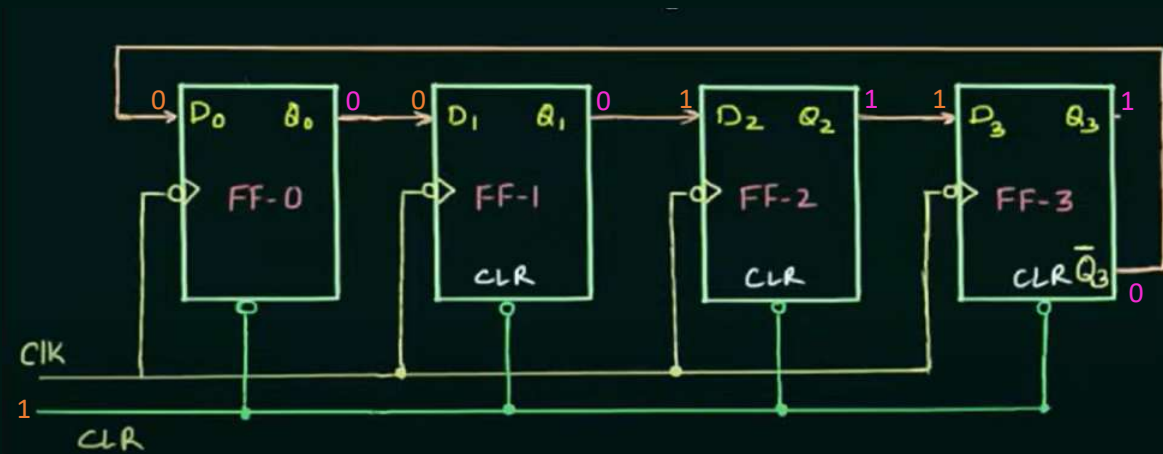
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1

Step 5: Now for 5th falling edge:



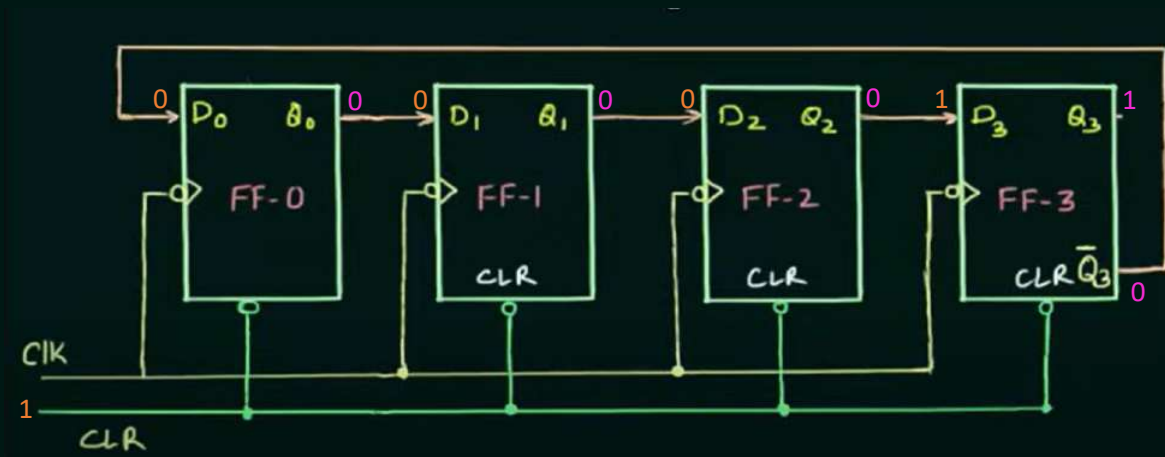
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1

Step 6: Now for 6th falling edge:



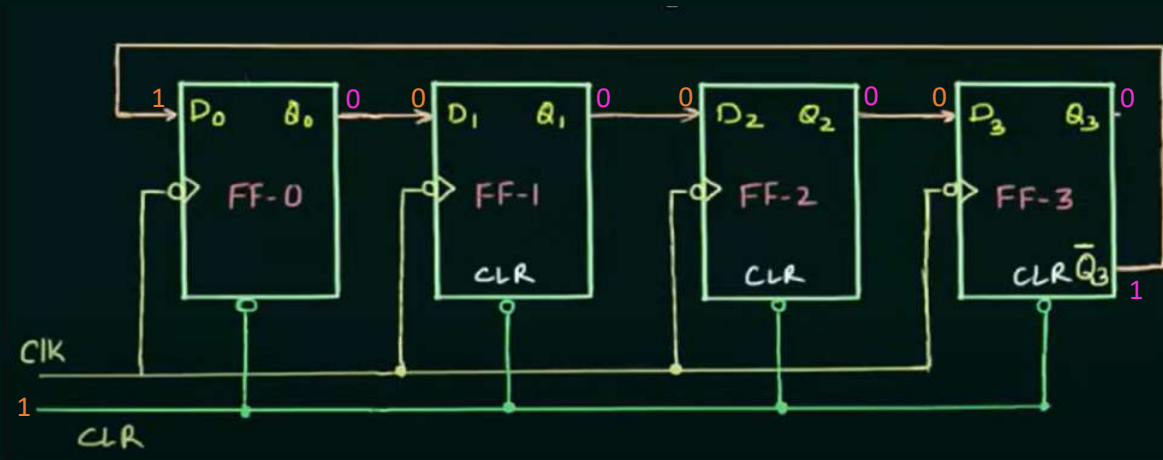
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1

Step 7: Now for 7th falling edge:



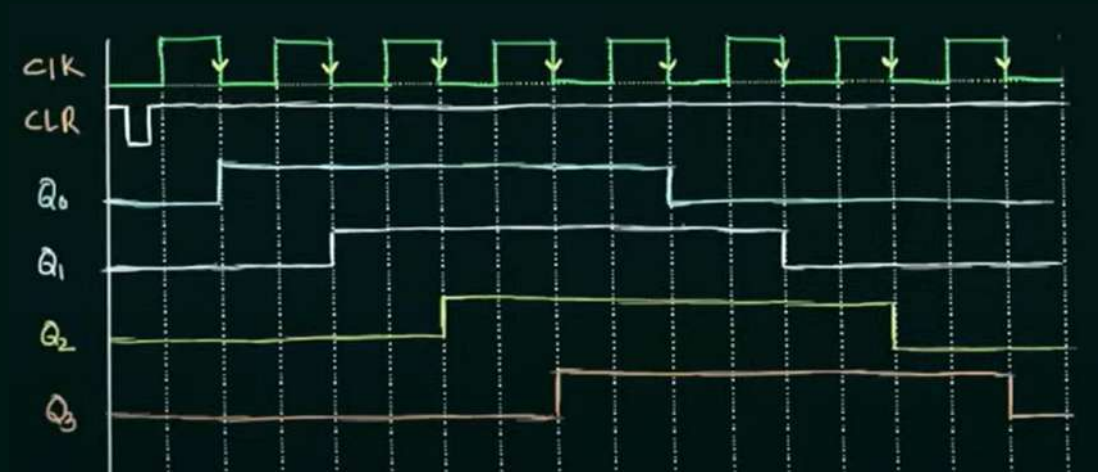
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	1

Step 8: Now for 8th falling edge:



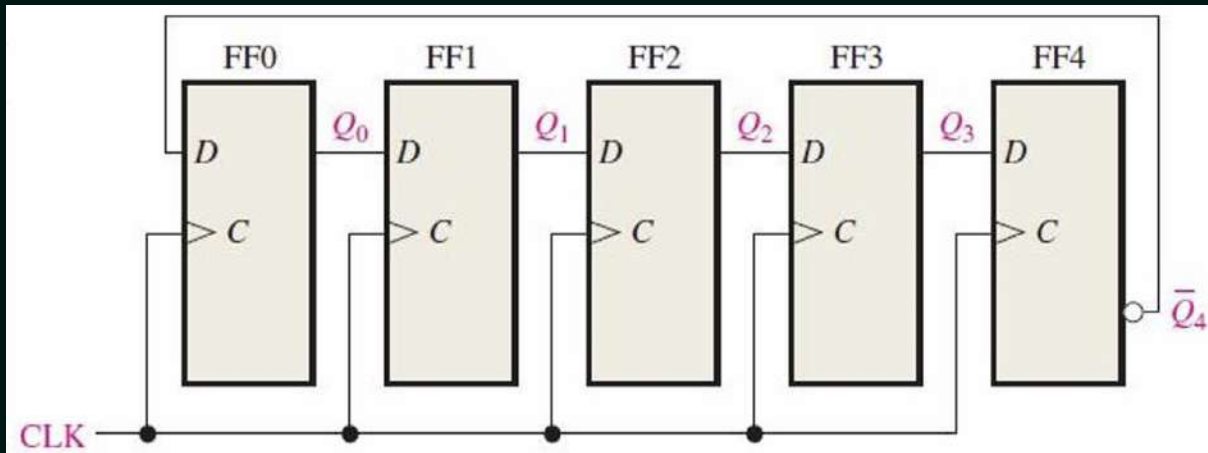
CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	0

Analyses the clock Diagram



5-bit Johnson counter

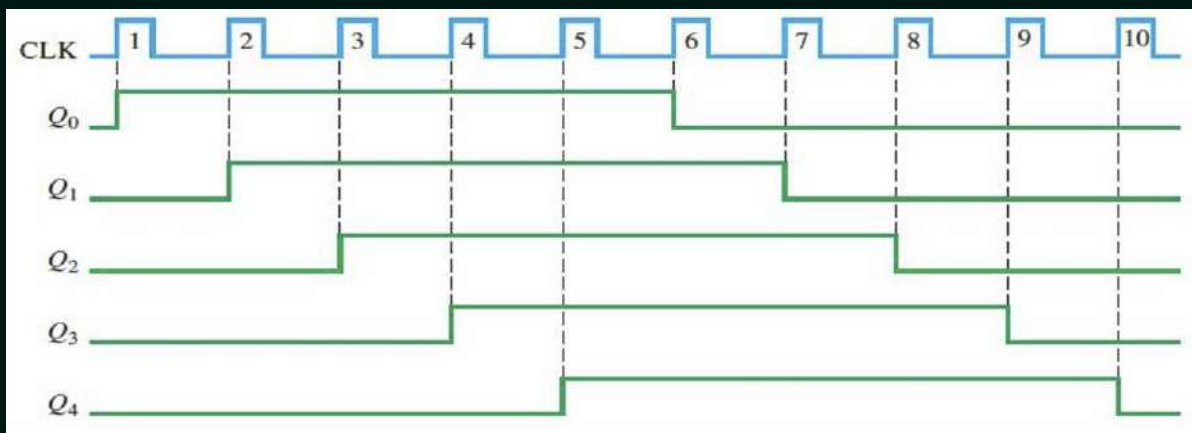
⇒ The circuit Diagram is:



⇒ The 5-bit Johnson_Counter Sequence is:

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4
0	0	0	0	0	0
1	1	0	0	0	0
2	1	1	0	0	0
3	1	1	1	0	0
4	1	1	1	1	0
5	1	1	1	1	1
6	0	1	1	1	1
7	0	0	1	1	1
8	0	0	0	1	1
9	0	0	0	0	1

Analyses the clock Diagram



DIGITAL SYSTEM DESIGN WITH FINITE STATE MACHINES (FSM)

Digital System Design with Finite State Machines (FSM)

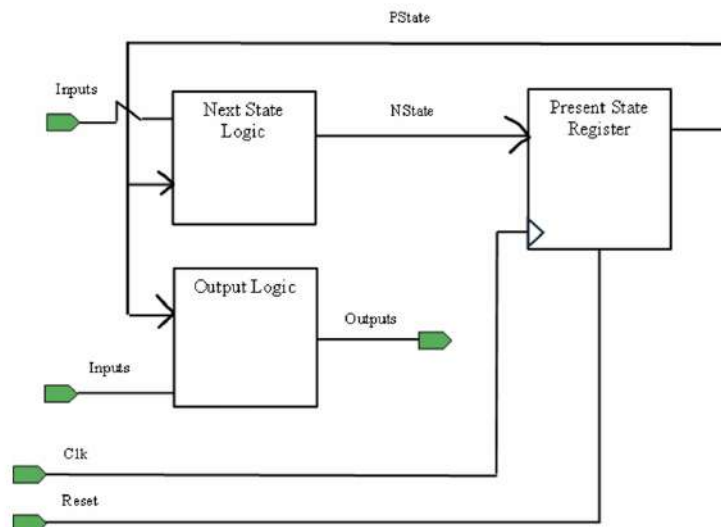
➔ **FSM/FSA:** Automata with a finite number of states; executes a predetermined sequence of operations.

- Self-regulating, automatic machine without external assistance.
- Computational model used in hardware/software for designing programs and circuits.
- Terminology:
 - "Finite State Automata (FSA)" in Mathematics/Computer Science.
 - "Finite State Machines (FSM)" in Electronics/Engineering.
- **Applications:**
 - Pattern recognition.
 - Modeling digital circuits.
 - Designing algorithms and programs.
 - Building compilers/interpreters.
 - Language models and machine learning.
- **Purpose:** Automate computational tasks.
- **FSM described by:**
 - Finite states.
 - Finite inputs.
 - State transitions.
 - Initial state.
 - Final state.
- In digital logic: FSMs design control circuits and represent sequential logic circuits.

FSM Architecture

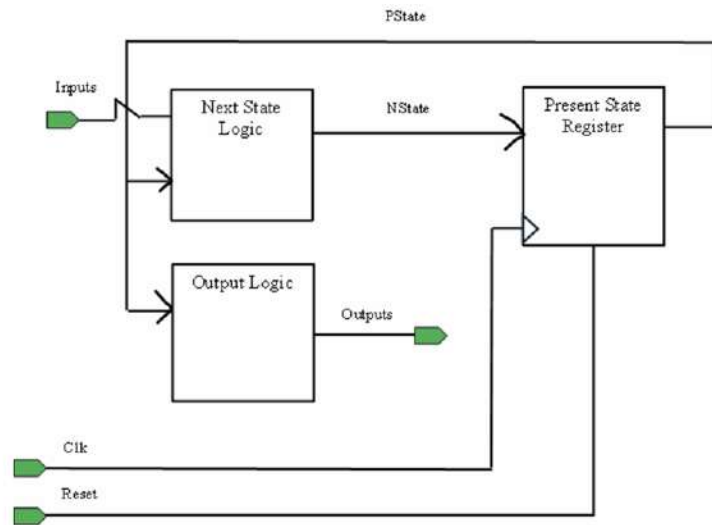
➔ **FSM Architecture Components:**

- **Present State Register:** Sequential logic.
- **Next State Logic:** Combinational logic.
- **Output Logic:** Combinational logic.



Moore vs Mealy FSMs

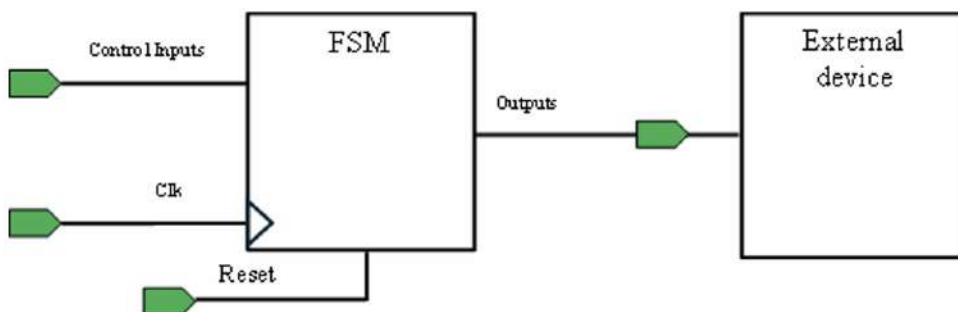
- **Moore FSMs:**
 - Outputs depend only on the present state.
 - Produce only Moore outputs.
 - Architecture focuses on state-based output generation.
- **Mealy FSMs:**
 - Produce one or more Mealy outputs.
 - Outputs depend on both the present state and inputs.
 - Architecture includes input-based output generation for faster response.



Control-dominated FSMs vs Data-dominated FSMs

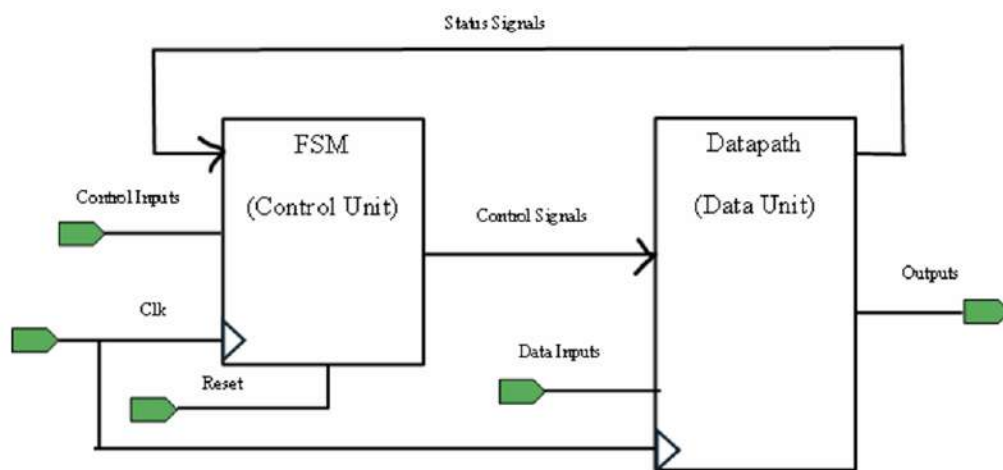
⇒ Control-Dominated Designs Using FSMs:

- Examples: Signal generators (e.g., traffic light controllers) and stepper motor controllers.
- FSMs manage and control the operation of external devices like LEDs or stepper motors.



⇒ Data-Dominated Designs Using FSMs:

- Examples: Microprocessors.
- FSMs control logical and arithmetic operations for computations.
- **Controlled Elements:**
 - Arithmetic and Logic Units (ALU).
 - Storage devices (registers).
 - Data steering circuits (multiplexers and demultiplexers).
- **Control Signals:**
 - ALU_Control signals, select signals, load/enable signals.
- FSM reads device output states as **status signals** for proper control.
- Computations occur in the **datapath** (data unit), while the FSM functions as the **control unit** to guide and sequence operations correctly.



✂ Example:

⇒ Design an FSM that:

- Starts in an idle state, generating an **output "Ready"** and evaluating the **input "Go."**
- If "Go" is low, it stays in the idle state.
- If "Go" is high, it generates three outputs (Red, Green, Yellow) consecutively for 3 seconds, 3 seconds, and 1 second, respectively.
- **The clock frequency is 1 Hz.**
- Use binary state encoding.

Show:

1. **FSM architecture**
2. **State chart**
3. **State table**
4. **Logic equations for next states and outputs.**

→Solution:

⇒ Is given:

$$f = 1\text{Hz}$$

$$\therefore T = \frac{1}{f} = 1\text{s} \quad [\text{that means each State is 1 sec long}]$$

FSM Design:

- **Objective:** The system needs to output "Ready," "Red," "Green," and "Yellow" in a sequence based on a 1 Hz clock.

States & Outputs:

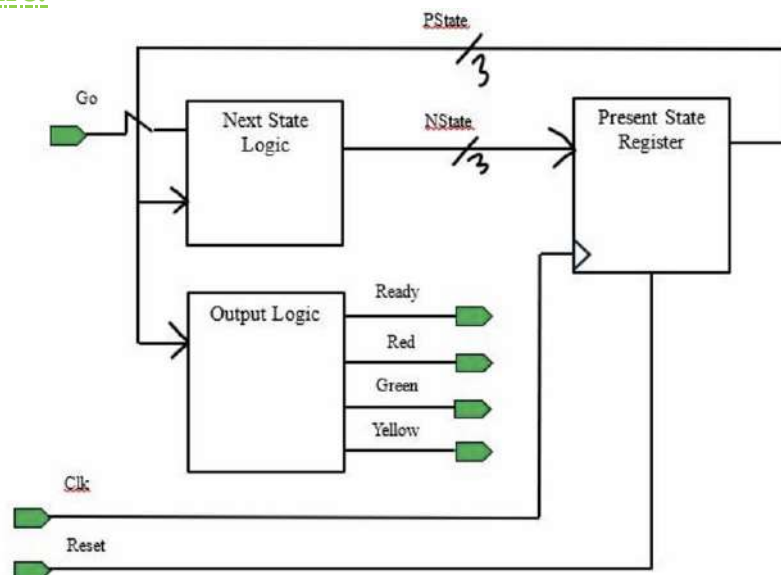
1. **Idle State (State 0):** Outputs "Ready" (only for 1 clock cycle).
2. **Red States (State 1, 2, 3):** Outputs "Red" for 3 clock cycles.
3. **Green States (State 4, 5, 6):** Outputs "Green" for 3 clock cycles.
4. **Yellow State (State 7):** Outputs "Yellow" for 1 clock cycle.

Total States:

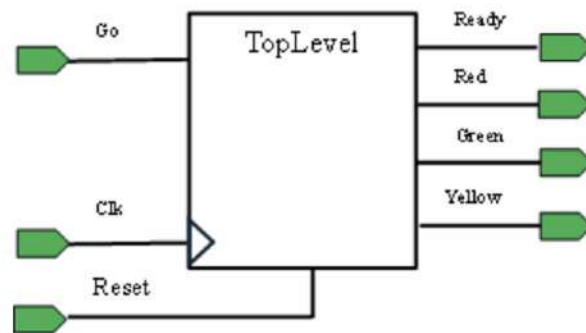
- The system **needs 8 states in total** (000 to 111 in binary), since **it has 8 distinct tasks**: 1 for Ready, 3 for Red, 3 for Green, and 1 for Yellow.

(a)

⇒ FSM architecture:

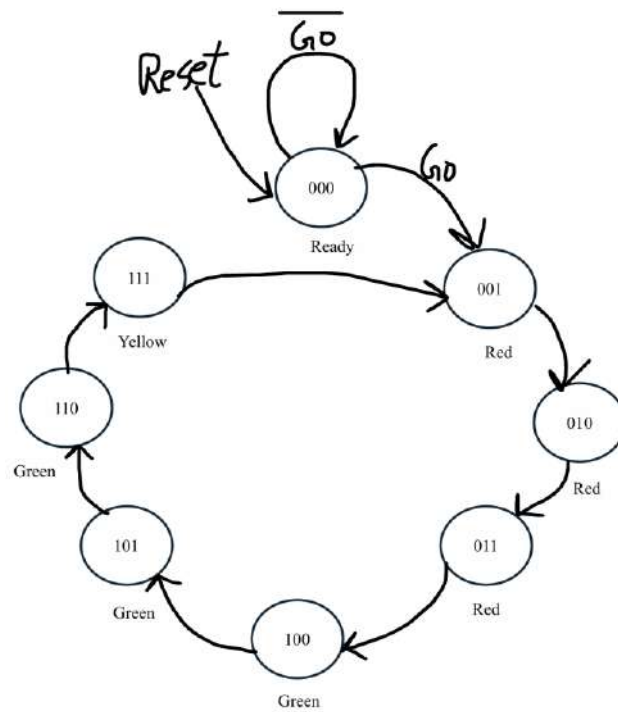


FSM Block Diagram:



(b)

⇒ State chart/State Diagram:



(c)

⇒ State Table:

Reset	Input	PState			NState			Outputs			
	Go	PS ₂	PS ₁	PS ₀	NS ₂	NS ₁	NS ₀	Ready	Red	Green	Yellow
1	X	X	X	X	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0
	1				0	0	1				
0	X	0	0	1	0	1	0	0	1	0	0
0	X	0	1	0	0	1	1	0	1	0	0
0	X	0	1	1	1	0	0	0	1	0	0
0	X	1	0	0	1	0	1	0	0	1	0
0	X	1	0	1	1	1	0	0	0	1	0
0	X	1	1	0	1	1	1	0	0	1	0
0	X	1	1	1	0	0	1	0	0	0	1

(d)

⇒ Logic equations for next states and outputs:

→ Let, NState = NS

PState = PS

∴ Equation for NState:

$$\rightarrow NS_0 = \overline{Reset}.(\overline{PS_2}.\overline{PS_1}.\overline{PS_0}.GO + \overline{PS_2}.PS_1.\overline{PS_0} + PS_2.\overline{PS_1}.\overline{PS_0} + PS_2.PS_1.\overline{PS_0} + PS_2.PS_1.PS_0)$$

$$\rightarrow NS_1 = \overline{Reset}.(\overline{PS_2}.\overline{PS_1}.PS_0 + \overline{PS_2}.PS_1.\overline{PS_0} + PS_2.\overline{PS_1}.PS_0 + PS_2.PS_1.\overline{PS_0})$$

$$\rightarrow NS_1 = \overline{Reset}.(\overline{PS_2}.PS_1.PS_0 + PS_2.\overline{PS_1}.\overline{PS_0} + PS_2.\overline{PS_1}.PS_0 + PS_2.PS_1.\overline{PS_0})$$

∴ Equation for Outputs:

$$\rightarrow Ready = \overline{PS_2}.\overline{PS_1}.\overline{PS_0}$$

$$\rightarrow RED = (\overline{PS_2}.\overline{PS_1}.PS_1 + \overline{PS_2}.PS_1.\overline{PS_0} + \overline{PS_2}.PS_1.PS_0)$$

$$\rightarrow Green = (PS_2.\overline{PS_1}.\overline{PS_0} + PS_2.\overline{PS_1}.PS_0 + PS_2.PS_1.\overline{PS_0})$$

$$\rightarrow Yellow = (PS_2.PS_1.PS_0)$$

Sub: _____

Time: _____

Date: / /

#hth
Timeo
He

555 Timer IC

⇒ It can be used as a Timer & Counter.

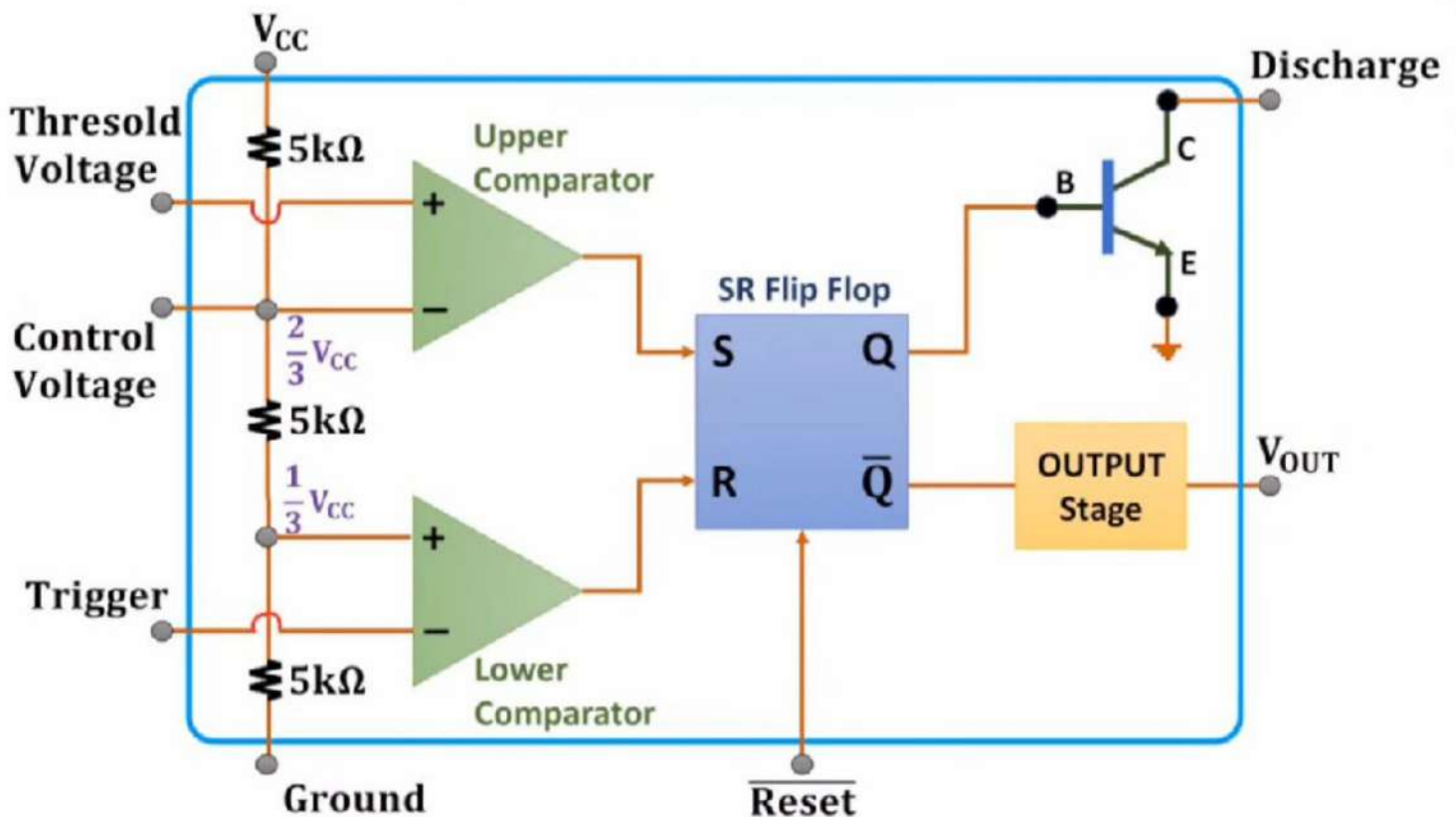
There are Three modes of operation:—

1' Astable Mode.

2' Monostable Mode.

3' Bistable Mode.

Block Diagram:—

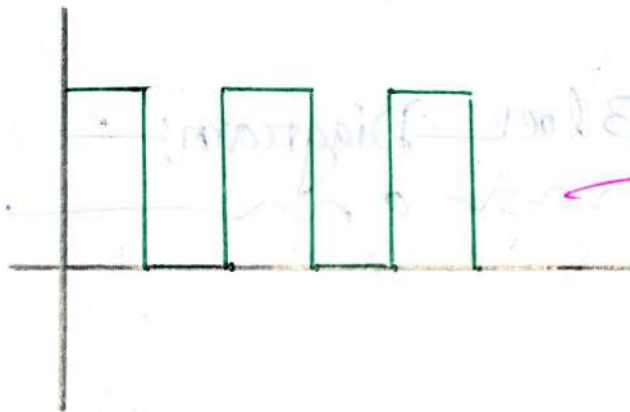


Astable Multivibrator

⇒ Astable Multivibrator has no stable state.

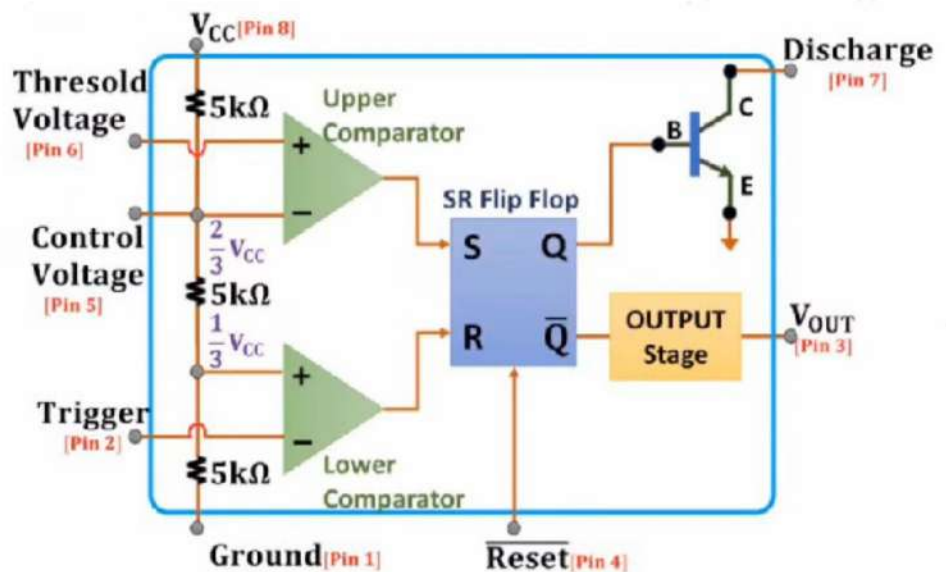
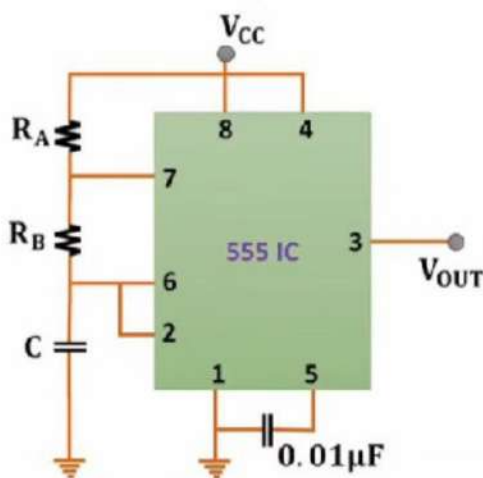
Both states are Quasi-Stable state. It can be used as a square wave generator.

Astable Multivibrator $\rightarrow V_{out}$

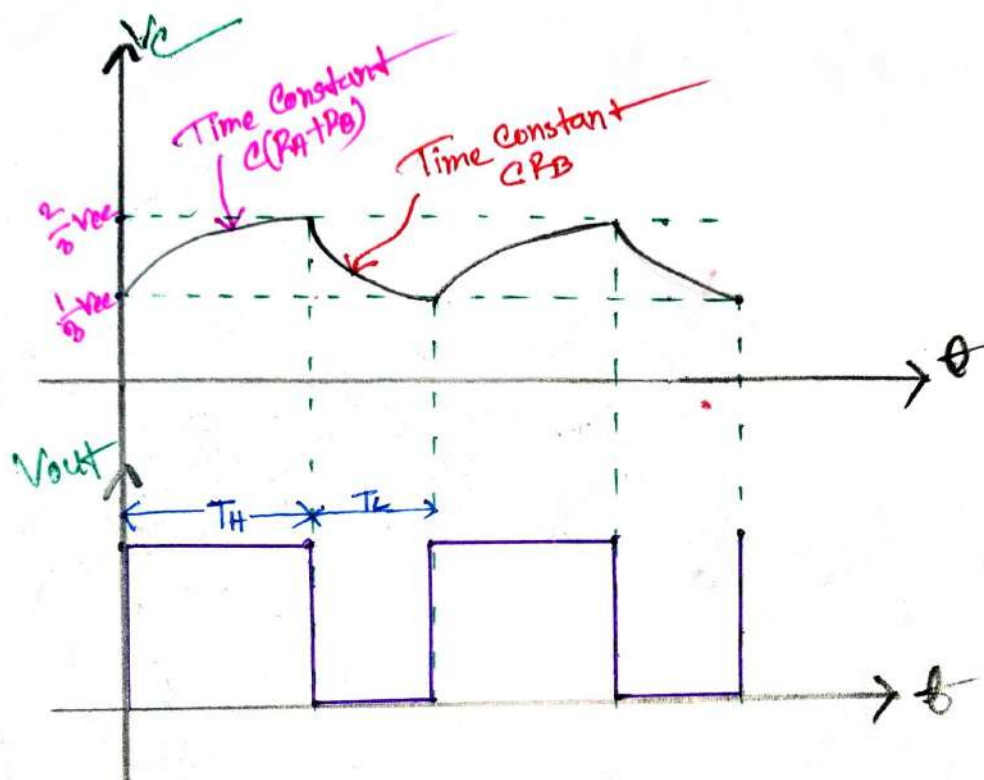


Quasi Stable State

Block Diagram



Waveforms of Astable Multivibrator



$$\therefore V_c = V_F + (V_I - V_F) e^{-t/R}$$

During Charging :-

$$t = T_H, R = R_A + R_B$$

$$\therefore \frac{2}{3} V_{cc} = V_{cc} + \left(\frac{1}{3} V_{cc} - V_{cc} \right) e^{-t/C(R_A+R_B)}$$

$$\Rightarrow -\frac{1}{3} = -\frac{2}{3} e^{T_H/C(R_A+R_B)}$$

$$\Rightarrow \frac{1}{2} = e^{-T_H/C(R_A+R_B)}$$

$$\therefore T_H = 0.693 C(R_A+R_B)$$

During Discharging:-

$$t = T_L, R = R_0$$

$$\therefore \Rightarrow \frac{1}{3} V_{ee} = 0 + \left(\frac{2}{3} V_{ee} - 0 \right) e^{-T_L / C R_0}$$

$$\Rightarrow \frac{1}{2} = e^{-T_L / C R_0}$$

$$\Rightarrow T_L = 0.693 C R_0$$

If the frequency is given:-

We know:-

$$T = T_H + T_L$$

$$\therefore F = \frac{1}{T}$$

$$F = \frac{1}{T_L + T_H}$$

$$= \frac{1}{0.693 C R_0 + 0.693 C (R_0 + R_A)}$$

$$\therefore F = \frac{1}{0.693 C (R_A + 2 R_0)}$$

Sub: _____

Day _____

Time: _____

Date: / /

For Duty Cycle:-

$$D.C = \frac{T_H}{T}$$

$$\therefore D.C = \frac{T_H}{T_H + T_L}$$

problem: 1:-

⇒ Design a oscillator for a frequency of 200 Hz with duty cycle of 78%. Determine

(a) Time period.

(b) High & Low time.

(c) R_2 & R_1 where $C_1 = C = 10 \mu F$

⇒ Is given:-

$$f = 200 \text{ Hz}$$

$$D.C = 78\% = 0.78$$

we know:-

(a)

$$T = \frac{1}{f} = \frac{1}{200} = 5 \times 10^{-3} \text{ sec}$$

Sub: _____

Day _____

Time: _____

Date: / /

(b)⇒ We know:—

$$D \cdot e = \frac{T_H}{T}$$

$$\therefore T_H = D \cdot e \times T$$

$$= 0.78 \times 5 \times 10^{-3}$$

$$= 3.9 \text{ ms}$$

$$T = T_H + T_L$$

$$\therefore T_L = T - T_H$$

$$= 5 \text{ ms} - 3.9 \text{ ms}$$

$$= 1.1 \text{ ms}$$

Ans:—(c)⇒ We know:—

$$T_H = 0.693 C (R_1 + R_2)$$

$$\therefore R_1 = (T_H / 0.693 C) - R_2 \quad \text{--- (1)}$$

$$\& \quad T_L = 0.693 C R_2$$

$$\therefore R_2 = T_L / 0.693 C$$

$$= \frac{1.1 \text{ ms}}{0.693 \times 10 \mu\text{F}}$$

$$= 158.73 \Omega$$

$$\left. \begin{array}{l} T_L = 1.1 \text{ ms} \\ C = 10 \mu\text{F} \end{array} \right\}$$

Sub: _____

Day _____

Time: _____

Date: / /

Now, put the value of R_2 in eq (1):-

$$\therefore R_1 = \frac{T_H}{0.693C} - R_2$$

$$= \frac{3.7ms}{0.693 \times 10\mu} - 158.73$$

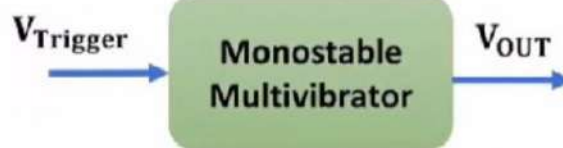
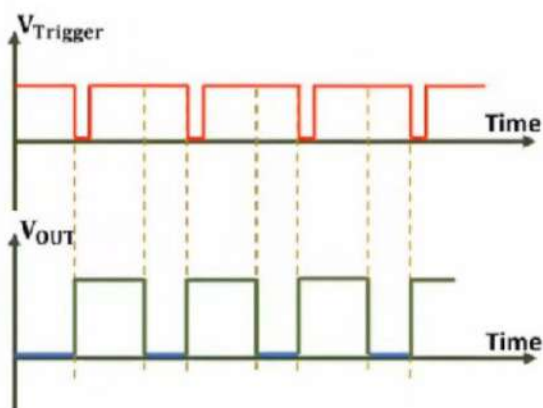
$$= 404.04 \Omega$$

$$\left. \begin{array}{l} T_H = 3.7ms \\ C = 10\mu F \\ R_2 = 158.73 \end{array} \right\}$$

Ans:-

Monostable Multivibrator:-

- Monostable has one stable state and the second state is not stable, it is Quasi stable State.
- Stable State: This state doesn't change. Stable State changes by external trigger.
- Quasi-stable State: This state changes based on the circuit. It changes automatically.



— Stable State
— Quasi Stable State

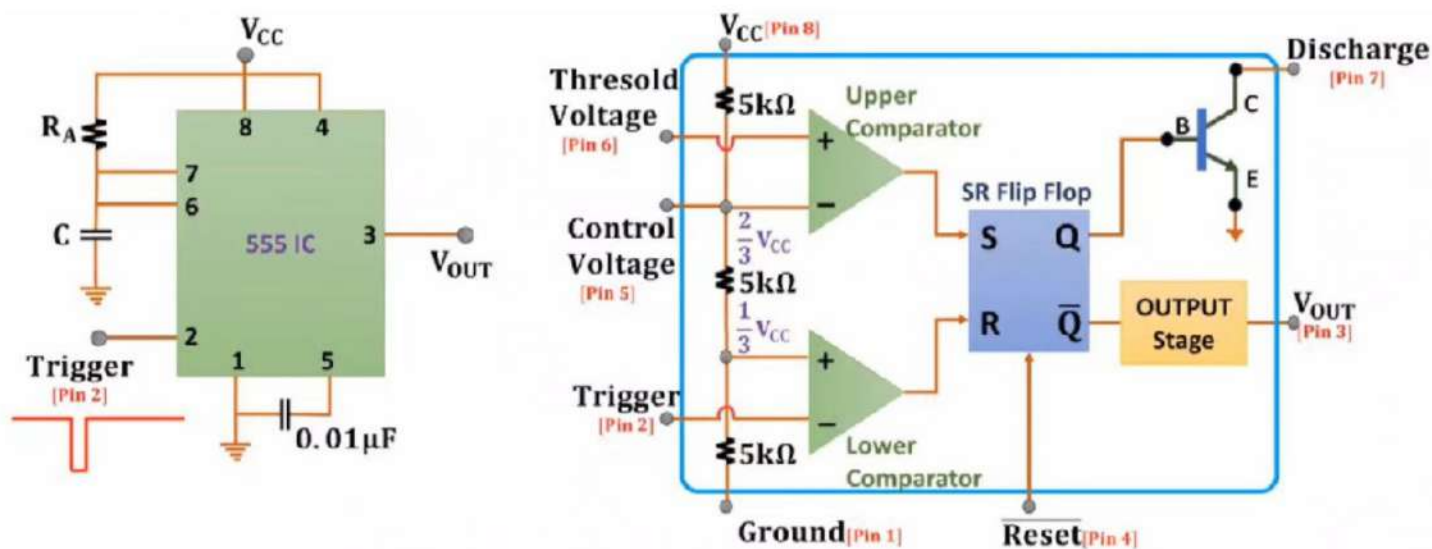
Sub : _____

Day _____

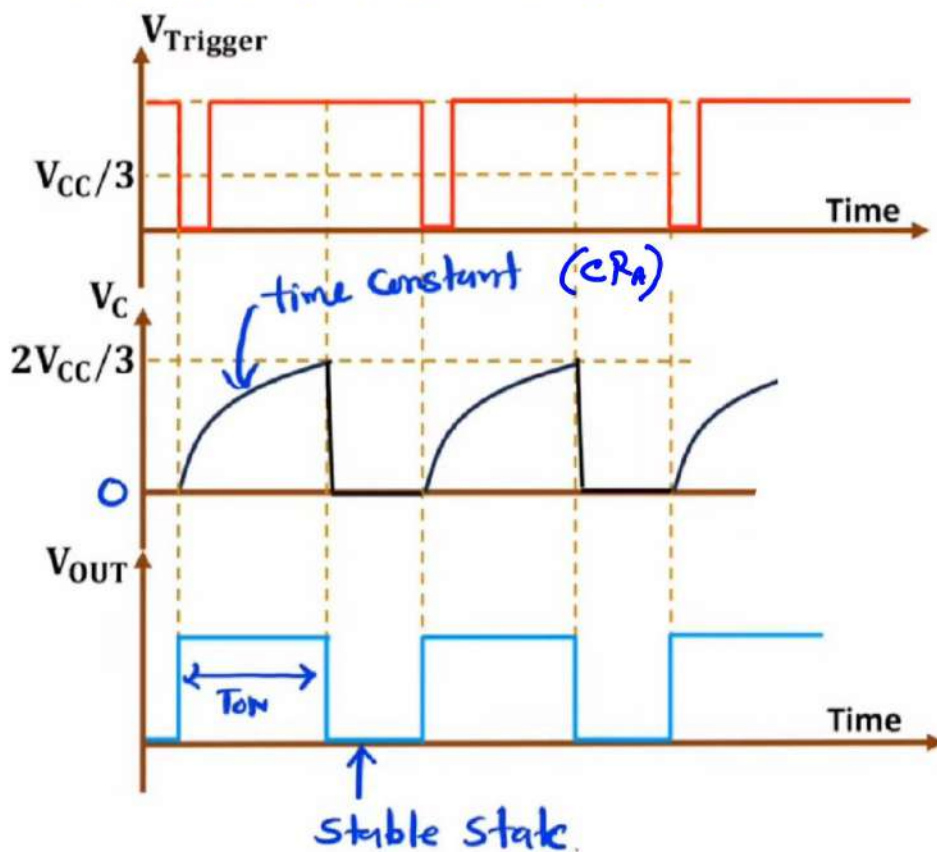
Time : _____

Date : / /

Block Diagram



Waveforms of Monostable Multivibrator



$$\therefore \Rightarrow V_C = V_F + (V_I - V_F) e^{-t/R}$$

During Charging:-

$$\Rightarrow t = t_H, R = R_A$$

$$\therefore \Rightarrow \frac{2}{3} V_{CC} = V_{CC} + (0 - V_{CC}) e^{-t_H/R_A}$$

$$\therefore \Rightarrow -\frac{1}{3} V_{CC} = -V_{CC} \cdot e^{-t_H/R_A}$$

$$\Rightarrow 1.1 = t_H / R_A$$

$$\therefore \boxed{t_H = 1.1 R_A C}$$

Problem 2:-

$\Rightarrow$ if the frequency of 400 Hz with 70% duty cycle. Determine

- Time period.
- High time if $C = 20 \mu F$
- Value of R_A

$\Rightarrow$ Is given:-

$$D.C = 70\% = 0.7$$

$$f = 200 + 200 = 400 \text{ Hz}$$

$$C = 20 \mu F$$

Sub: _____

Day _____

Time: _____

Date: / /

(a)⇒ We know:-

$$T = \frac{1}{f} = \frac{1}{400} = 2.5 \text{ ms}$$

Ans:-(b)⇒ We know:-

$$D \cdot e = \frac{T_H}{T}$$

$$\begin{aligned} \therefore T_H &= D \cdot e \times T \\ &= 0.7 \times 2.5 \text{ ms} \\ &= 1.75 \text{ ms} \end{aligned}$$

(c)⇒ We know:-

$$T_H = 1.1 C R_A$$

$$\therefore R_A = \frac{T_H}{1.1 \times 20 \mu F}$$

$$= \frac{1.75 \text{ ms}}{1.1 \times 20 \mu}$$

$$= 79.54 \Omega$$

Sub :

Time :

Date :

Digital to Analog Converter (DAC)

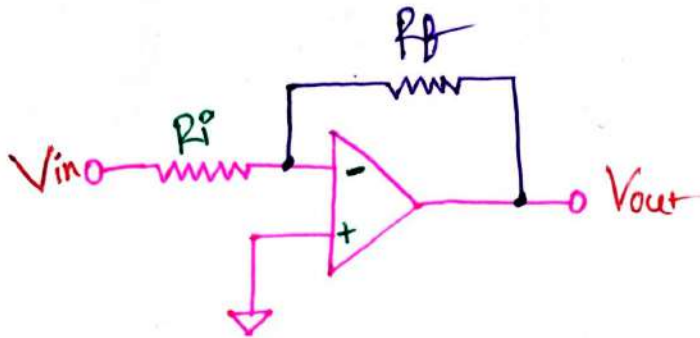
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Day _____

Time: _____

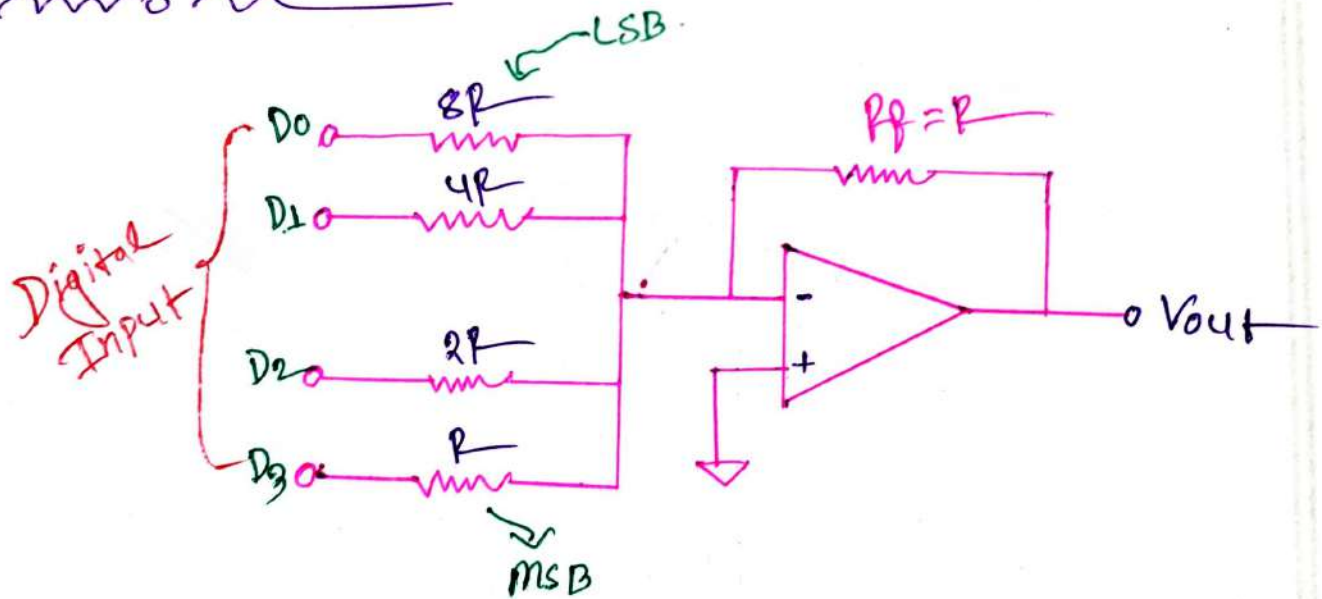
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Binary Weighted DAC



$$\therefore V_{out} = -V_{in} \left(\frac{R_f}{R_i} \right)$$

Example, 1:-



$$\therefore V_{out} = - \left(\frac{R}{8R} V_0 + \frac{R}{4R} V_1 + \frac{R}{2R} V_2 + \frac{R}{R} V_3 \right)$$

$$\Rightarrow V_o = - \left(\frac{1}{8} V_0 + \frac{1}{4} V_1 + \frac{1}{2} V_2 + \frac{1}{1} V_3 \right)$$

$$\Rightarrow V_o = - \left(\frac{V_0}{8} + \frac{V_1}{4} + \frac{V_2}{2} + \frac{V_3}{1} \right)$$

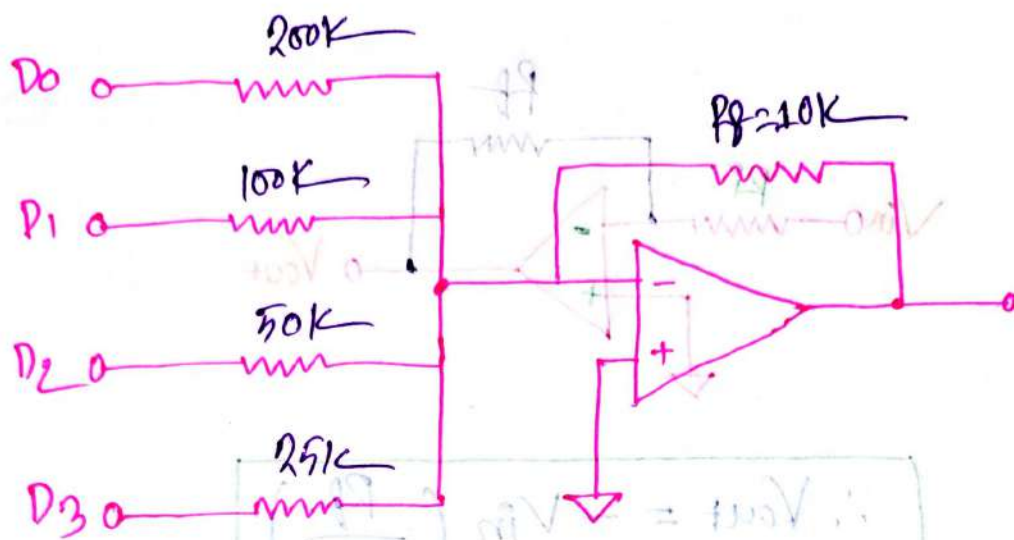
Sub: _____

Day _____

Time: _____

Date: / /

Example 1: —



⇒ Determine the output of the DAC if waveform representing a sequence of 4-bit numbers are applied to the inputs. D_0 is the LSB.

⇒ Since 4-bit,

There are 16 combinations.

For: 0001: —

We know: —

$$V_{out} = - \left(\frac{V_{0R}}{R_0} + \frac{V_{1R}}{R_1} + \frac{V_{2R}}{R_2} + \frac{V_{3R}}{R_3} \right) \quad \text{--- (1)}$$

In given: —

$$R_f = 10K \quad \left(\begin{array}{l} V_0 = 5V \\ V_1 = 0 \\ V_2 = 0 \\ V_3 = 0 \end{array} \right) \rightarrow \left(\because 0001 \right)$$

$$\left(\begin{array}{l} R_0 = 200K \\ R_1 = 100K \\ R_2 = 50K \\ R_3 = 25K \end{array} \right)$$

Sub: _____

Day _____

Time: _____

Date: / /

put the values in eq (1) $\Rightarrow$

$$\therefore V_{out} = - \left(\frac{5V \times 10k}{250k} + 0 + 0 + 0 \right) = -0.25V$$

Now, for: 0010:-

$$V_0 = 0, V_1 = 5V, V_2 = 0V, V_3 = 0V$$

$$\begin{aligned} \therefore V_{out} &= - \left(\frac{V_1 \times R_B}{R_1} + 0 + 0 + 0 \right) = \\ &= - \left(\frac{5 \times 10k}{100k} + 0 + 0 + 0 \right) = -0.5V \end{aligned}$$

Now, for: 0100:-

$$V_0 = 0, V_1 = 0V, V_2 = 5V, V_3 = 0V$$

$$\therefore V_{out} = - \left(\frac{5 \times 10k}{50k} + 0 + 0 + 0 \right) = -1V$$

Now, for: 1000:-

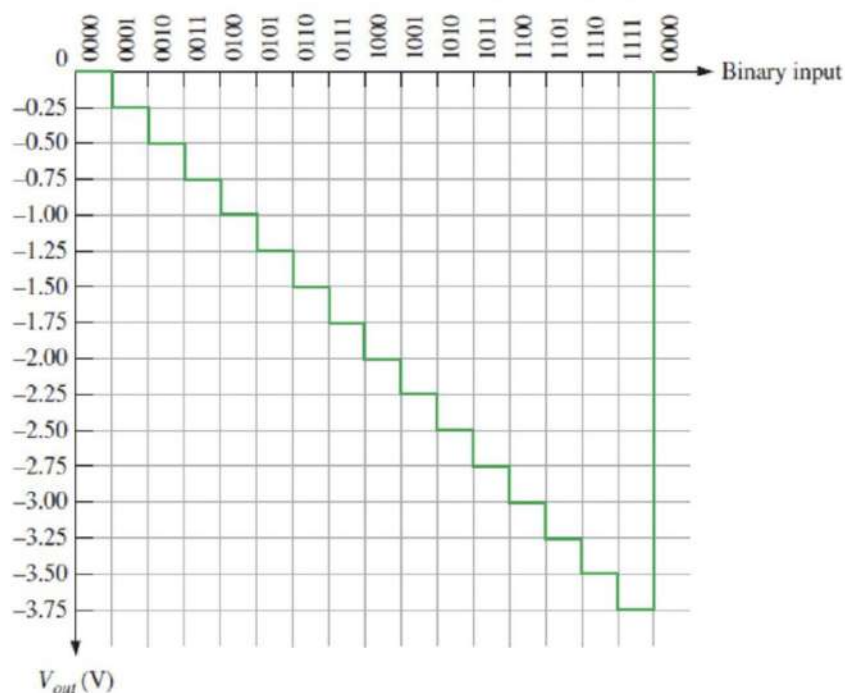
$$V_0 = 0, V_1 = 0, V_2 = 0, V_3 = 0.5V$$

$$\therefore V_{out} = - \left(\frac{5 \times 10k}{25k} + 0 + 0 + 0 \right) = -2V$$

So, Now you know the game,
Rest of the combinations are similar.

#Data Table:—

D ₃	D ₂	D ₁	D ₀	V _{out} (V)
0	0	0	0	0
0	0	0	1	-0.25
0	0	1	0	-0.50
0	0	1	1	-0.75
0	1	0	0	-1.00
0	1	0	1	-1.25
0	1	1	0	-1.50
0	1	1	1	-1.75
1	0	0	0	-2.00
1	0	0	1	-2.25
1	0	1	0	-2.50
1	0	1	1	-2.75
1	1	0	0	-3.00
1	1	0	1	-3.25
1	1	1	0	-3.50
1	1	1	1	-3.75



Sub :

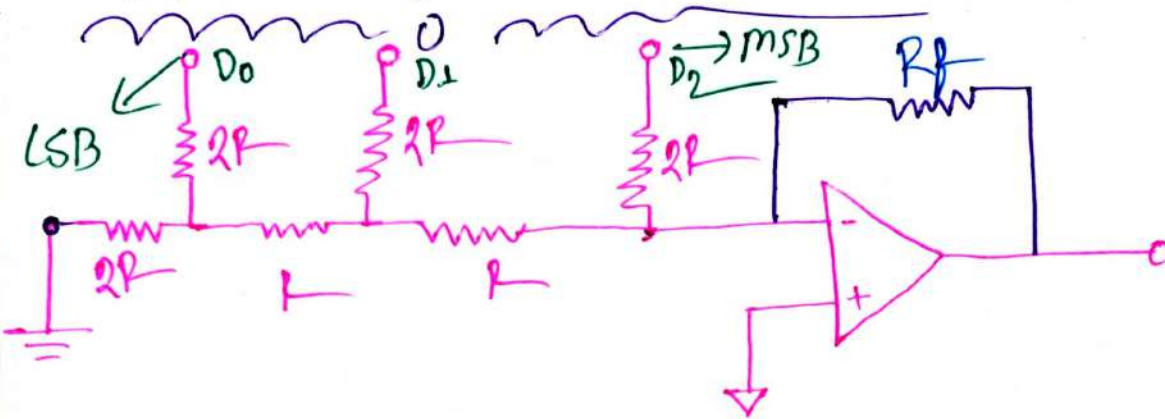
Day

Time :

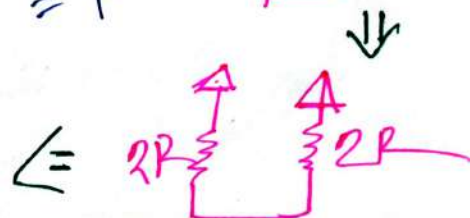
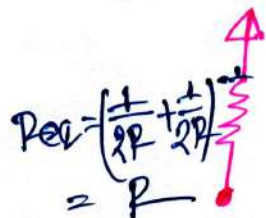
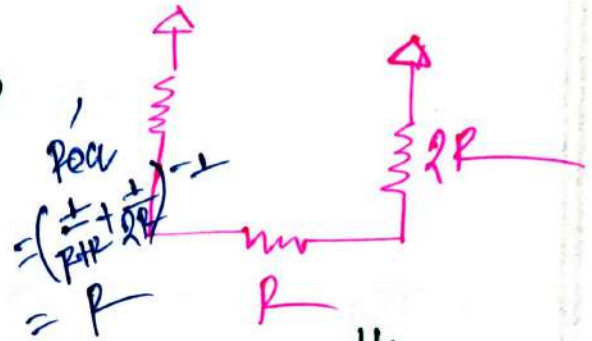
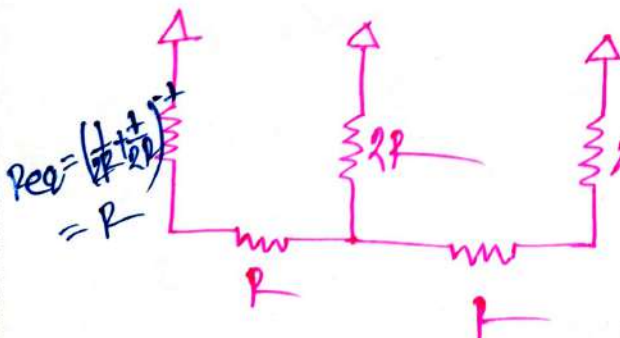
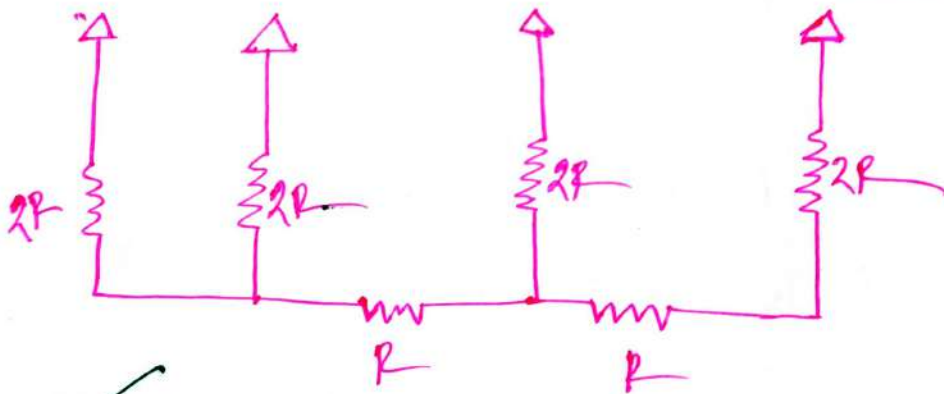
Date : / /

R-2R Ladder DAC

3 bit R-2R Ladder:



Find the impedance of network



Sub: _____

Day _____

Time: _____

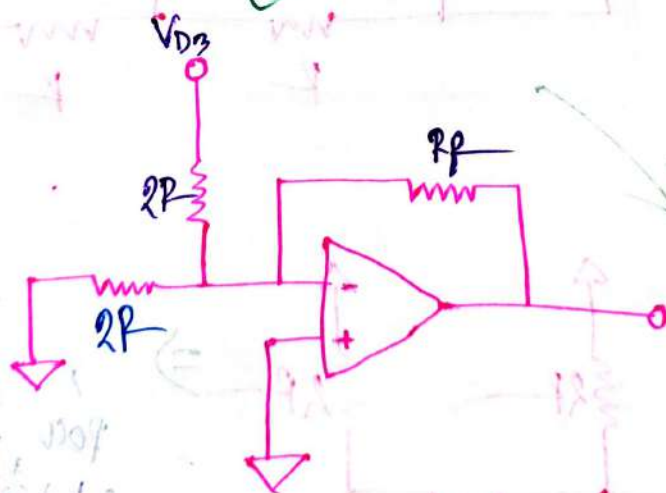
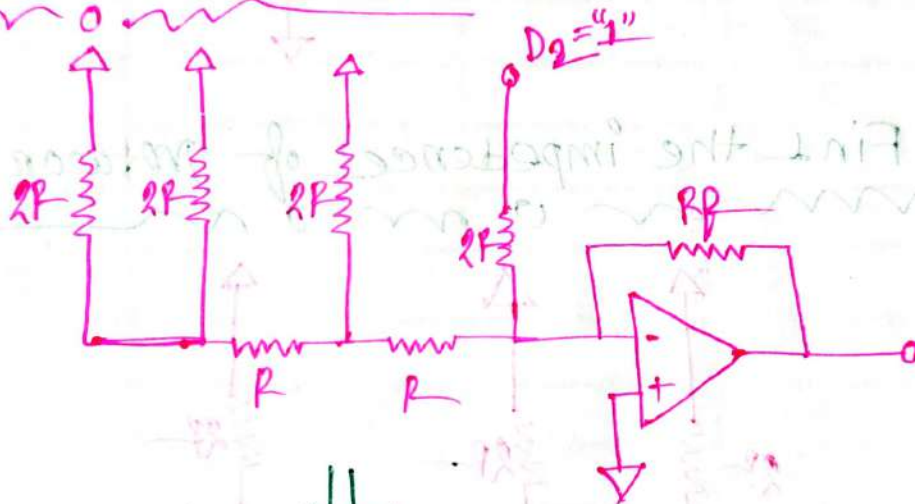
Date: ____/____/____

Note - the impedance of network is R , regardless of number of bits.

Since 3 bit, so there are 8 combinations! —

$\Rightarrow$ For: $D_2=1, D_1=D_0=0$

$\Rightarrow$ Redraw the circuit: —



$$\therefore R_{Th} = \left(\frac{1}{2R} + \frac{1}{2R} \right)^{-1} = R$$

$$\therefore V_{Th} = V_{02} \times \left(\frac{2R}{2R+2R} \right) = \frac{V_{02}}{2}$$

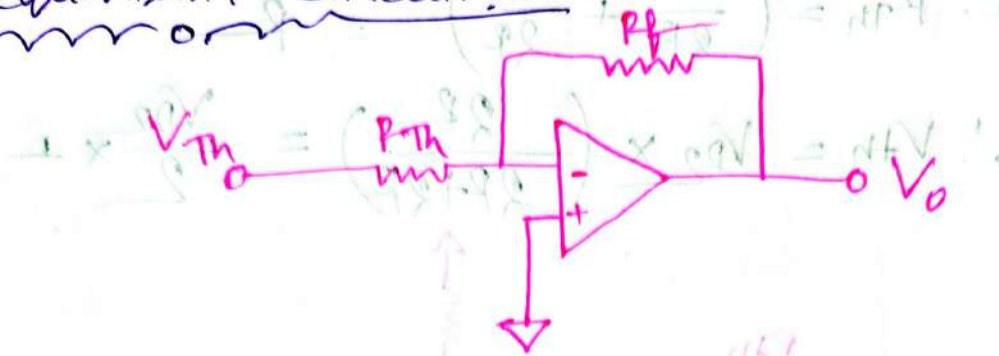
Sub: _____

Day _____

Time: _____

Date: / /

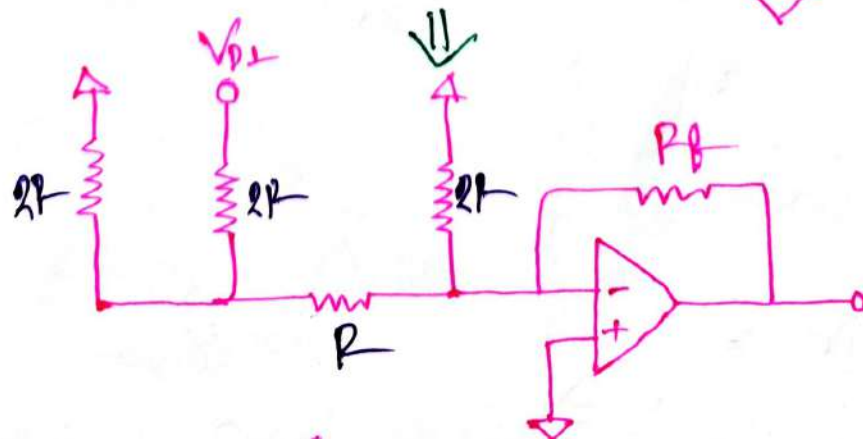
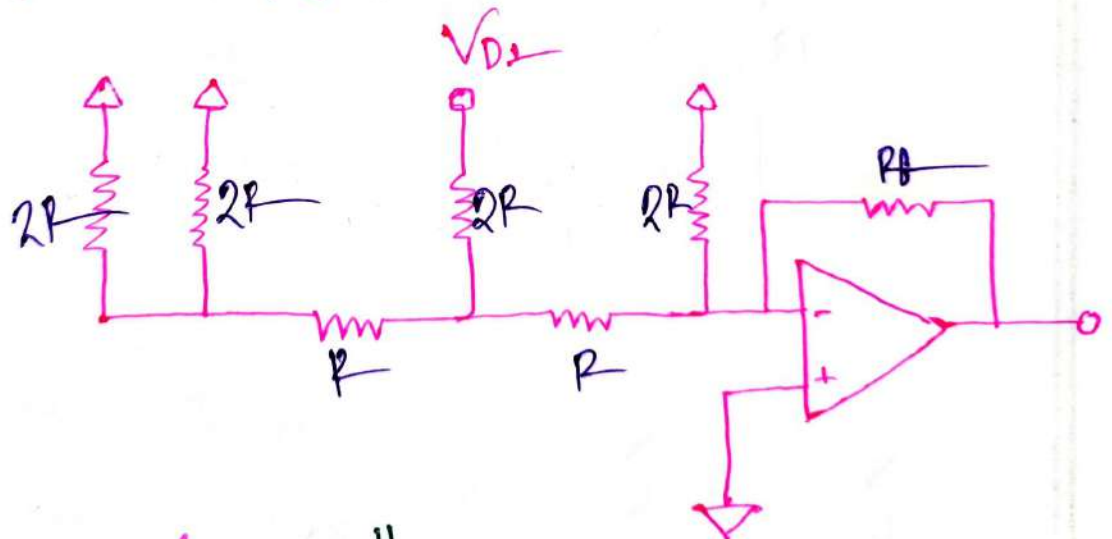
$\therefore$ equivalent circuit:-



$$\therefore V_o = - \left(\frac{V_{th} \times R_f}{R_{th}} \right) = - \left(\frac{V_{D2}}{2} \times \frac{R_f}{R} \right)$$

$\Rightarrow$ For $D_1 = 1$, $D_2 = D_0 = 0$:-

$\Rightarrow$ Redraw the circuit:-



Sub: _____

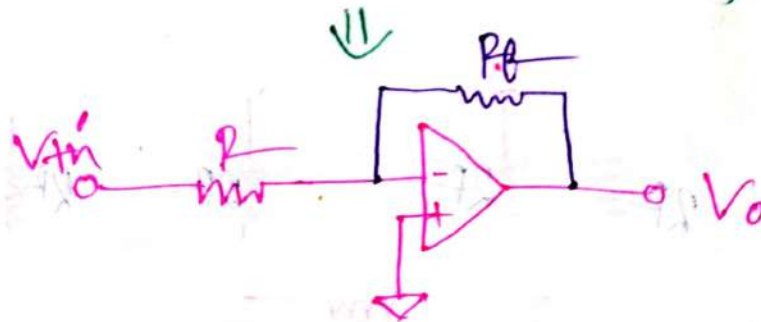
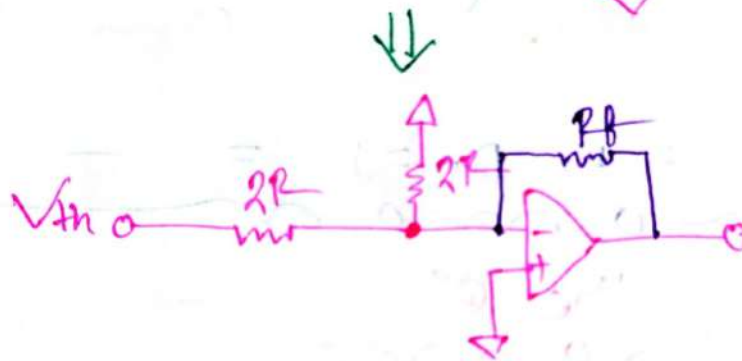
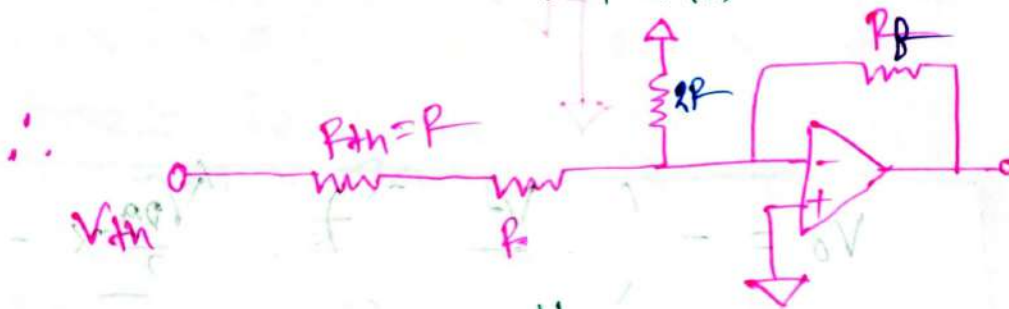
Day _____

Time: _____

Date: / /

$$\therefore R_{th} = \left(\frac{1}{2R} + \frac{1}{2R} \right)^{-1} = R$$

$$\therefore V_{th} = V_{D2} \times \left(\frac{2R}{2R+2R} \right) = \frac{V_{D2}}{2} \times 1$$



$$\therefore V_{th}' = \frac{\left(+\frac{V_{D1}}{2} \right) \times 2R}{2R} = \frac{V_{D1}}{4}$$

$$R_{th2} = \left(\frac{1}{2R} + \frac{1}{2R} \right)^{-1} = R$$

$$\therefore V_o = - \left(\frac{V_{th} \times R_F}{R} \right) = - \left(\frac{V_{D2}}{4} \times \left(\frac{R_F}{R} \right) \right)$$

$\therefore$ For $D_0 = 1$, $D_1 = D_2 = 0V$:-

$$V_o = - \left(\frac{V_{D0}}{8} \times \frac{R_F}{R} \right)$$

Sub: _____

Day _____

Time: _____

Date: / /

∴ The generalised eq:—
info mon

$$V_0 = -\left(\frac{R_f}{R}\right) \times V \left(\frac{D_2}{2} + \frac{D_1}{4} + \frac{D_0}{8} \right)$$

$$\therefore V_0 = -V \left(\frac{D_2}{2} + \frac{D_1}{4} + \frac{D_0}{8} \right) \quad [\because R_f = R]$$

∴ Digital data to Analog
mon

D ₂	D ₁	D ₀	Analog V ₀
0	0	0	0
0	0	1	-V/8
0	1	0	-2V/8
0	1	1	-3V/8
1	0	0	-4V/8
1	0	1	-5V/8
1	1	0	-6V/8
1	1	1	-7V/8

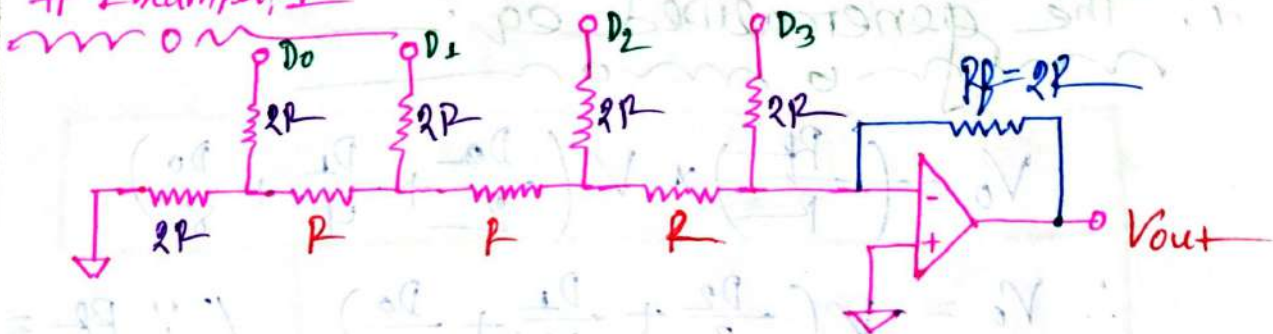
Sub: _____

Day _____

Time: _____

Date: / /

Example 1:



⇒ Find the value of V_o for (i) $D_3 = 1, D_2 = D_1 = D_0 = 0$

(ii) $D_2 = 1, D_3 = D_1 = D_0 = 0$

(iii) $D_1 = 1, D_3 = D_2 = D_0 = 0$

(iv) $D_0 = 1, D_3 = D_2 = D_1 = 0$

⇒ we know: _____

$$V_o = -\left(\frac{R_f}{R}\right) \left(\frac{D_3}{2} + \frac{D_2}{4} + \frac{D_1}{8} + \frac{D_0}{16} \right) \times V$$

∴ Now, Case 1: _____

$D_3 = 1, D_2 = 0, D_1 = 0, D_0 = 0$

$$\begin{aligned} \therefore V_o &= -\left(\frac{2R}{R}\right) \times \left(\frac{1}{2}\right) V \\ &= -(2) \times \left(\frac{1}{2}\right) \times 5 \\ &= -5V \end{aligned}$$

hence,

$V = 5V$

$D_3 = 1$

$R_f = 2R$

$R = R$

∴ Now Case 2: _____

$D_3 = 0, D_2 = 1, D_1 = 0, D_0 = 0$

Sub: _____

Day _____

Time: _____

Date: / /

$$\therefore V_0 = - \left(\frac{2R}{R} \right) \times \left(\frac{1}{4} \right) \times 5$$
$$= -2.5V$$

Since,
 $D_2 = 1$

Now, Case 3:—

$$D_3 = 0, D_2 = 0, D_1 = 1, D_0 = 0$$

$$\therefore V_0 = - \left(\frac{2R}{R} \right) \times \left(\frac{1}{8} \right) \times 5$$
$$= -1.25V$$

Since,
 $D_1 = 1$

$\therefore$ Now Case 4:—

$$D_3 = 0, D_2 = 0, D_1 = 0, D_0 = 1$$

$$\therefore V_0 = - \left(\frac{2R}{R} \right) \times \left(\frac{1}{16} \right) \times 5$$
$$= -0.625V$$

Since,
 $D_0 = 1$

Sub: _____

Day _____

Time: _____

Date: / /

Resolution of DAC

Resolution of n-bit DAC:

$$RDAC = \frac{1}{2^n - 1} \times 100$$

problem 1:-

⇒ Determine the resolution expressed as a percentage of the following:-

(a) An 8-bit DAC

(b) A 12 bit DAC

⇒ We know:-

$$RDAC = \frac{1}{2^n - 1} \times 100$$

(a)

For 8-bit DAC:-

$$RDAC = \frac{1 \times 100}{2^8 - 1} = 0.392\%$$

(b)

For 12-bit DAC:-

$$RDAC = \frac{1}{2^{12} - 1} \times 100 = 0.0244\%$$

Sub : _____

Day

Time : _____

Date : / /

Analog
To DIGITAL
CONVERSION (ADC)

Sub: _____

Day

Time: _____

Date: / /

Flash ADC

⇒ Flash ADC is fastest among all ADC. It takes only one clock cycle to have the conversion.

Structure of Flash ADC

⇒ Consist three major components.

1. High speed comparator $[2^n - 1]$

2. Resistive voltage divider Network $[2^n]$

3. Priority Encoder $[1]$

8 bit Flash ADC

⇒ Is given:

$$n = 8.$$

So, 1. Need comparator $= 2^n - 1 = 2^3 - 1 = 8 - 1 = 7$

2. No. of Resistor $= 2^n = 2^3 = 8$

3. No. of Encoder $= 8$ to 3 line priority Encoder
[∵ 8 bit]

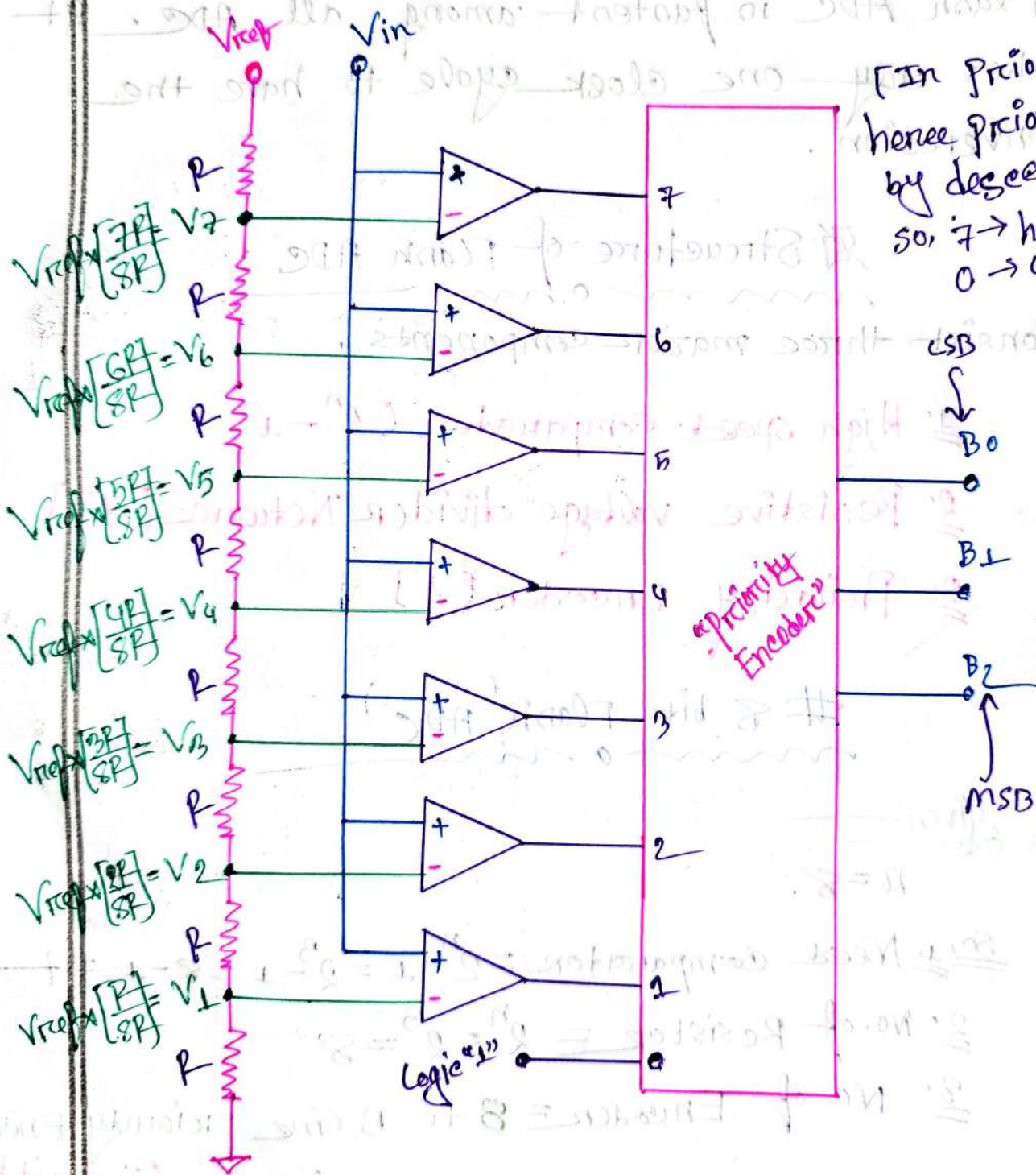
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Day _____

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Date: / /

Logic Circuit:



[In Priority encoder, hence priority is given by descending order. so, 7 → highest 0 → lowest]

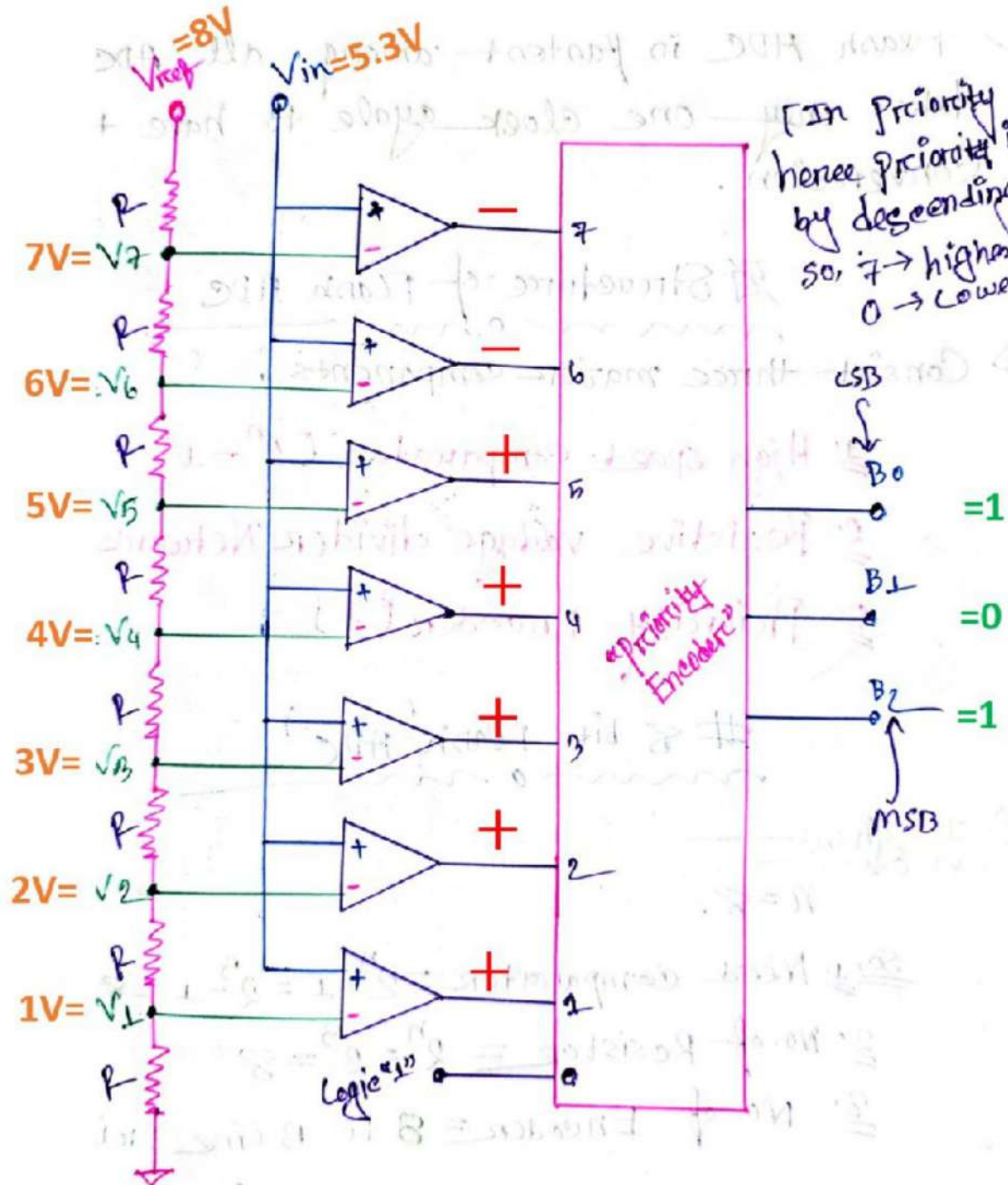
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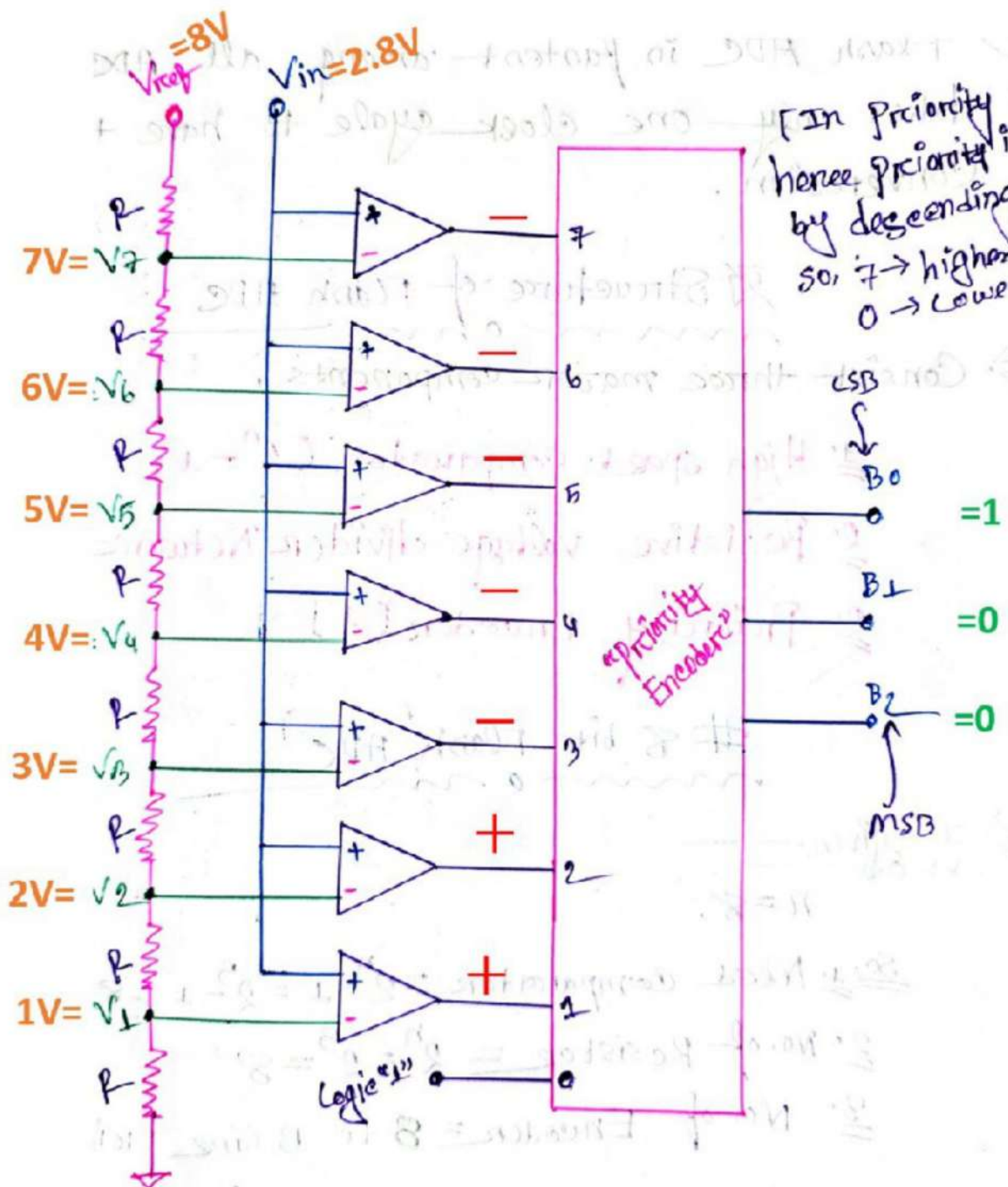
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Example 1:



[In Priority encoder, hence priority is given by descending order. so, 7 $\rightarrow$ highest 0 $\rightarrow$ lowest]

Example 2:



[In Priority encoder,
hence priority is given
by descending order.
so, 7 $\rightarrow$ highest
0 $\rightarrow$ lowest]

Sub : _____

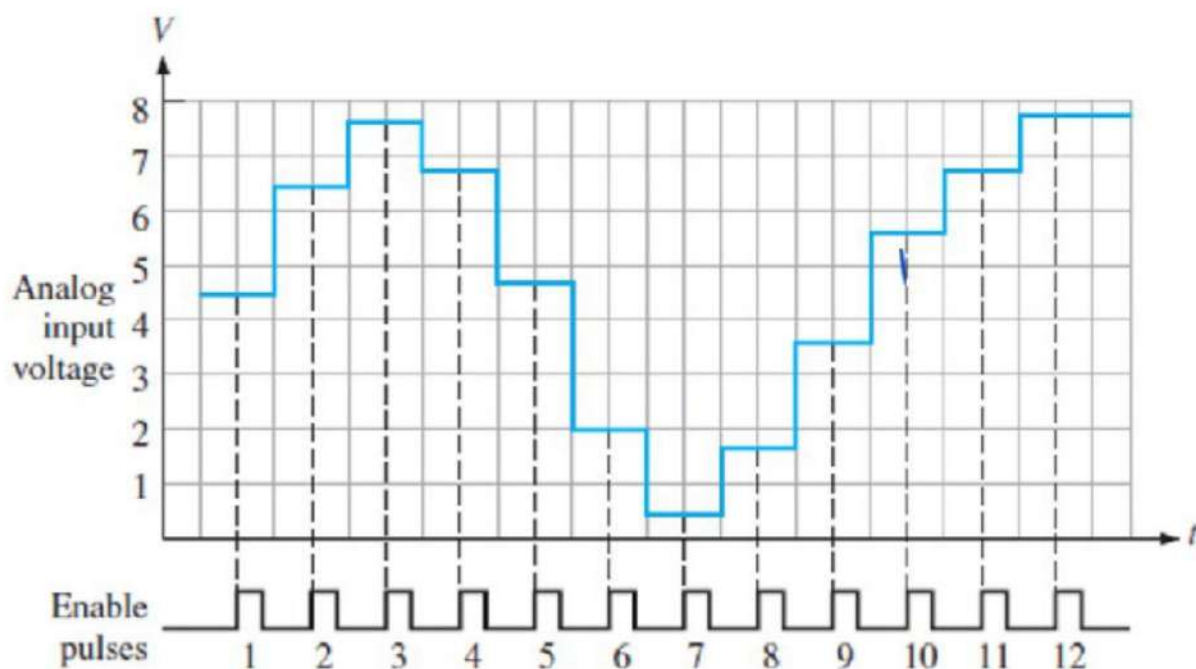
Day _____

Time : _____

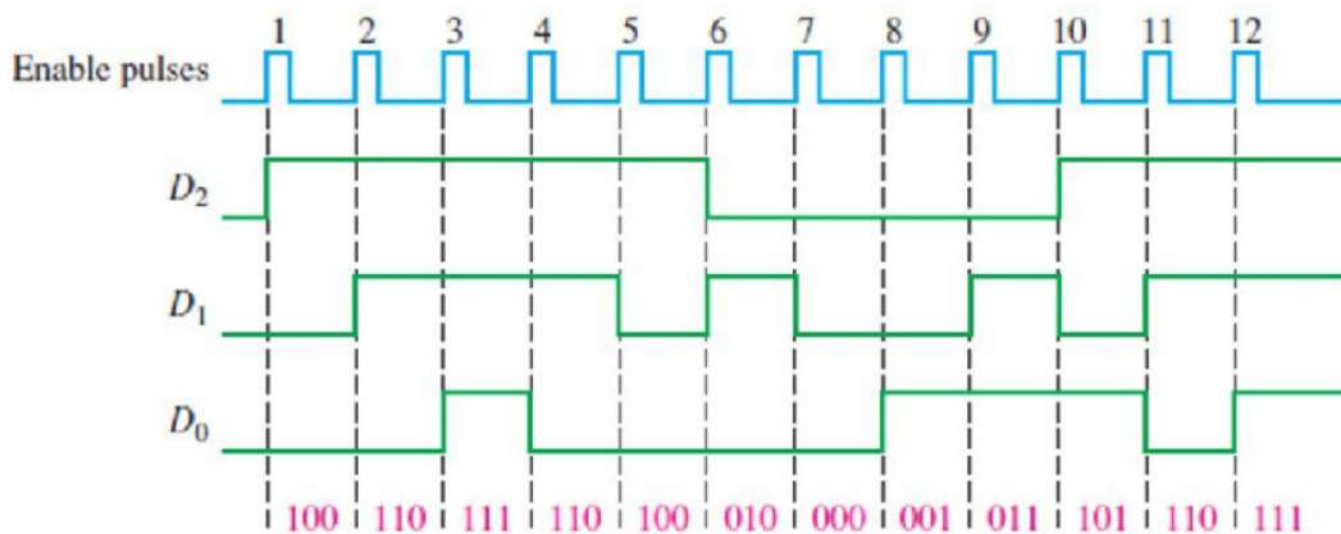
Date : / /

✂ Example 3:

⇒ Determine the binary code output of a **3-bit ADC** for the input signal and encoder enable pulses shown. For this example, $V_{REF} = +8V$



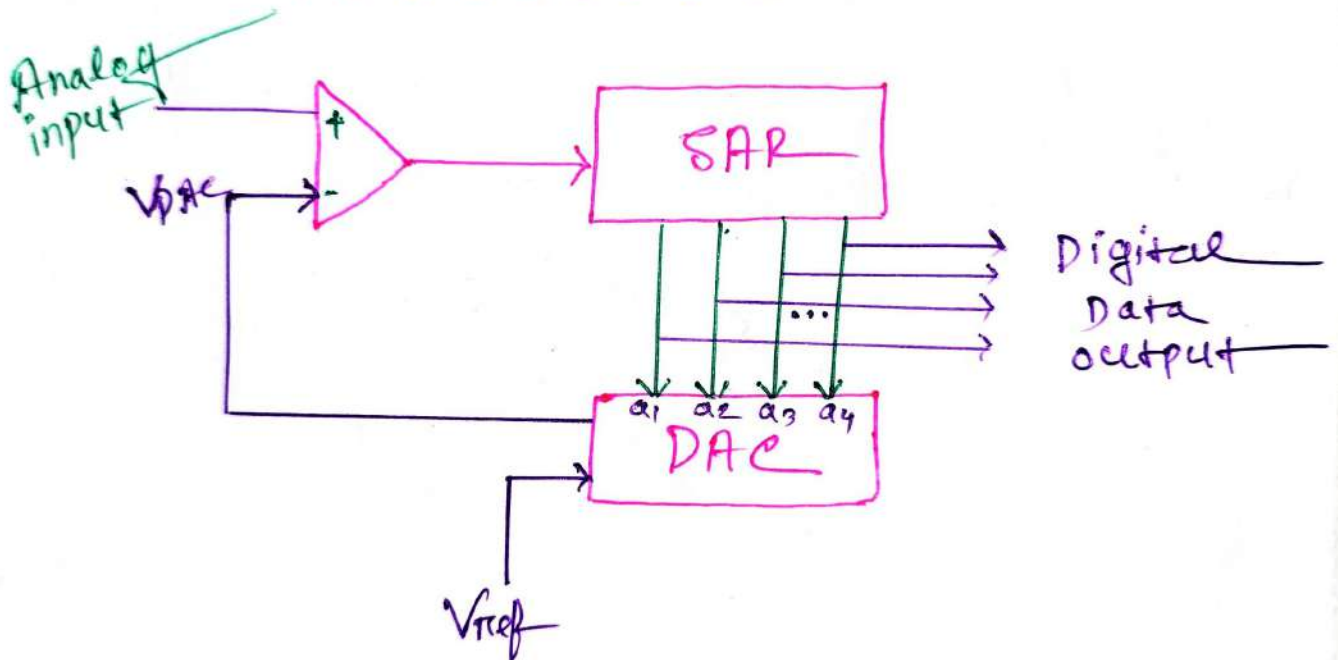
→ Solution:



Successive Approximation ADC

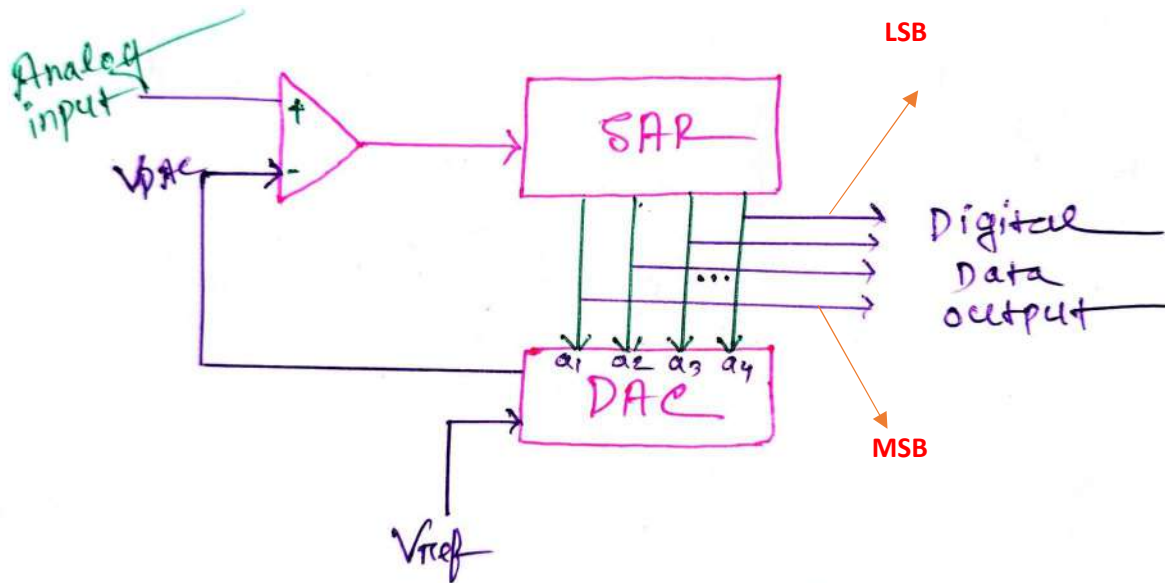
⇒ This ADC reduces conversion time and its conversion time does not depend on input analog voltage.

Structure of Successive A/D ADC



$$\therefore V_{DAC} = V_{ref} \left[\frac{a_1}{2} + \frac{a_2}{4} + \frac{a_3}{8} + \frac{a_4}{16} \right]$$

3 Bit Structure of Successive A.ADC



$$\therefore V_{DAC} = V_{ref} \left[\frac{a_1}{2} + \frac{a_2}{4} + \frac{a_3}{8} + \frac{a_4}{16} \right]$$

- It starts by MSB, SAR = 1000
- In next clock pulse, if $V_{in} > V_{DAC}$ then SAR = 1100
- In next clock, if $V_{in} < V_{DAC}$ then SAR = 0100 & Continue

Problem 1:

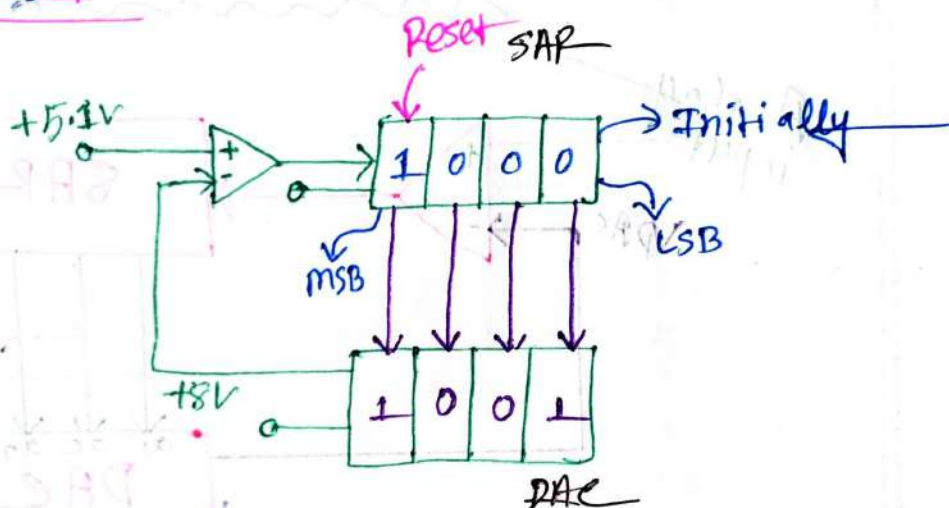
⇒ Construct 6-3 = 3 bit analog to digital converter (ADC) using successive approximation method. Show the step wise conversion of an analog signal of 5.1V into 3 bit digital binary value.

⇒ For 1st clk pulse:

$$V_{in} < V_{DAC}$$

$$5.1 < 8$$

So, Reset

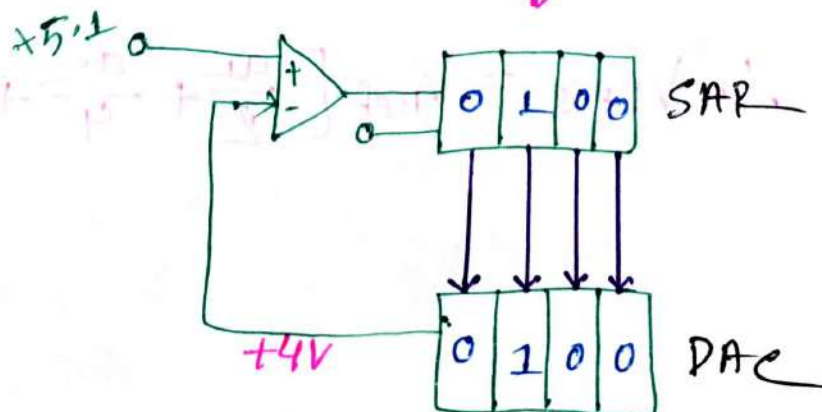


⇒ For 2nd clk-pulse:

$$V_{in} > V_{DAC}$$

$$5.1 > 4V$$

So, keep



Sub: _____

Day _____

Time: _____

Date: / /

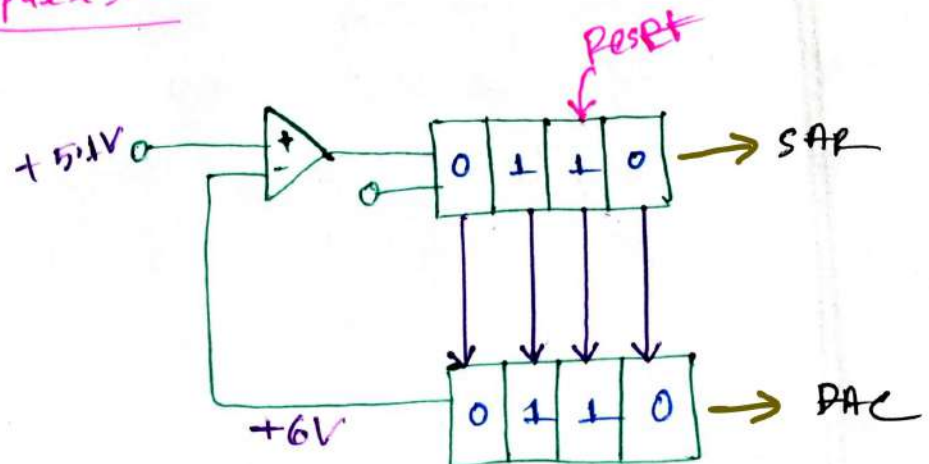
For 3rd clk-pulse

$V_{in} < V_{DAC}$

$5.1 < 6$

So

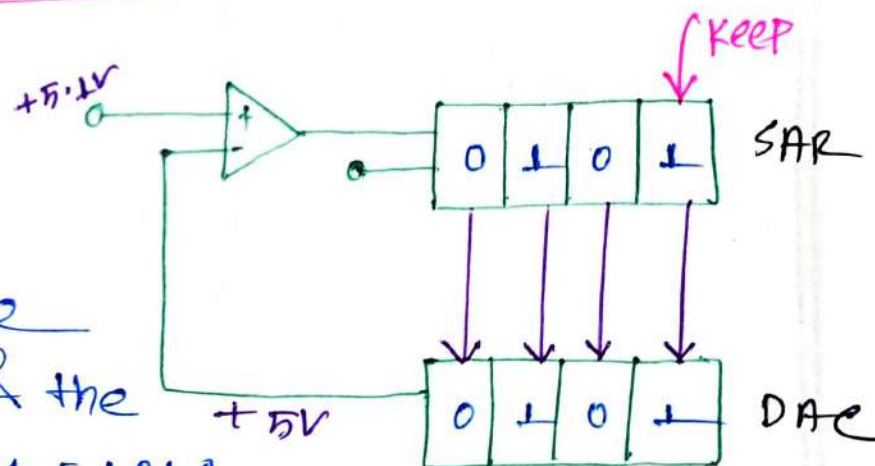
Reset



For 4th clk-pulse

$V_{in} = V_{DAC}$

$5.1 = 5$



So After 4 clk pulse
All bits are tested & the
Digital binary for $5.1 = 1010$,
representing the 5V.